

IMMUNOLOGICAL & BIOLOGICAL PRODUCTS

Introduction: Biological products

 They are **medicinal products derived from biological sources**

 ***Biological products:*** products produced through

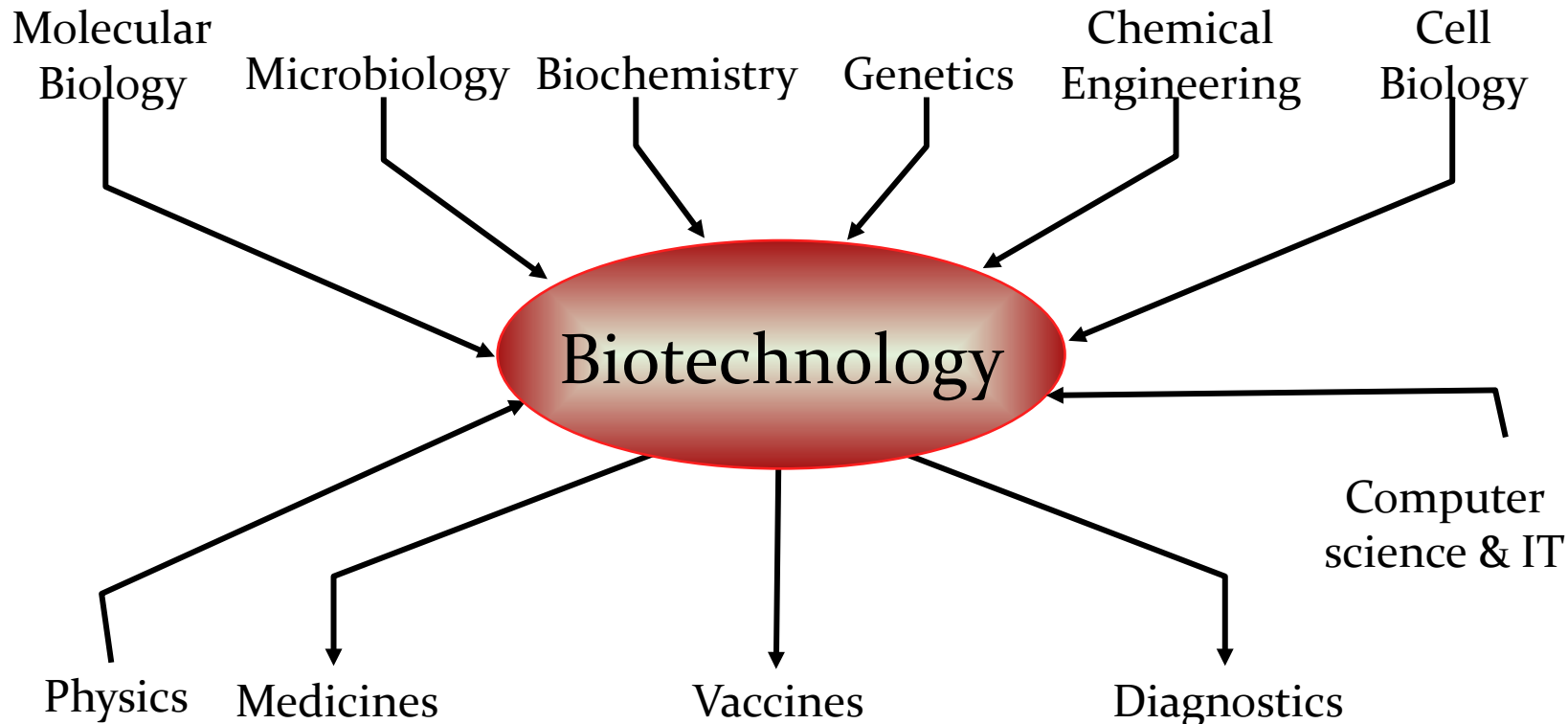
biotechnology/genetic engineering. These include:

- ✓ Protein based pharmaceuticals: Protein and peptide drugs
- ✓ Cytokines (glycoproteins: interleukins and interferones)
- ✓ Hormones: growth hormones, follicle stimulating hormones
- ✓ Enzymes and enzyme activators: plasminogen activator
- ✓ Antibody based therapeutics and diagnostics: monoclonal and polyclonal antibodies
- ✓ Nucleic acid based therapeutics: oligonucleotides and antisense molecules
- ✓ Recombinant vaccines
- ✓ Recombinant toxins and antitoxins, etc

Biotechnology

Definition: the application of biological organisms, processes or systems to the manufacturing and service industries.

◆ Biotechnology is an applied science that uses the knowledge of other applied sciences and basic sciences.



Areas of interests in biotechnology

- ◆ Recombinant DNA technology/gene cloning
- ◆ Hybridoma technology; for the production of monoclonal antibodies
- ◆ Enzyme and biocatalysts technique
- ◆ Plant and animal cell technology
- ◆ Fermentation technology
- ◆ Waste treatment and utilization technology
- ◆ Process engineering
- ◆ Production of biopharmaceuticals

Journey from biotechnology to pharmaceutical biotechnology

1917: The term biotechnology was defined for the first time

1943; Large scale production of penicillin using fermentation technology was started

1944: Genetic material and DNA for all organisms except viruses was identified

1953: Structure of DNA was revealed (Watson & Crick)

1973: Recombinant DNA technology was developed

1975: Monoclonal antibodies were developed

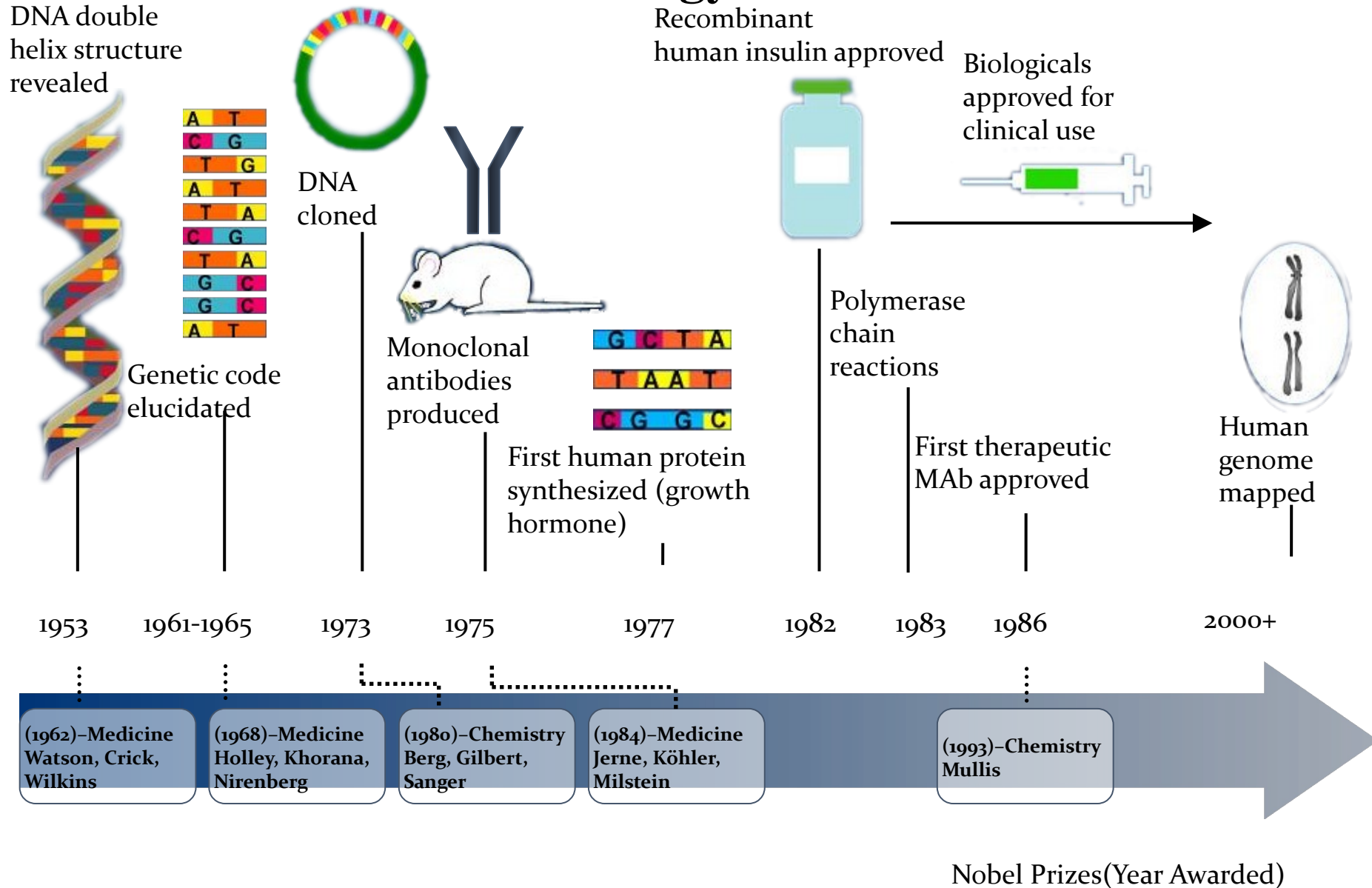
1978: Recombinant human insulin was produced

- ✓ The 1st biopharmaceutical product

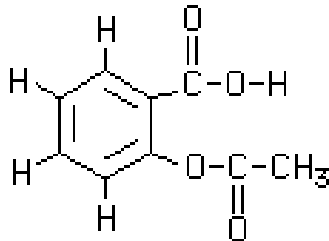
- ✓ The 1st genetic engineering derived pharmaceutical

1983: Polymerase Chain Reaction(PCR) was discovered.

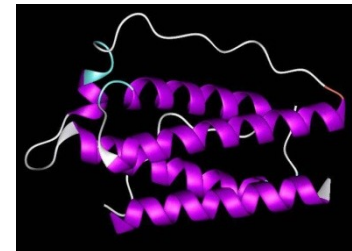
Evolution of Biotechnology



Comparing Traditional Medical Discovery with Biotechnology



- **Traditional Medicines**
- Synthetic small molecules produced through chemical processes
- Identified through high-throughput screens of >10,000 randomly generated compounds
- Function predominantly intracellularly
- Low specificity for targets

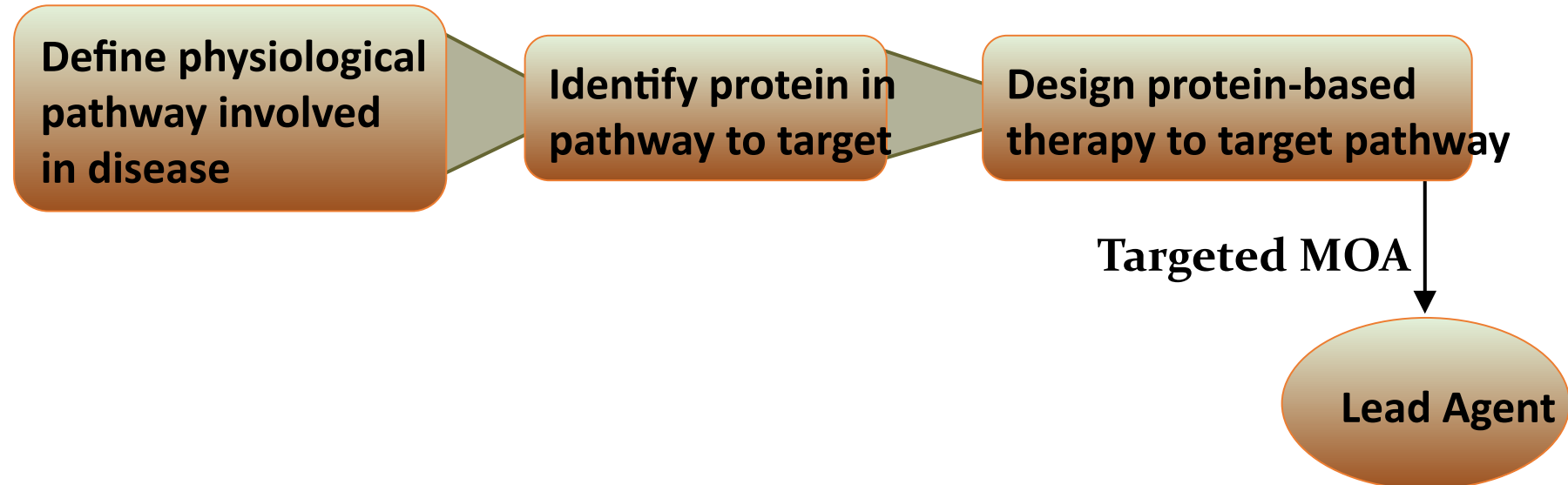


Biopharmaceuticals

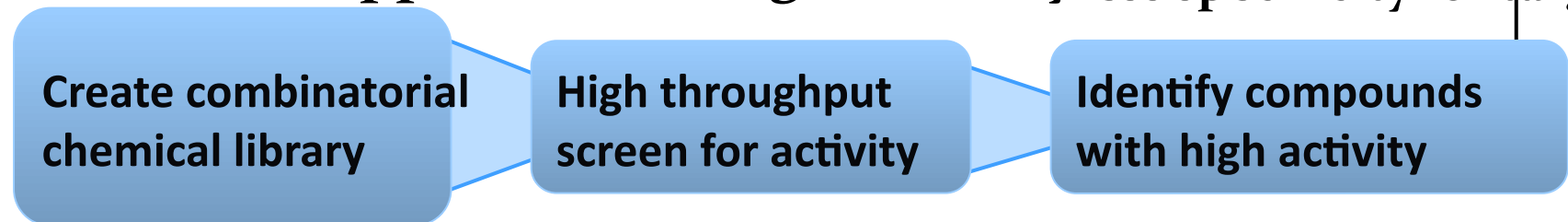
- Biological macromolecules produced by living systems
- Created from amino acids and nucleic acids
- Rely on defined biological pathways
- Function predominantly extracellularly
- Have a targeted mechanism of action

Unique Approaches to Medical Discovery

Biotechnology approach to drug discovery



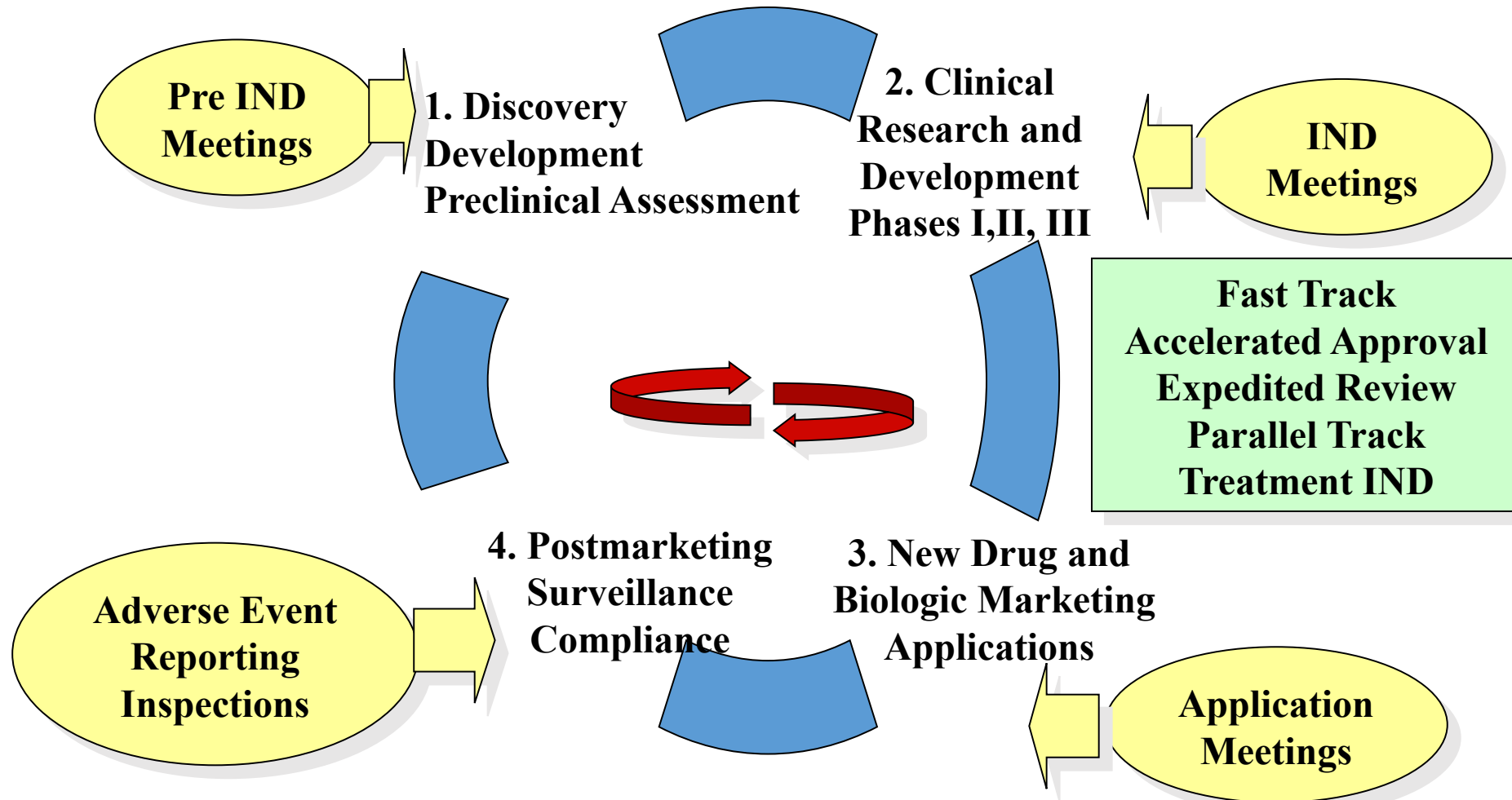
Traditional approach to drug discovery



Discovery of biopharmaceuticals

- The discovery of virtually all biopharmaceuticals was a knowledge basedone
- Continuing advances in the molecular sciences towards understanding of the molecular mechanisms that underline health and disease
- An understanding at the molecular level of how the body functions in health and deviations during development of a disease
- The ‘discovery’ of biopharmaceuticals, in most cases, merely relates to the logical application of rapidly increasing knowledge of the biochemical basis of how the body functions
- The impact of genomics and related technologies enabling the complete sequencing of entire DNA complement of the cell and to map genome arrangement

Drug and Biologic Development Cycle



Human Clinical Trial Phases

Phase I studies

- Assess the safety of a drug or device.
- take several months to complete, healthy volunteers (20 to 100)
- The study is designed to determine the effects of the drug or device on humans including how it is absorbed, metabolized, and excreted.

Phase II studies

- ❖ test the efficacy of a drug or device.
- ❖ last from several months to two years, and involves several hundred patients.
- ❖ About one-third of experimental drugs successfully complete both Phase I and Phase II studies.

Phase III studies

- ✓ involve several hundred to several thousand patients.
- ✓ understanding of the effectiveness of the drug or device, the benefits and the range of possible adverse reactions.
- ✓ 70% to 90% of drugs that enter Phase III studies successfully complete this phase of testing.

Phase IV studies,

- Post Marketing Surveillance Trials, are conducted after a drug or device has been approved for consumer sale.
 - (1) to compare a drug with other drugs already in the market;
 - (2) to monitor a drug's long-term effectiveness and impact on a patient's quality of life; and
 - (3) to determine the cost-effectiveness of a drug therapy relative to other traditional and new therapies.

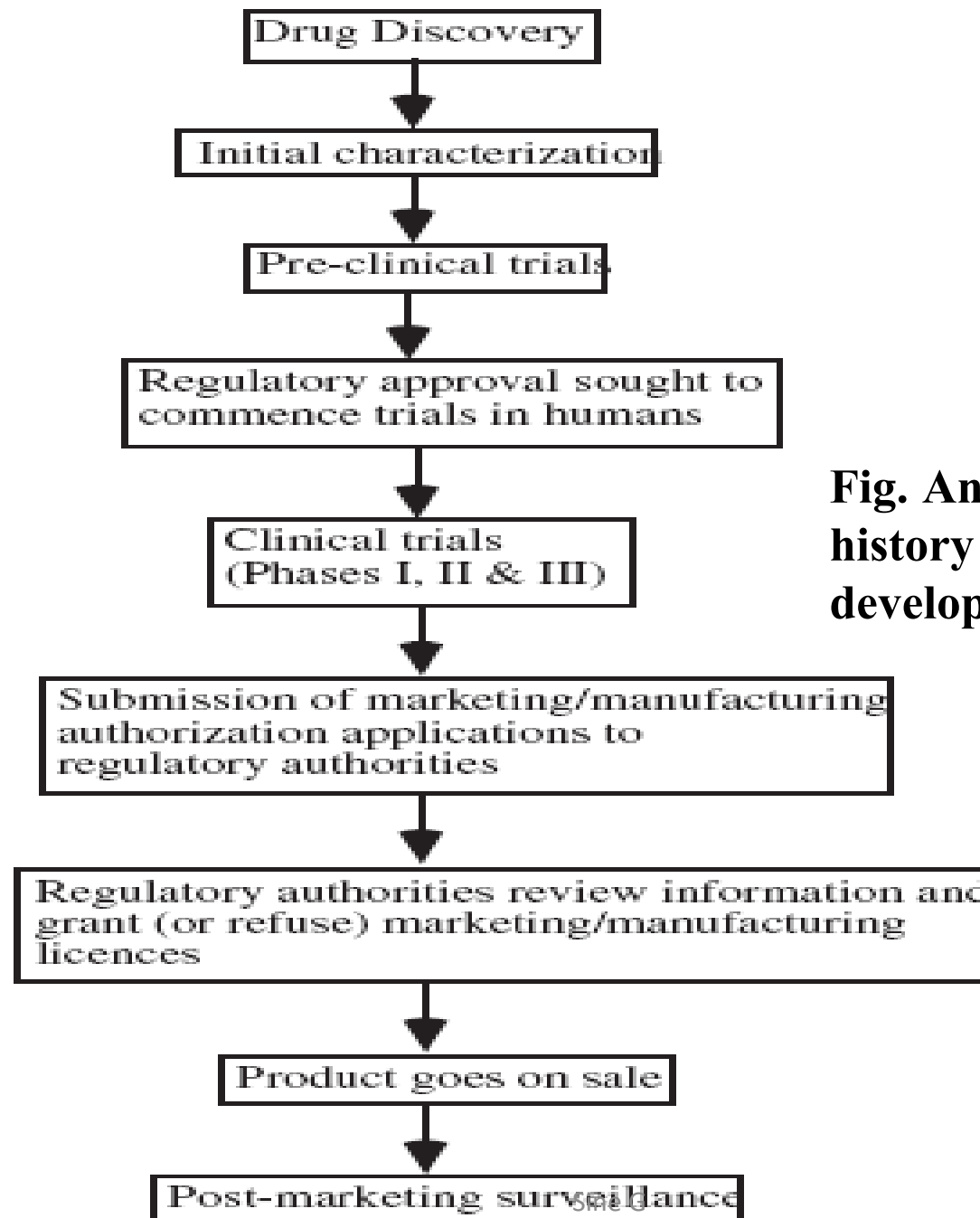


Fig. An overview of the life history of a successful drug development.

Sources of biopharmaceuticals

- The bulk of biopharmaceuticals currently on the market are produced by genetic engineering using various recombinant expression systems
- Most of the recombinant proteins that have gained marketing approval are produced either in recombinant *E. coli* or in recombinant mammalian cell lines

Expression systems used for the production of recombinant biopharmaceuticals
<i>E. coli</i> (and additional prokaryotic systems, e.g. bacilli)
Yeast (particularly <i>S. cerevisiae</i>)
Fungi (particularly aspergilli)
Animal cell culture (particularly CHO and BHK cell lines)
Transgenic animals (focus thus far is upon sheep and goats)
Plant-based expression systems (various)
Insect cell culture systems

Some biopharmaceuticals currently on the market which are produced by genetic engineering in either *E. coli* or animal cells

Biopharmaceutical	Source	Biopharmaceutical	Source
tPA	<i>E. coli</i> , CHO	FSH	CHO
Insulin	<i>E. coli</i>	IFN- β	CHO
		EPO	CHO
IFN- α	<i>E. coli</i>	Glucocerebrosidase	CHO
IFN- γ	<i>E. coli</i>	Factor VIIa	BHK
IL-2	<i>E. coli</i>		
G-CSF	<i>E. coli</i>		
hGH	<i>E. coli</i>		

E. coli displays a number of advantages

- ✓ *E. coli* has long served as the model system for studies relating to prokaryotic genetics.
- ✓ High levels of expression of heterologous proteins can be achieved in recombinant *E. coli*
- ✓ *E. coli* cells grow rapidly on relatively simple and inexpensive media, and the appropriate
- ✓ fermentation technology is well established

Recombinant therapeutic proteins approved for general medical use that are produced in *S. cerevisiae*.

Trade name	Description	Use
Novolog	Engineered short-acting insulin	Diabetes mellitus
Leukine	Granulocyte macrophage CSF (GM-CSF)	Bone marrow transplantation
Recombivax, Comvax, Enderix B, Tritanrix-HB, Infanrmix, Twinrix, Primavax, Hexavax	All vaccine preparations containing rHBsAg as one component	Vaccination
Revasc, Recludan	Hirudin	Anticoagulant
Fasturtec	Urate oxidase	Hyperuricaemia
Regranex	PDGF	Diabetic ulcers

Yeast cells (particularly *Saccharomyces cerevisiae*) display a number of characteristics such as: long history of industrial application, grow relatively quickly in inexpensive media, their tough outer wall protects them from physical damage, suitable for industrial-scale fermentation, possess the ability to carry out post-translational modifications of proteins.

Proteins of actual/potential therapeutic use that have been produced in the milk of transgenic animals

Protein	Animal species	Expression level in milk (mg l ⁻¹)
tPA	Goat	6000
IL-2	Rabbit	0.5
Factor VIII	Pig	3
Factor IX	Sheep	1000
α_1 -Antitrypsin	Goat	20 000
Fibrinogen	Sheep	5000
EPO	Rabbit	50
Antithrombin III	Goat	14 000
Human α -lactalbumin	Cow	2500
IGF-I	Rabbit	1000
Protein C	Pig	1000
GH	Rabbit	50

- ✓ *The generation of transgenic animals is undertaken by **directly microinjecting exogenous DNA into an egg cell***
- ✓ *This DNA will be stably integrated into the genetic complement of the cell*
- ✓ *After fertilization, the ova may be implanted into a surrogate mother*
- ✓ *Each cell of the resultant transgenic animal will harbour a copy of the transferred DNA*

WAP (whey acid protein), the most abundant protein found in mouse milk.

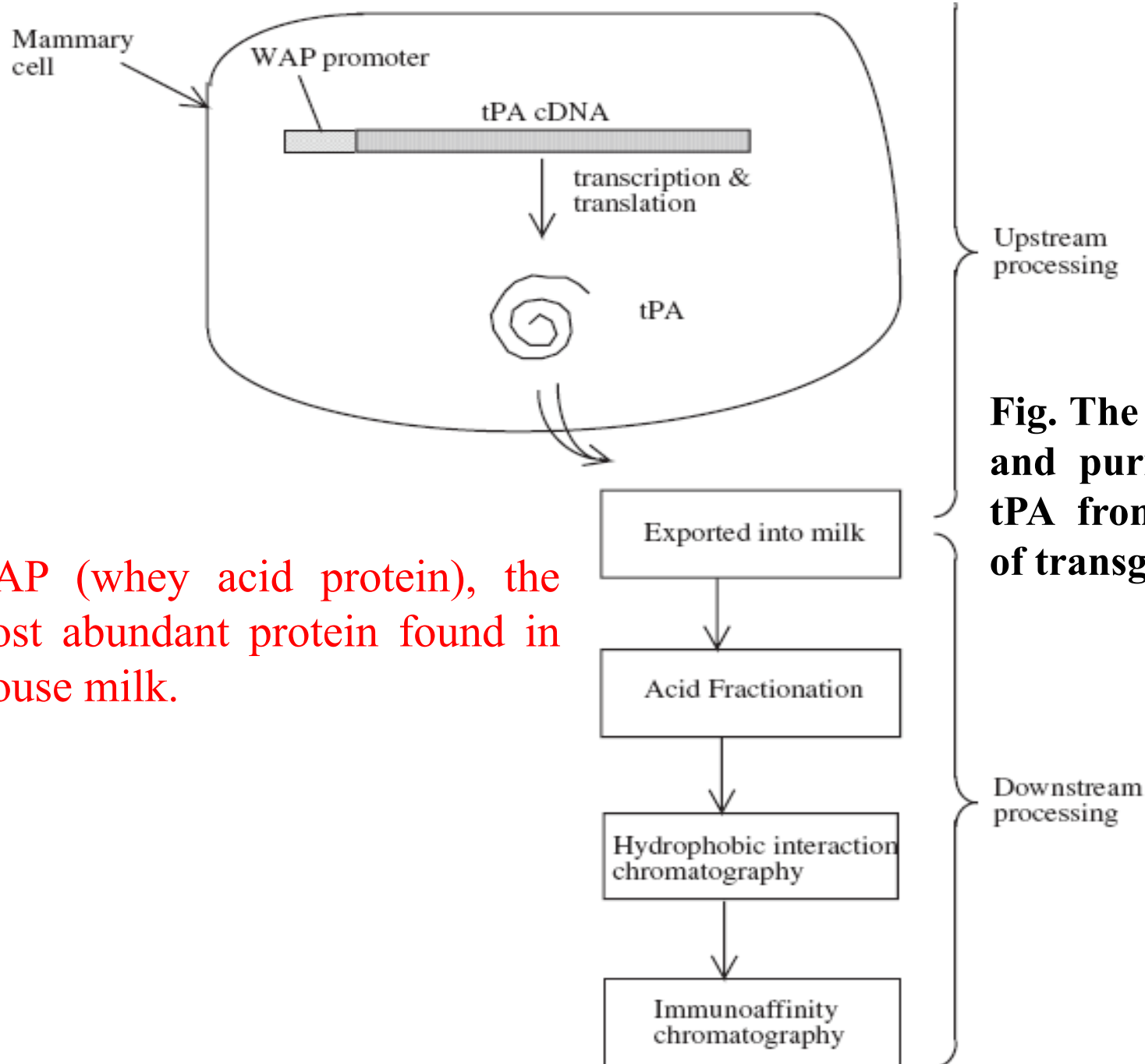


Fig. The production and purification of tPA from the milk of transgenic goats

Biological products from natural sources

☐ Traditionally, protein and nucleic acid based drugs have been extracted from their natural sources

☐ products from **the natural sources have limitations:**

- ❖ Natural sources are often rare and expensive
- ❖ Difficult to keep up with demand
- ❖ Hard to isolate the pure product
- ❖ Lead to immune reactions and are usually species specific
- ❖ Prone to viral & pathogen contamination

Role of recombinant DNA technology and monoclonal antibody technologies

▣ the advent of recombinant DNA technology and Monoclonal antibody technology boosted the development and production of potentially usable biological products.

▣ The potential advantages:

- Increased availability for increasing demand
- Improved product safety through high scale purification and modification and/or removing of potentially toxic portions
- Facilitated large scale production of clinically important biopharmaceuticals

Mechanism of production of recombinant biopharmaceuticals

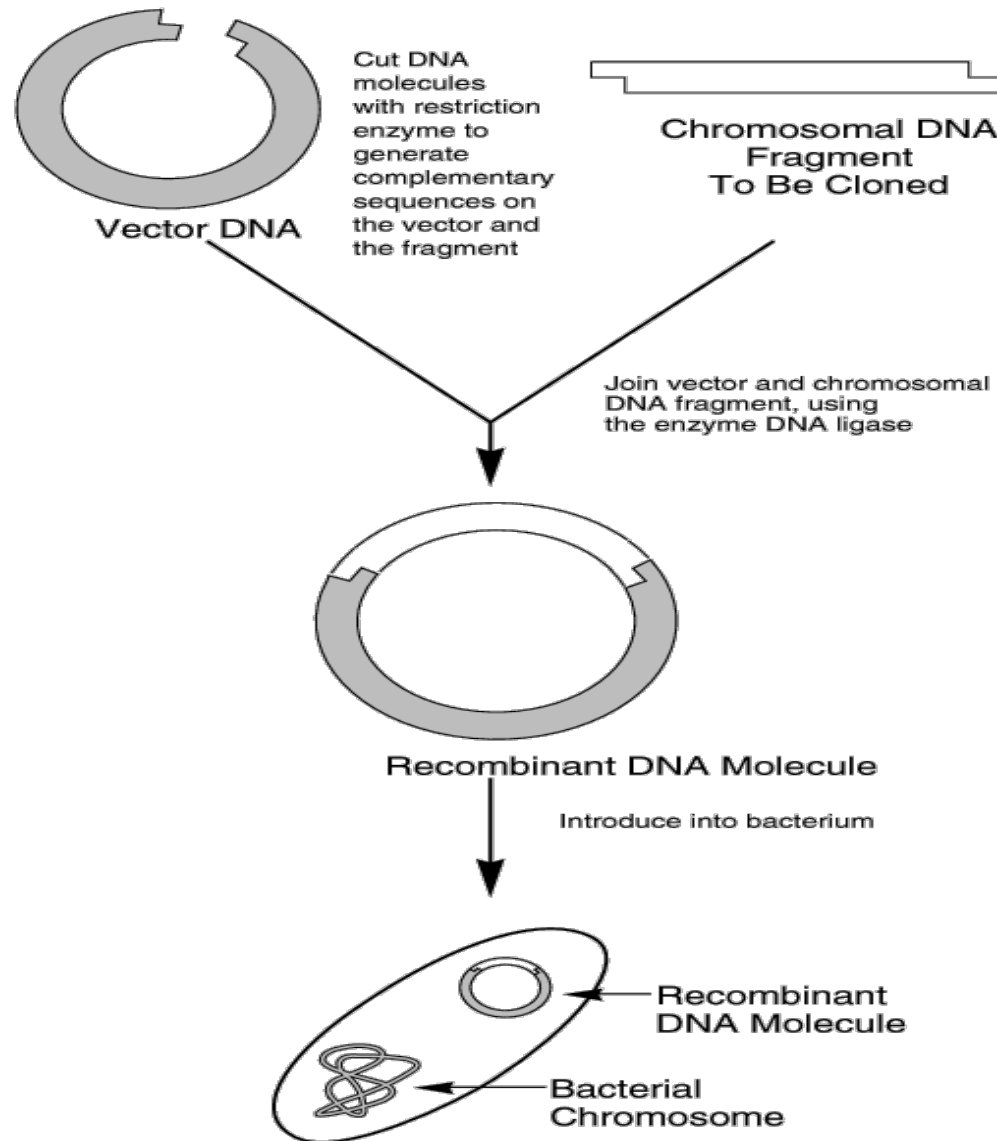
◆ These methods were first developed in 1970's & 1980's where notably, **Paul Berg (1973) found restriction enzymes** and **Herbert Boyer from Genentech (1978) developed the method of cloning human insulin into E. coli.**

◆ Two general approaches for the production are:

- **Expression in isolated cells**

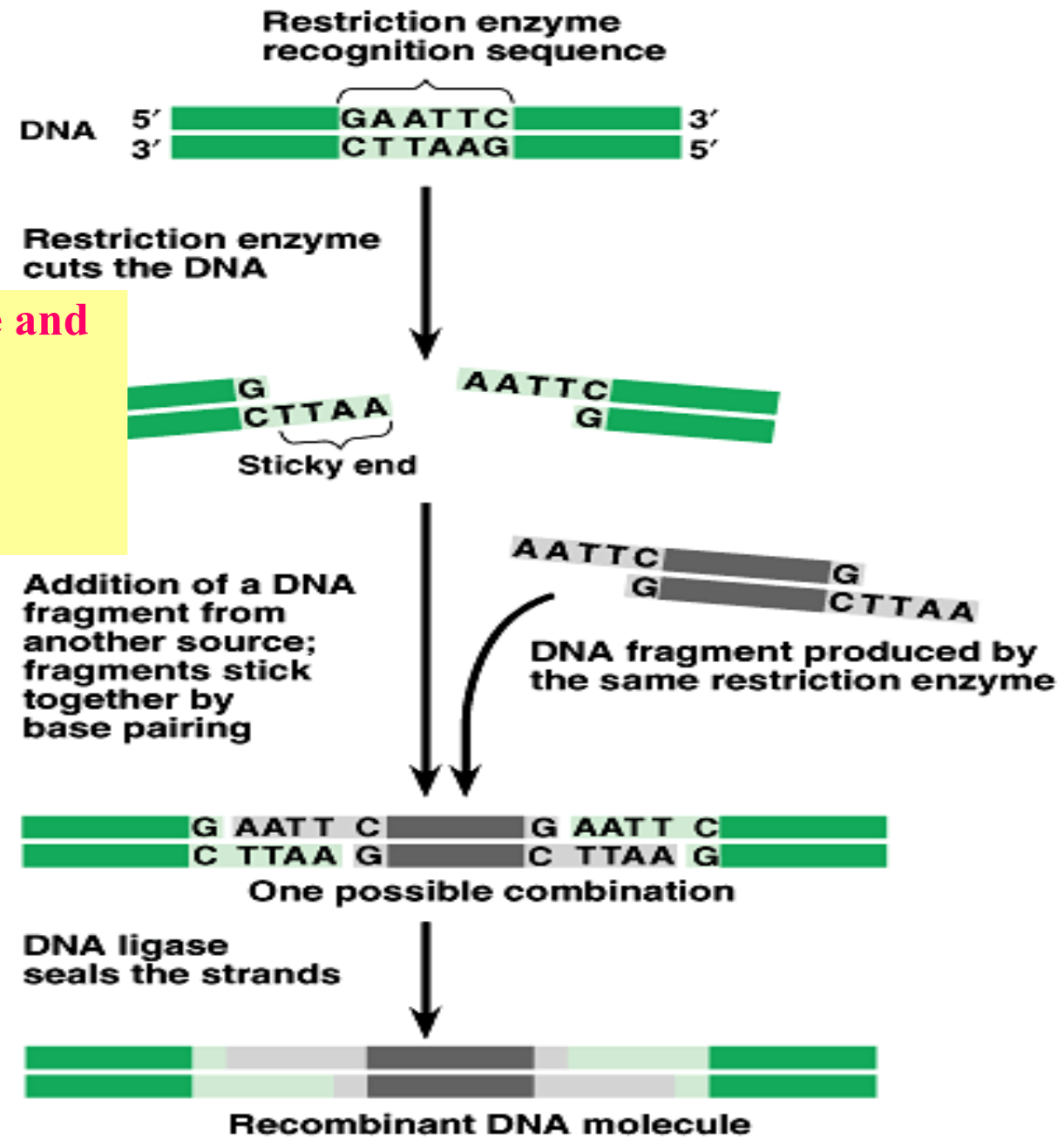
- **Expression in transgenic plants/animals**

Six steps of the recombinant process:



- ✓ *Isolation of gene of interest*
- ✓ *Introduction of gene to expression vector*
- ✓ *Transformation into host cells*
- ✓ *Growth of cells through fermentation*
- ✓ *Isolation & purification of protein*
- ✓ *Formulation of protein product*

Using a restriction enzyme and DNA Ligase to make recombinant DNA



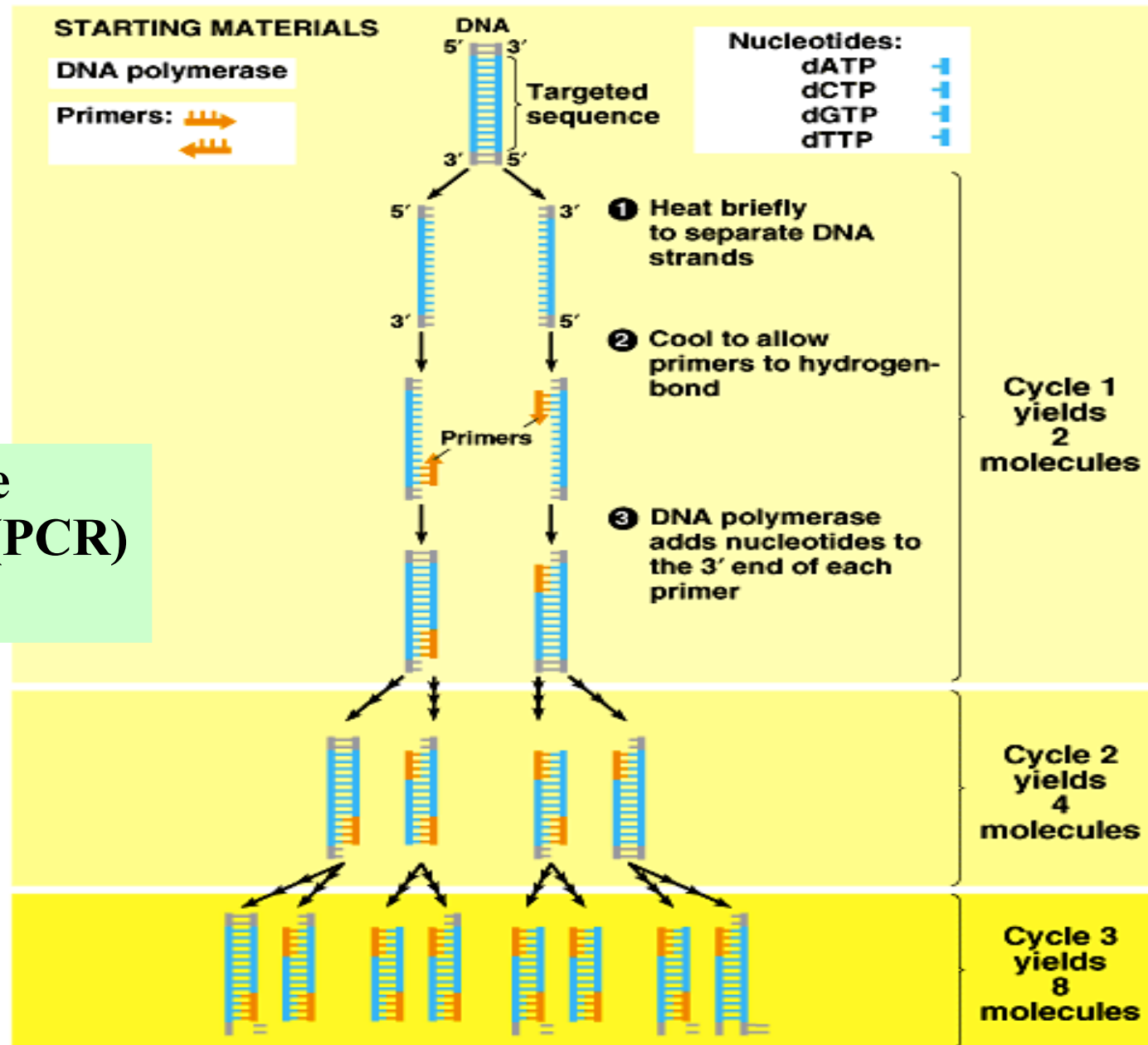
Polymerase chain reaction (PCR)

It is used to amplify DNA that lies between two regions of known sequences.

The PCR utilizes the following steps:

- 1. a DNA preparation containing the desired segment to be amplified**
- 2. Two nucleotide primers which are complementary to the 3' boarder of the desired segment**
- 3. Deoxyneucleotide triphosphates**
- 4. Heat stable DNA polymerase**
- 5. Magnesium salts used as co-enzymes**

The polymerase chain reaction (PCR)



Applications of PCR

- ➡ Amplify specific gene fragments present in genetic material
- ➡ Genetic diagnosis or detection of mutation
- ➡ Paternity testing i.e. to identify the father of a child
- ➡ Industrial quality control
- ➡ Forensic analysis at scene of crime
- ➡ Determine sex of embryo

Note: *PCR is an efficient, time saving and fully automated technique and doesn't require the use of restriction endonucleases*

Roles of Pharmaceutical biotechnology

1. Genetic improvement/modification of microorganisms for the production of medically and pharmaceutically important compounds
2. Development of recombinant vaccines for bacterial and viral diseases
3. Gene therapy for the treatment of genetic disorder
4. Development of noble drug delivery systems for protein and peptides, nucleic acid based therapeutics: vaccines, genes, biopharmaceuticals



Potential Applications

- Genetically modify mammals to produce **therapeutic peptide drugs** (e.g. insulin,)
- Isolate and purify drug from the milk
- Potentially a more cost effective method to produce pharmaceuticals

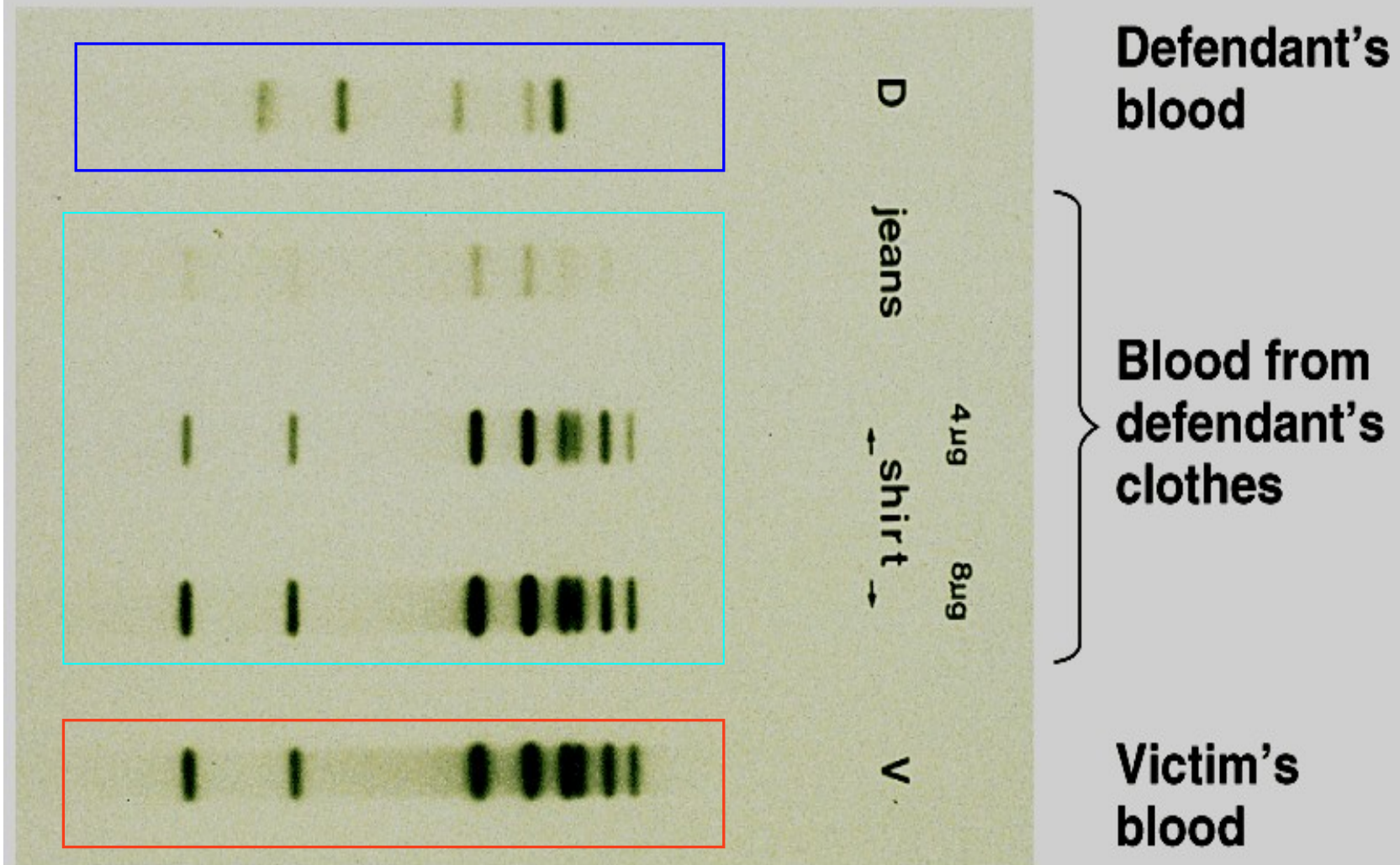


Transgenic Rice

- Genetically modify plants to produce **beta-carotene**
- **Beta Carotene** is converted to **vitamin A in humans**
- Vitamin A deficiency leads to poor vision and high susceptibility to disease

~70% of children <5 years old in SE Asia suffer from vit. A deficiency

DNA fingerprints from a murder case

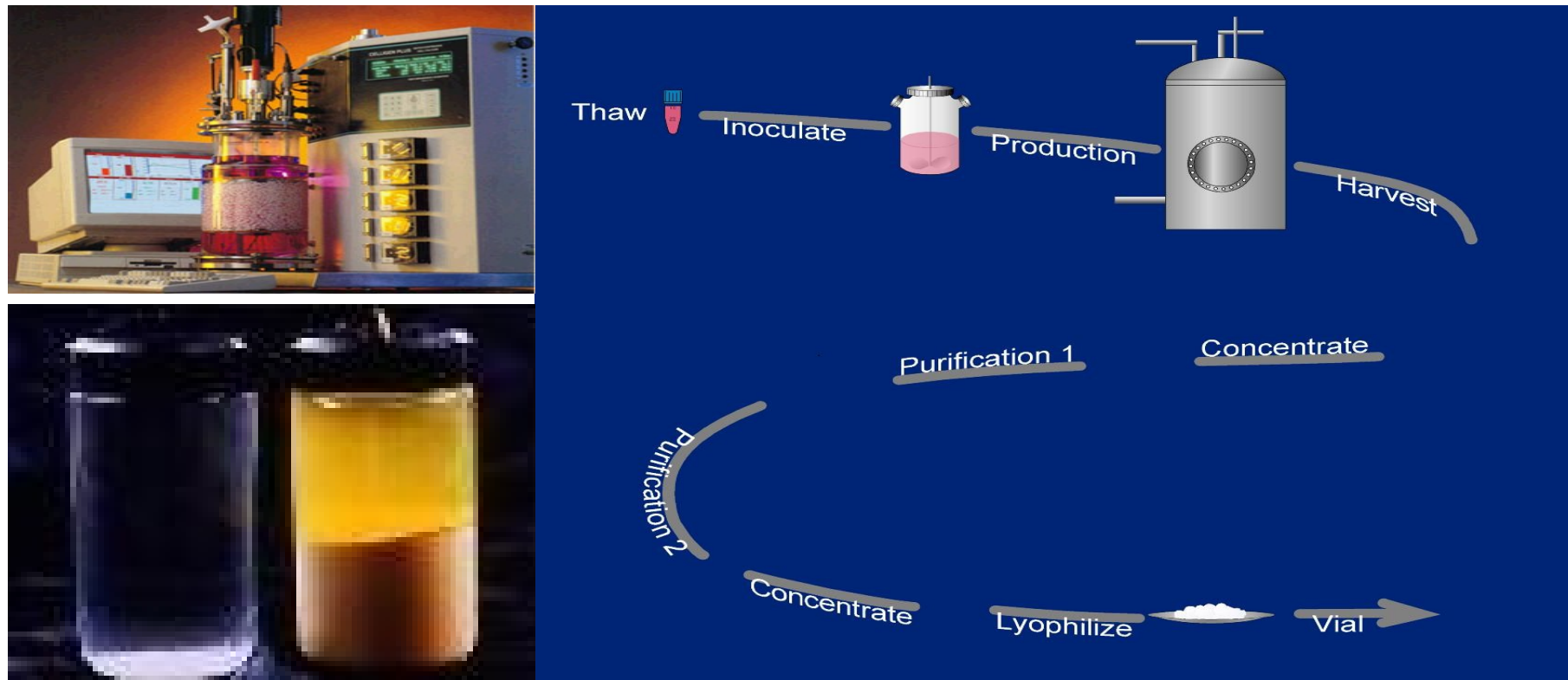


Whose blood is on the defendant's clothing?

Production of recombinant biopharmaceuticals

Upstream processing: All Cells can be grown in incubators and fermentors for mass cultivation

Downstream processing: the cultured and grown preparations pass through successive filtration and purification steps and finally dried and used for manufacturing.



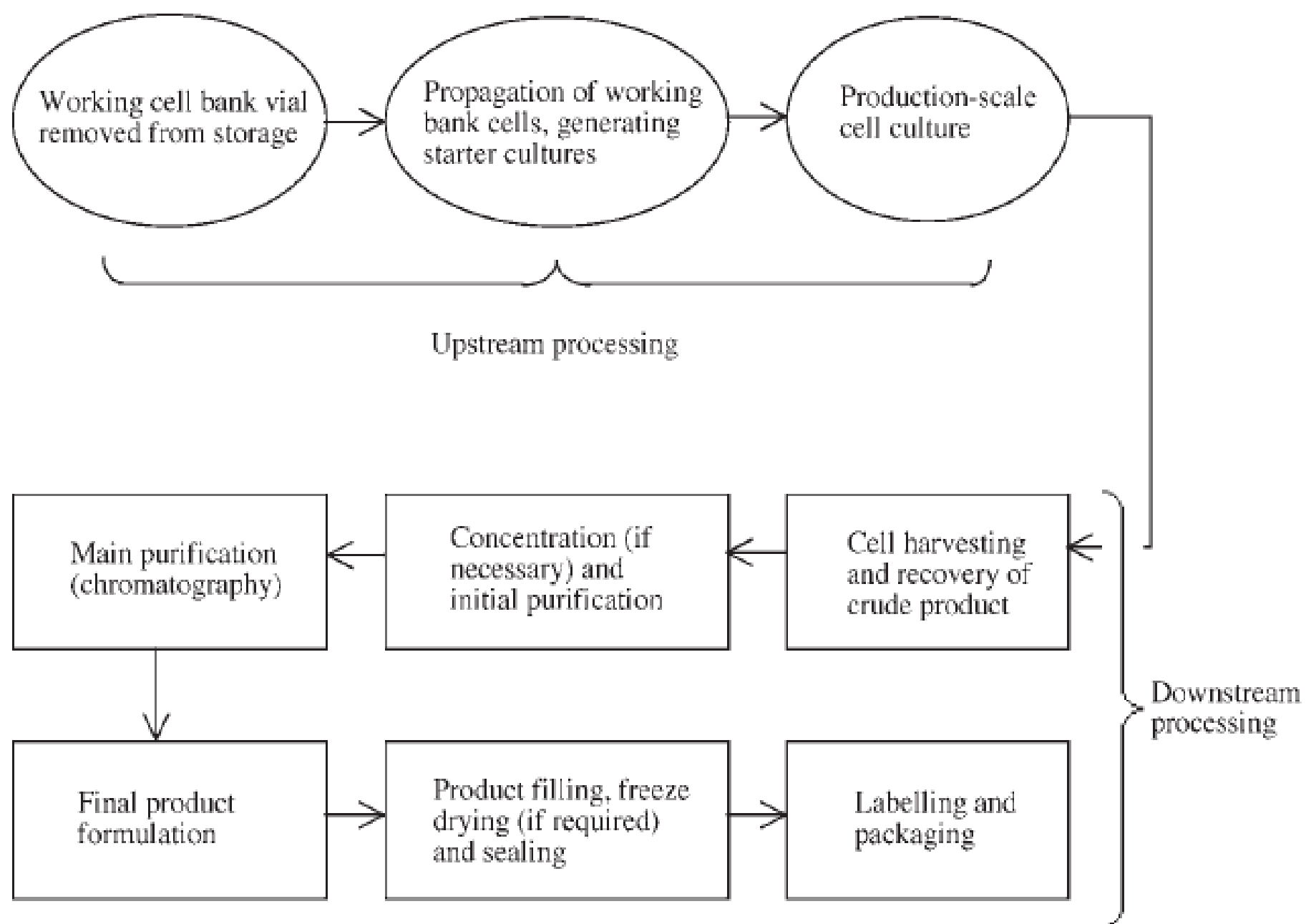
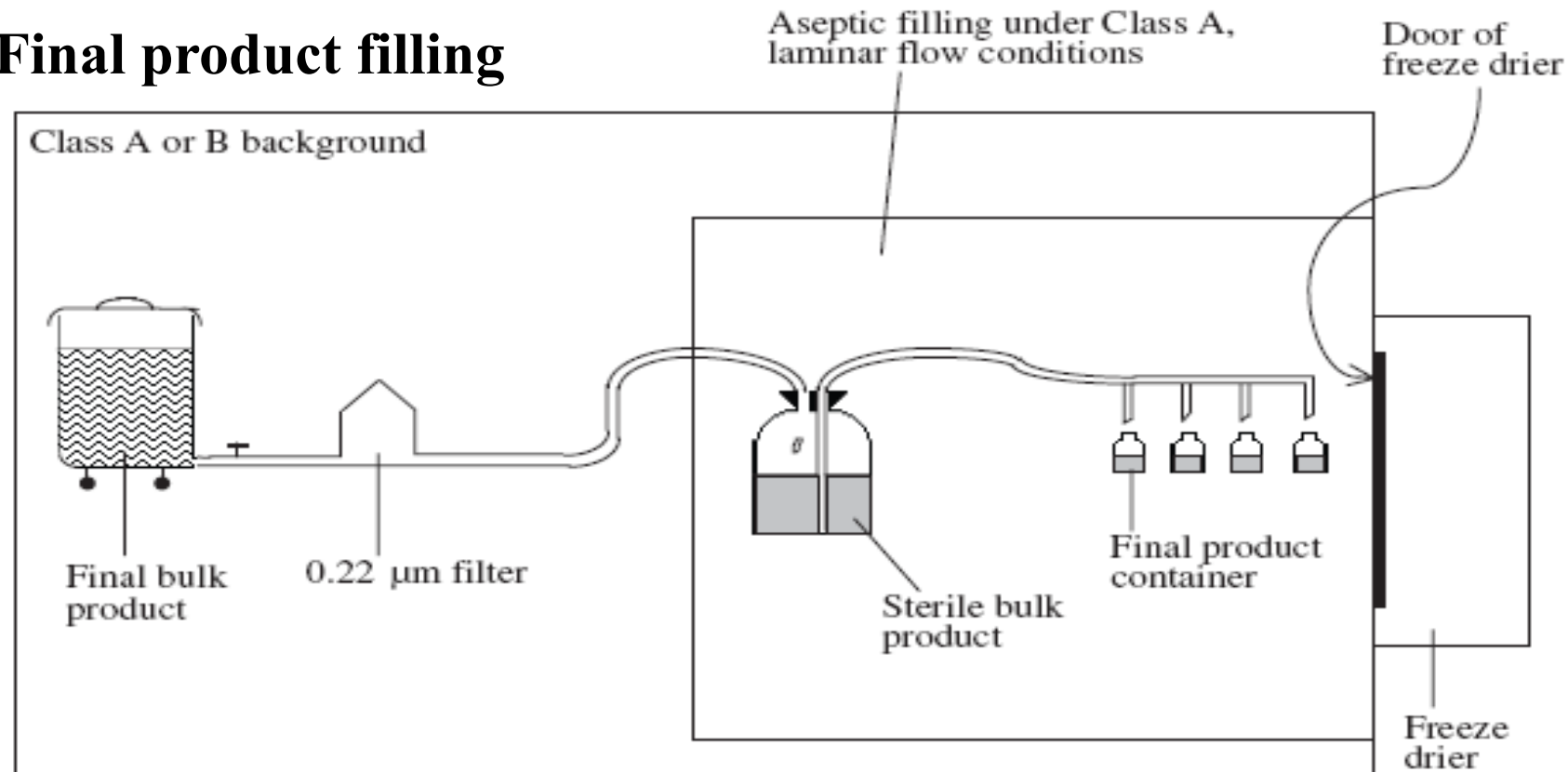


Fig. Overview of the production process for a biopharmaceutical product

Final product filling



- ✓ The final bulk product is filter sterilized by passing through a 0.22 µm filter
- ✓ The sterile product is aseptically filled into pre-sterile final product containers under grade A laminar flow conditions
- ✓ The product container is either sealed (by an automated aseptic sealing system), or freeze-dried first, followed by sealing

Storage of immunological/biological products

- Preservation of the potency of immunological/biological products involves:
 - ✓ *Maintaining the viability of living cells*
 - ✓ *Preventing denaturation of proteins(antigens and antibodies)*
- The cause of denaturation and/or loss of viability are usually:
 - ✓ *Chemical interactions/deterioration*
 - ✓ *Environmental factors like temperature and oxygen*
- For most products, the optimum storage temperature is just above the freezing point between 2 and 10 °C
- A number of viral vaccines like smallpox and oral poliomyelitis are more stable at or below their freezing point
- Bacterial vaccines and antitoxins seriously deteriorate if stored below freezing point because of:
 - ✓ *Mechanical damage by ice crystal*
 - ✓ *Adverse effects of high local concentration of inorganic salts and preservative bactericides*

- Freeze-dried vaccines are relatively more stable compared to corresponding liquid formulations but still needs careful storage.

- ✓ *Freeze-dried smallpox vaccine should be stored below 10 °C*

- ✓ *Freeze-dried yellow fever vaccine should be stored at about 0 °C*

- Light accelerate decomposition of biological products and hence should be protected

- Light resistant glasses are not usually used because they cause difficulties to identify:

- ✓ *Slight color changes*

- ✓ *Faint precipitates*

Dispensing and handling biotech products

- Pharmacists have been responsible for preparation and/or dispensing of biopharmaceuticals
- Biotechnology products are proteins subjected to denaturation and this require special handling techniques
- The shelf-life of biotech drugs is usually shorter
- These products need to stored at low temperature (refrigerator)
- Most biotech products need also to be shipped under ice box containers
- Once reconstituted, they should be kept under refrigerator until use

Dispensing and handling....

- Most biological products may adhere to plastics (PVC) or glass containers like syringes and infusion bags which may reduce their effectiveness
 - ✓ *Human serum albumin can be added to the solutions to prevent adherence*
- Biotech products should not be shaken while adding any diluent
 - ✓ *May cause the product to break down or aggregate and subsequent phase separation*
- Biotech products should be stored and shipped in a sealed container
- The products should also be protected from freezing
 - ✓ *There may be mechanical damage and denaturation*

Challenges for practicing pharmacists

- Many pharmacists are reluctant to provide pharmaceutical care service to patients who require therapy with biotechnology drugs for variety of reasons:
 - ✓ Most of biotech drugs are newer to the market
 - ✓ Lack of knowledge about the tools of biotechnology
 - ✓ Lack of understanding of the therapeutic aspects of recombinant protein products
 - ✓ Unfamiliarity of the side effects and patient counseling information on biotech drugs
 - ✓ Unfamiliarity with the storage, handling and reconstitution of biotech drugs
 - ✓ Difficulty of handling reimbursement issues

Responsibilities of practicing pharmacists

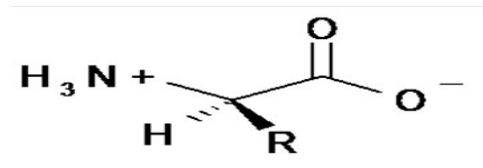
- To give effective pharmaceutical service to patients on biotech pharmaceuticals, pharmacists need to engage in areas of study like:
 - ✓ Basics of biotechnological and immunological methods
 - ✓ Protein chemistry
 - ✓ Pharmacological and therapeutic applications of biotechnology products including their indication, mechanism of action, side effects and adverse reaction management
 - ✓ Storage, handling, reconstitution and administration of biotechnology products
 - ✓ Pharmaceutical care for patients receiving biotech drug products

PROTEINS AND PEPTIDES AS THERAPEUTIC PRODUCTS

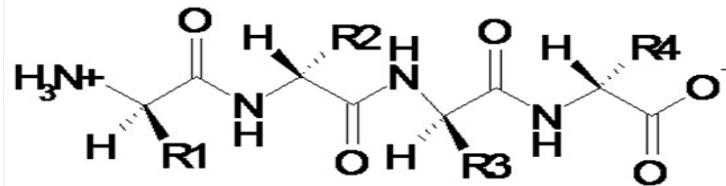
◆ From the different biopharmaceuticals, protein and peptide based products are the current and future focus of many research institutes and manufacturing industries because of their association with the anatomic and physiologic activities of the human body.

✓ *Polypeptides composed of covalently linked amino acids*

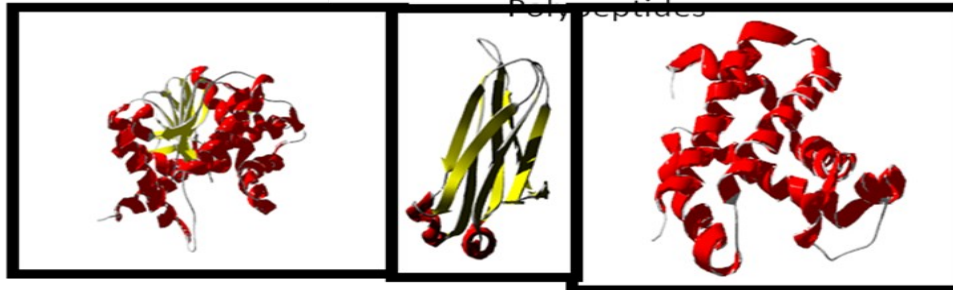
✓ *Polypeptides with <40 amino acids are called peptides whereas polypeptides with >40 amino acids are called proteins*



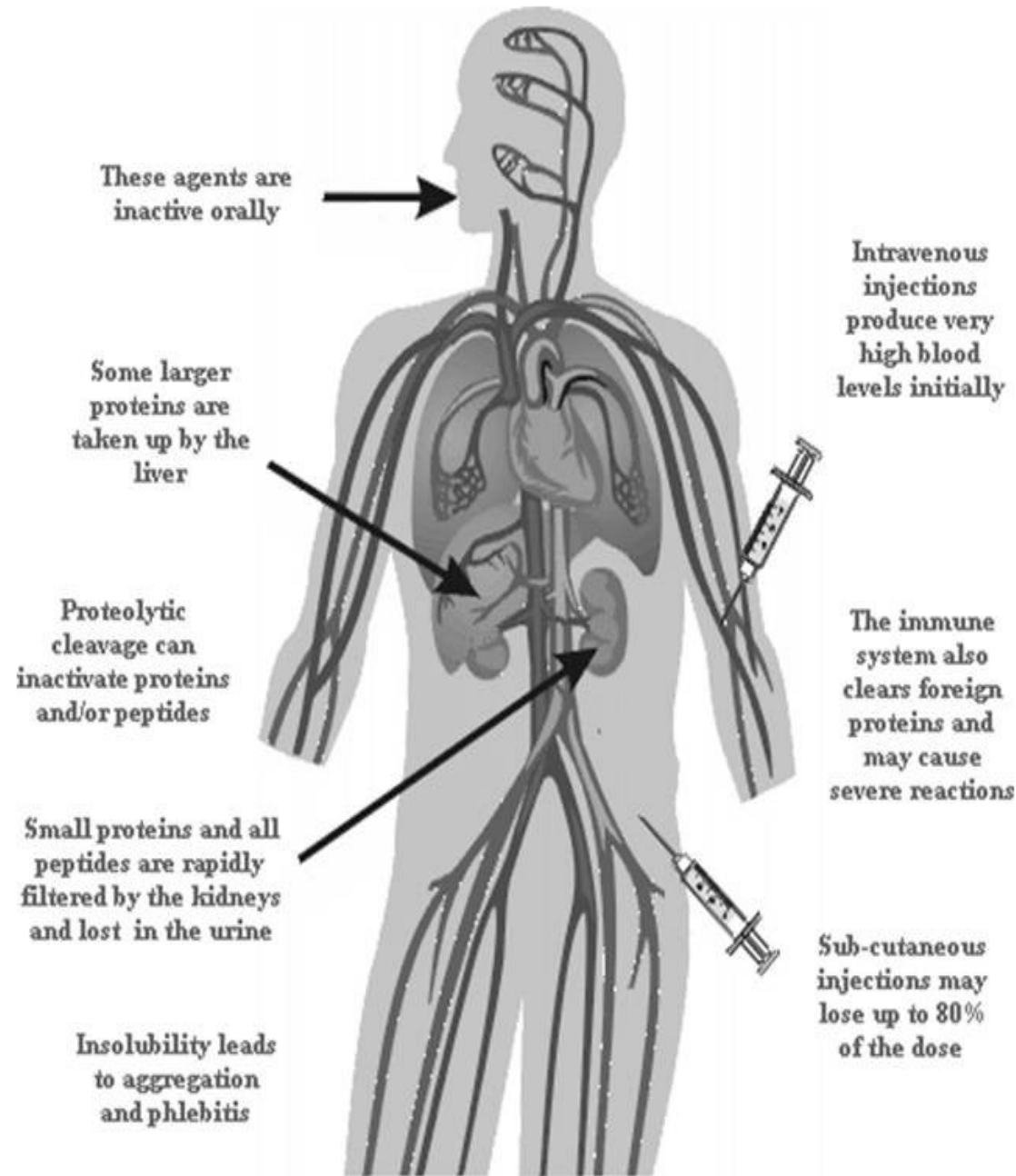
Aminoacid



Polypeptides



Sine G



The Problems with Proteins

- ✓ *Very **large and unstable molecules***
- ✓ *Easily **destroyed/eliminated by the body***
- ✓ *Hard to obtain in large quantities*
- ✓ *Held together by weak **noncovalent forces***
- ✓ *Destroyed by relatively mild storage conditions*

***In vivo* challenges of proteins**

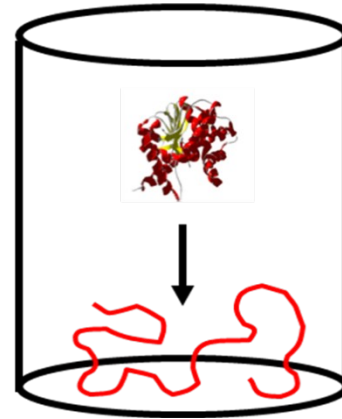
- ➡ Elimination by B and T cells
- ➡ Proteolysis by endo/exo peptidases
- ➡ Loss due to insolubility/adsorption (the accumulation and adhesion of molecules to a surface, but without surface penetration occurring).
- ➡ Small proteins (<30 kD) filtered out by the kidneys very quickly
- ➡ Unwanted allergic reactions may develop (even toxicity)

In vitro challenges of proteins

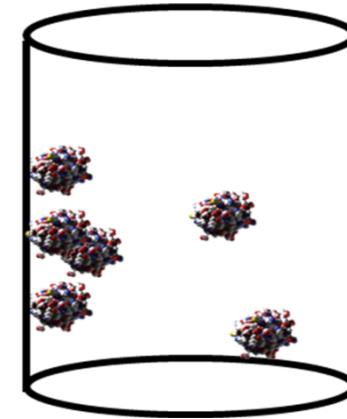
◆ These properties are challenging during handling, packaging and storage of protein based drugs.

Physical stability problems

- Denaturation
- Aggregation
- Precipitation
- Adsorption



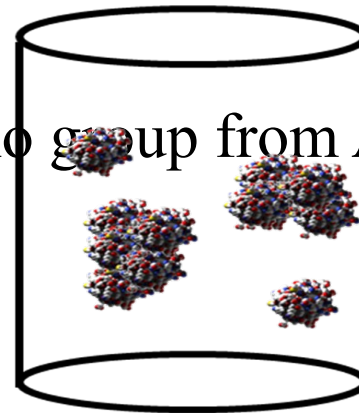
Denaturation



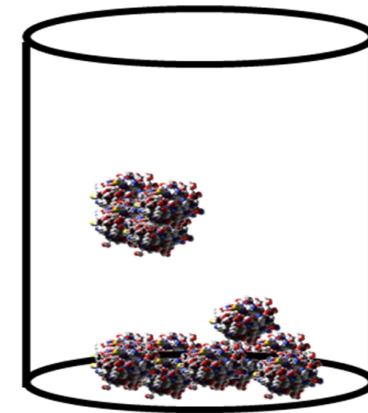
Adsorption

Chemical stability problems

- Deamidation (removal of amino group from AA)
- Oxidation
- Disulfide exchange

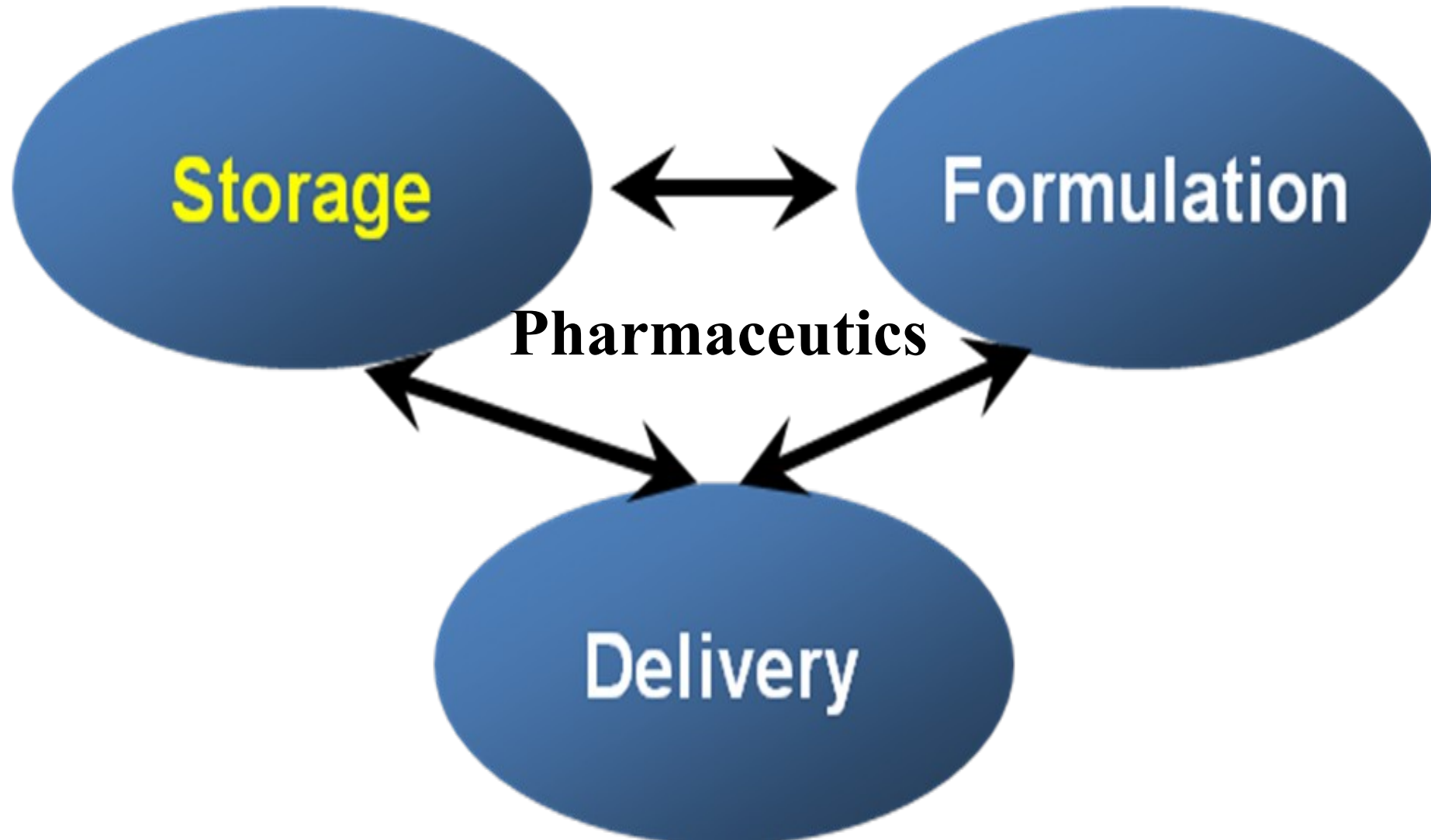


Aggregation



Precipitation

These challenges can be significantly minimized by:



Handling and storage of protein and peptide based pharmaceuticals

- Most of the biopharmaceuticals are recommended to be handled and stored **under low temperature and optimum storage conditions**
 - Low temperature reduces microbial growth and metabolism
 - Low temperature reduces thermal or spontaneous denaturation
 - Low temperature reduces adsorption
 - Freezing is best for long-term storage; however, freeze/Thaw can denature proteins

Freeze drying: is commonly used during the production and manufacturing as:

- It is the only **cost-effective means to prepare solid**, chemically active proteins
- It is **best for long term storage**
- It removes a considerable amount of water from protein lattice for **stabilization**; however, **removal of so much water may lead to deactivation of some proteins**

Packaging of biopharmaceuticals

- Biopharmaceuticals are recommended to be packaged using/in:
 - **Smooth glass** walls best to **reduce adsorption or precipitation of products**
 - **Dark, opaque walls** to **reduce product oxidation**
 - **Air-tight containers or argon atmosphere** to **reduces air oxidation**
 - **Polystyrene** or containers with **silanyl** or **plasticizer coatings** should be avoided

Formulation additives

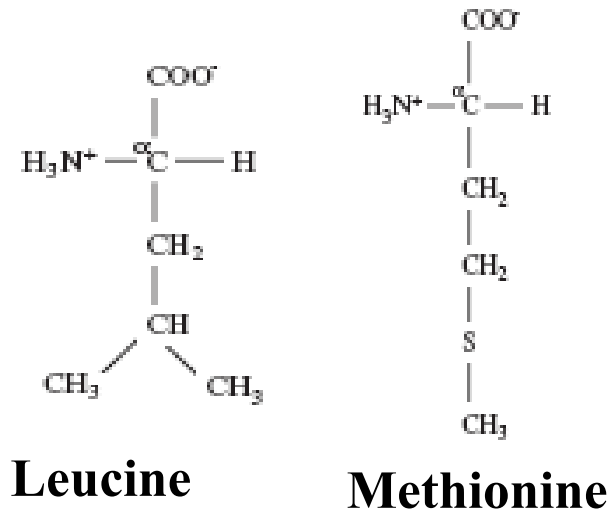
- The following classes of additives can be incorporated to stabilize the product or its performance:
- @ Stabilizing salts or ions (eg. Zn^{+} for insulin) can be added to reduce solubility and prolong its effect
- @ Polyols (glycerol and/or polyethylene glycol) can be added to solubilize the product
- @ Sugars or dextran can be included to displace water or reduce microbial growth
- @ Surfactants can be used to reduce adsorption and aggregation of proteins

Protein Formulation

- These products are highly vulnerable for environmental, physical and chemical factors and are also very difficult to deliver for their therapeutic purpose.
- These *in vitro* or *in vivo* difficulties can be avoided or minimized through the formulation process. The formulation modification can include;
 - Protein sequence modification (site directed mutagenesis)
 - Microsphere/Nanosphere encapsulation
 - PEGylation
 - Proteinylation
 - Formulating with permeabilizers

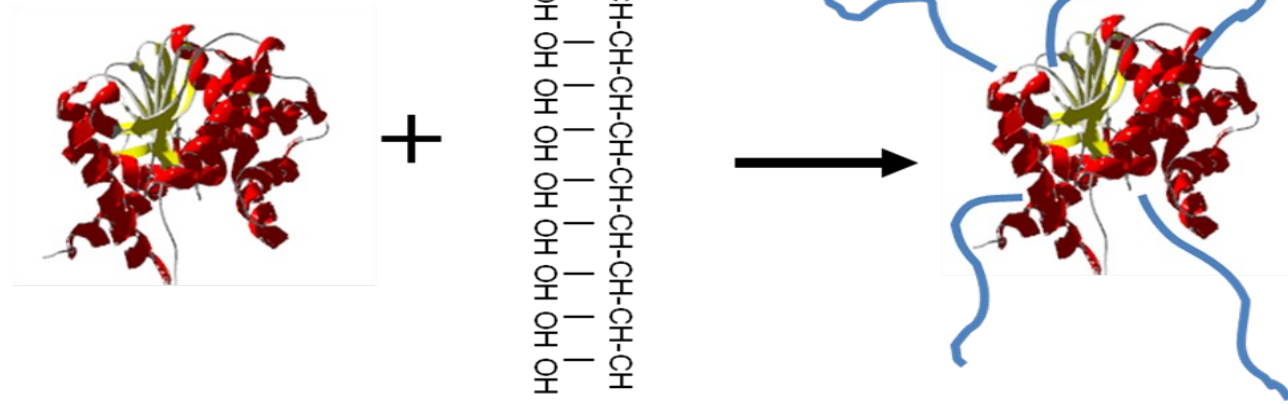
Site directed mutagenesis

- Allows amino acid substitutions at specific sites in a protein: Eg. substituting a Met to a Leu will reduce the likelihood of oxidation



- Strategic placement of cysteines can be used to produce disulfides to increase Tm
- Protein engineering (size, shape, etc.) can be employed to increase performance or to reduce toxicity

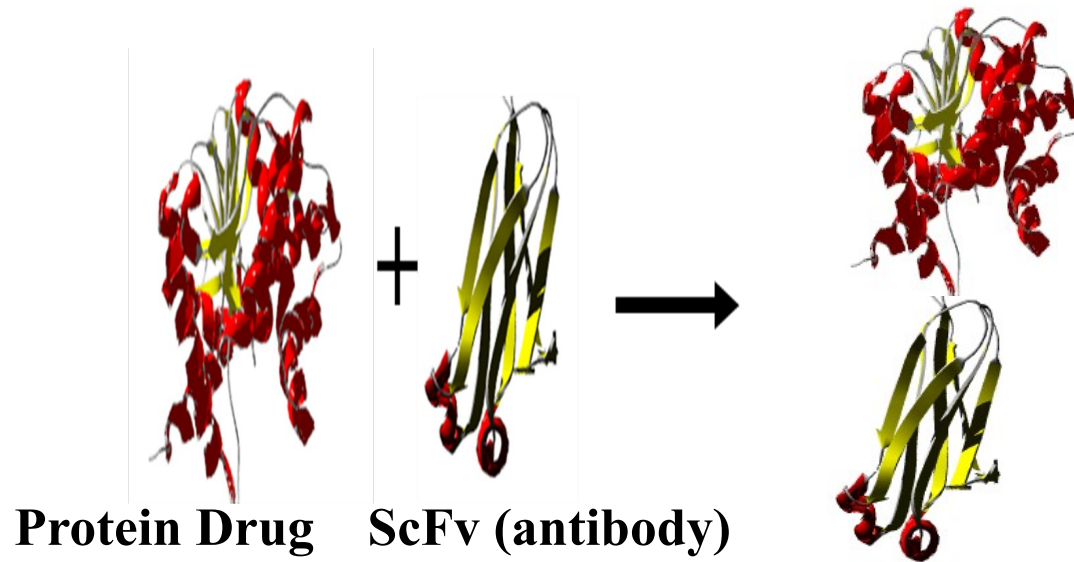
PEGylation



Polyethylene glycols (PEGs) are **non-toxic**, **hydrophilic**, FDA approved, **uncharged polymers** usually used as **cosolvents** and other formulation additives for pharmaceuticals.

- ✓ They are used to **increases in vivo half life (4-400X) of protein and peptide drugs**
- ✓ They can also **decrease immunogenicity**
- ✓ They **reduce enzymatic degradation by increasing protease resistance**
- ✓ **Increases solubility & stability of proteins**
- ✓ **Reduces depot loss at injection sites**

Proteinylation



- ◆ It is the attachment of additional or secondary (nonimmunogenic) proteins for *in vivo* protection
- ◆ Increases *in vivo* half life of proteins (10X)
- ◆ It is done by cross-linking with Serum Albumin or connecting by protein engineering with antibody fragments

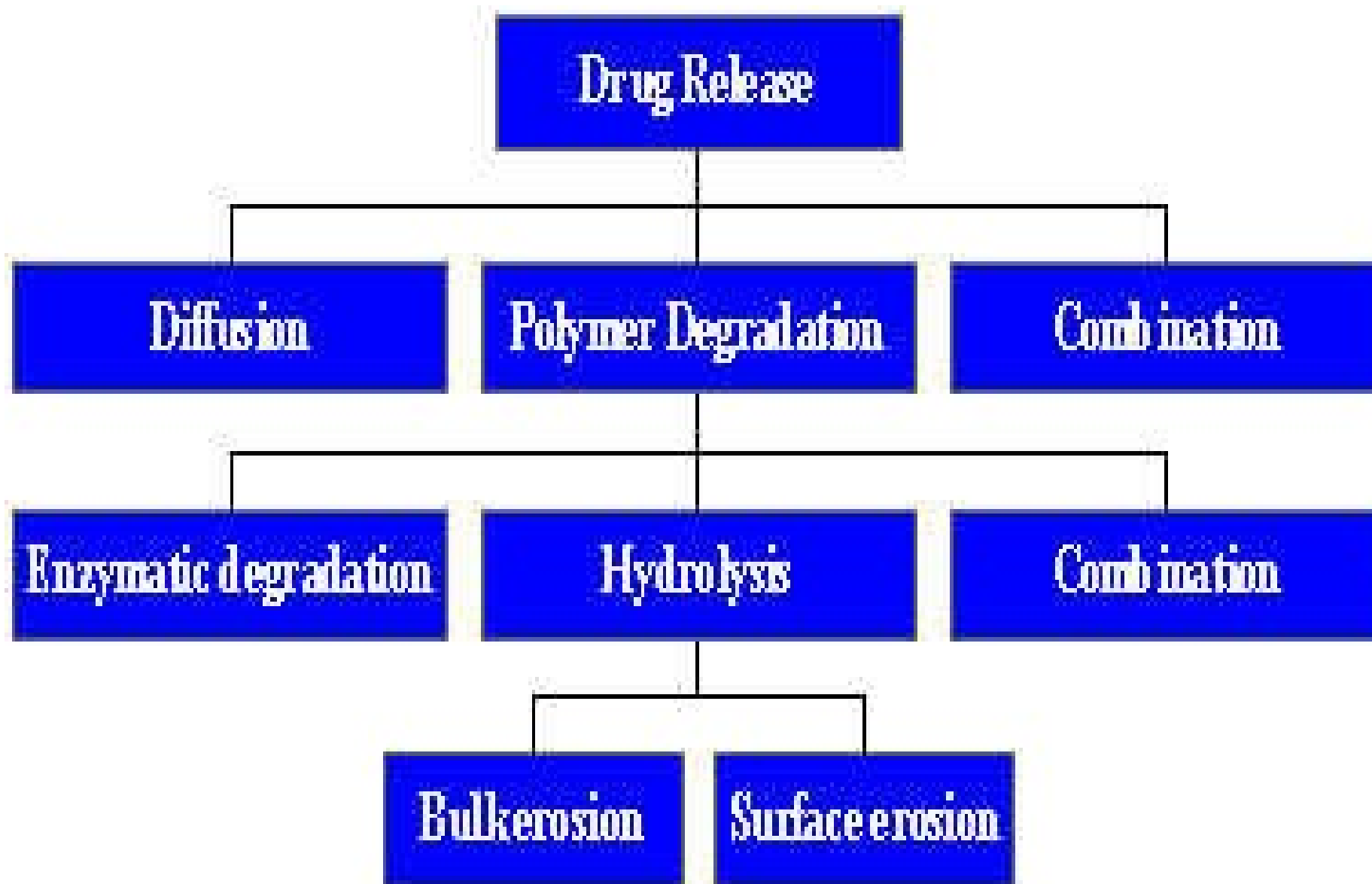
Encapsulation of protein and peptide drugs

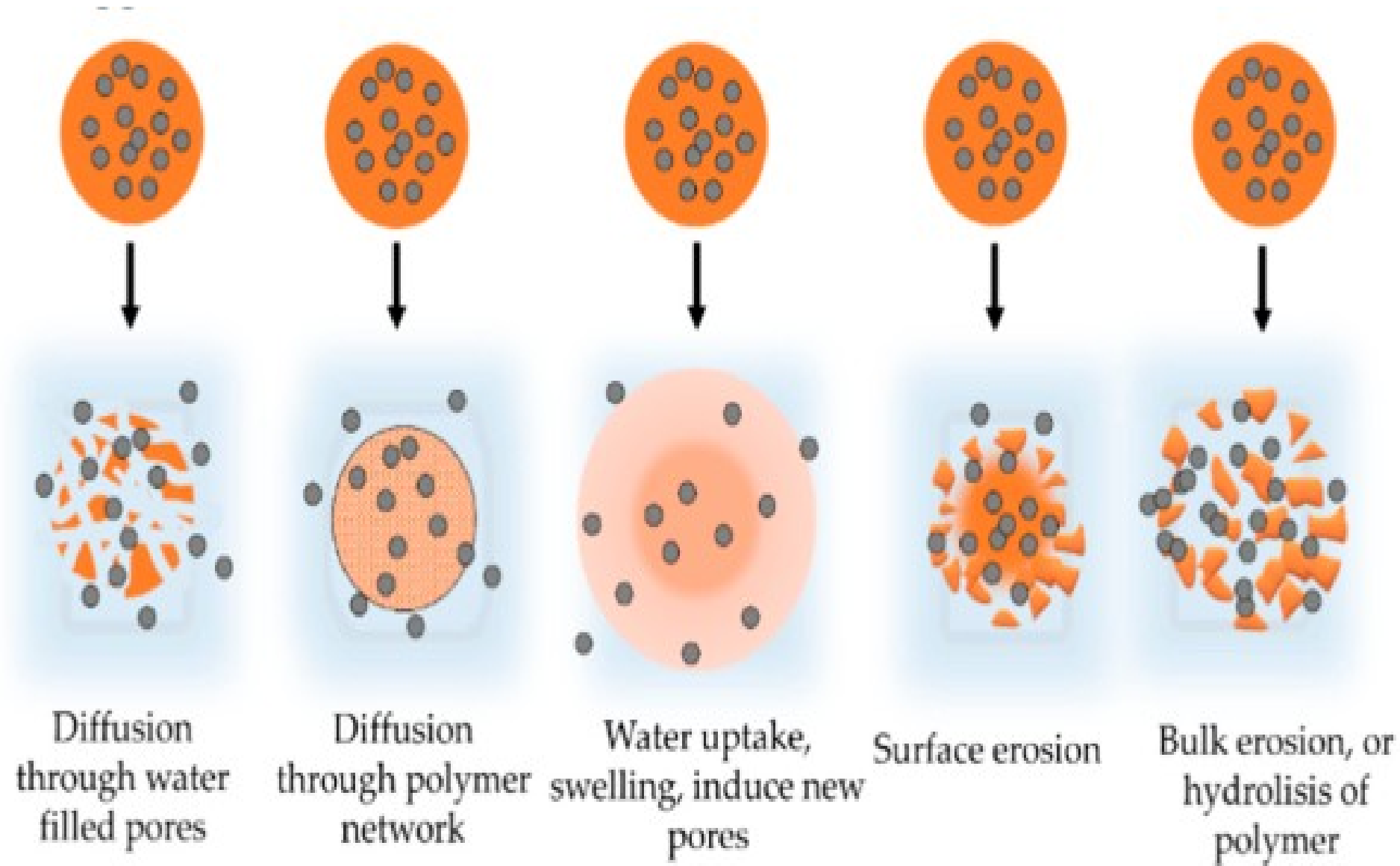
- ◆ These processes involve encapsulating protein or peptide drugs in small porous particles for **protection from “insults”** and/or for **sustained release**.
- ◆ These can be done using nonbiodegradable or biodegradable polymers.

Nonbiodegradable polymers include: ceramic particles, polyethylene co-vinyl acetate, polymethacrylic acid etc.

Biodegradable (**the preferred polymers**) can be **gelatin**, polylactic-co-glycolic acid (PLGA) and related copolymers.

Drug Release Mechanisms from encapsulated systems





Quality Control of Biological/Immunological Products

- Pharmacopoeial standards outline tests applicable to official immunological products to ensure safety and efficacy in use:

Identification tests: to establish the identity of the preparation

Toxicity tests: to confirm that the active constituents (organisms, toxoids or toxins) can be safely administered

Sterility tests: to demonstrate freedom from extraneous micro-organisms

Potency test: to make sure the product is satisfactory in immunizing or as a diagnostic agent

Keys for quality control and the product life cycle of biopharmaceuticals

- ❖ Adherence to standard regulations
- ❖ Control and maintenance of proper documentation
- ❖ Ensuring quality of suppliers, components and raw materials
- ❖ Maintaining reliability and consistency throughout the process
- ❖ Monitoring/auditing of the manufacturing process
- ❖ Correction and prevention of deviations, unexpected situations
- ❖ Enforcing lot release/rejection decision making
- ❖ Establishing customer complaints handling system
- ❖ Striving for continuous improvement

Regulatory/Environment Challenges

- ◆ Very strict cGMP requirements of the 21st Century particularly for biological and immunological products
- ◆ The need for capital intensive aseptic processing requirements
- ◆ Difficult risk management
- ◆ Increased inspection of product insert claims
- ◆ Focus on patient safety
- ◆ Sophisticated Process Analytical Technology (PAT) requirement

Major process challenges

- ◆ Sterile and aseptic system
 - ✓ Requires the application of microbiological contamination control to prevent infectious organisms from the sterile product

- ◆ Process control while technical complexity increases
 - ✓ Characterization to identify variability components
 - ✓ Application of science and new technologies

- ◆ Maintenance of cell lines
 - ✓ Contamination risks

- ◆ Personnel as “incubators”
 - ✓ Major source of microbial load

Biopharmaceutical drugs reportable cases

- ➡ Raw materials that failed specifications used in manufacturing
- ➡ Aseptic processing not performed according to procedures
- ➡ Stability testing not performed at required intervals
- ➡ Missing information in label (product type, lot number, storage temperature, concentration)
- ➡ Distributed product prior to completion of required testing
- ➡ Product release prior to validation of manufacturing process
- ➡ Biological Drug Substance stored at the incorrect temperature

Delivery of biopharmaceuticals

- The vast majority of biopharmaceuticals approved for general medical use are administered by direct injection via intravenous, subcutaneous or intramuscular routes.
- Administration via the s.c. or i.m. route is **generally followed by slow release of the drug from its depot site into the bloodstream**
- In the case of products **administered frequently/daily**, non-parenteral delivery routes would be preferred.
- Such routes would be more **convenient, less invasive, less painful** and generally would achieve **better patient compliance**
- Alternative potential delivery routes include **oral, nasal, transmucosal, transdermal** or **pulmonary** routes.
- However, routine administration of biopharmaceuticals by such means has been technically challenging due to:
 - **Their high molecular mass,**
 - **Susceptibility to enzymatic inactivation and**
 - **Potential to aggregate.**

Parenteral delivery of biopharmaceuticals

- ◆ It is the route of delivery for most of the biological products
- ◆ It *allows rapid and complete absorption and requires smaller dose size (less waste)*
- ◆ Avoids first pass metabolism and prevents proteins from “unfriendly zones”

However, parenteral administration is associated with several challenges like:

- ✓ Problems with **overdosing, necrosis**
- ✓ Local tissue **reactions/hypersensitivity** reactions
- ✓ It is an **invasive** methods as dependent on hypodermic needles

Oral delivery systems

- Oral drug delivery is usually preferred owing to **its convenience and high level patient compliance**
- Biopharmaceutical delivery via this route has proven problematic for a number of reasons:
 - ✓ Inactivation due to **stomach acid**
 - ✓ Inactivation due to **digestive enzymes**
 - ✓ Their (relatively) **large size** and **hydrophilic** nature renders **difficulty to cross the intestinal mucosa**
 - ✓ Orally absorbed drugs are subjected to **first-pass metabolism**

It is usual that bioavailabilities below 1 per cent are often recorded in the context of oral biopharmaceutical drug delivery.

Strategies to improve oral delivery

- ✓ Physically protecting the drug **via encapsulation and formulation as microemulsions/microparticulates**
- ✓ Inclusion of enzyme inhibitors (**protease inhibitors**) and absorption enhancer (**permeability enhancers**).

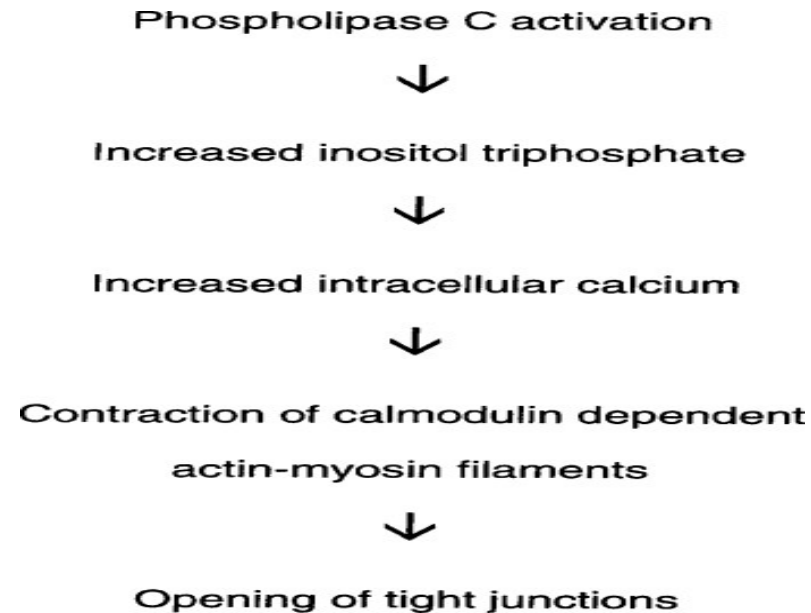
E.g. Bestatin, amastatin And Aprotinin And Protease Inhibitors.

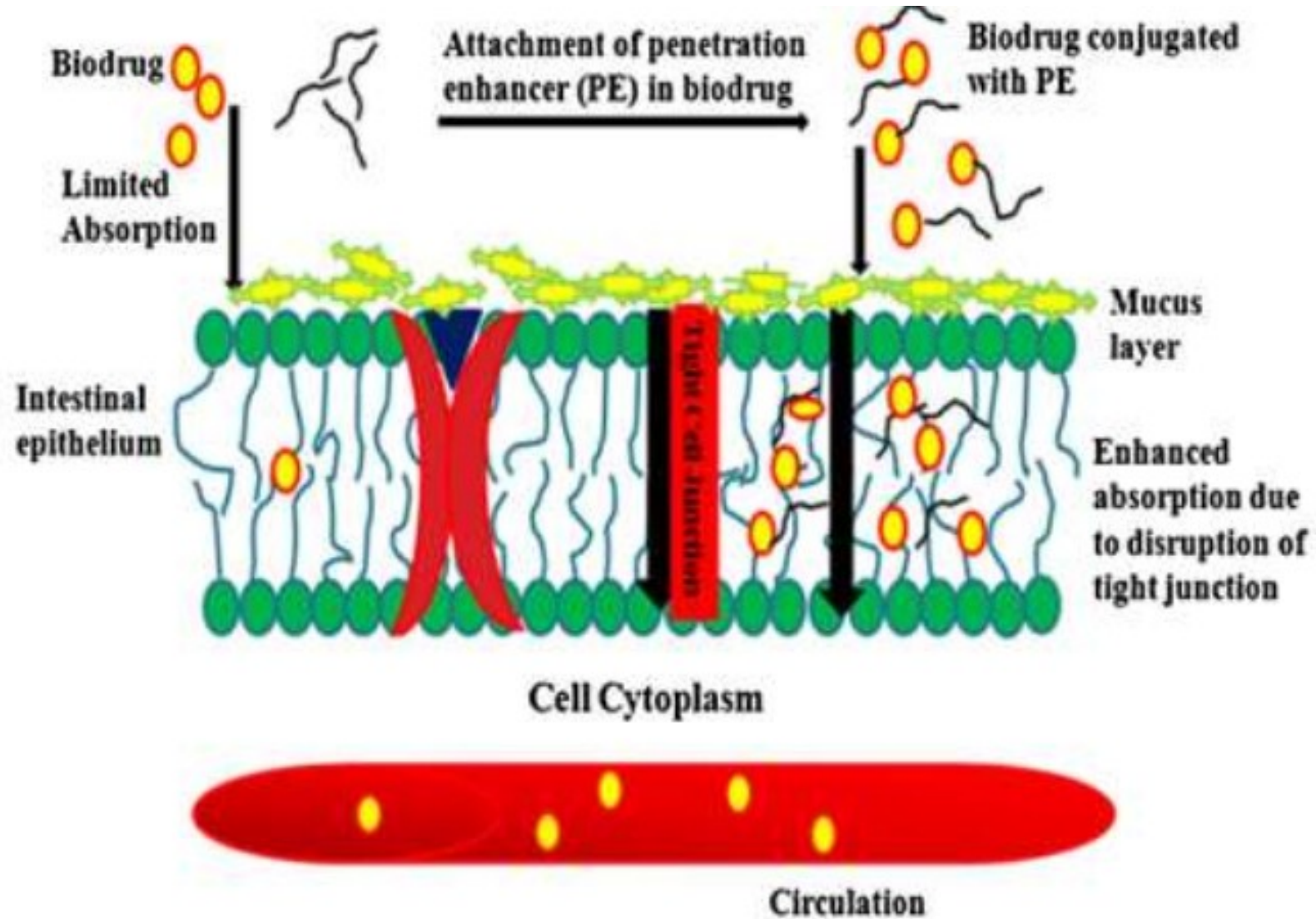
- ✓ Encapsulation within an **enteric coat** (resistant to low pH values) **protects the product during stomach transit.**
- ✓ Delivery systems based upon the use of **liposomes and cyclodextrin-protective coats** (a minute spherical sac of phospholipids molecules enclosing a water droplet especially as formed artificially to carry drugs).
- ✓ The use of ‘**mucoadhesive delivery systems**’ in which the biopharmaceutical is formulated **with/encapsulated in molecules that interact with the intestinal mucosa membranes.**

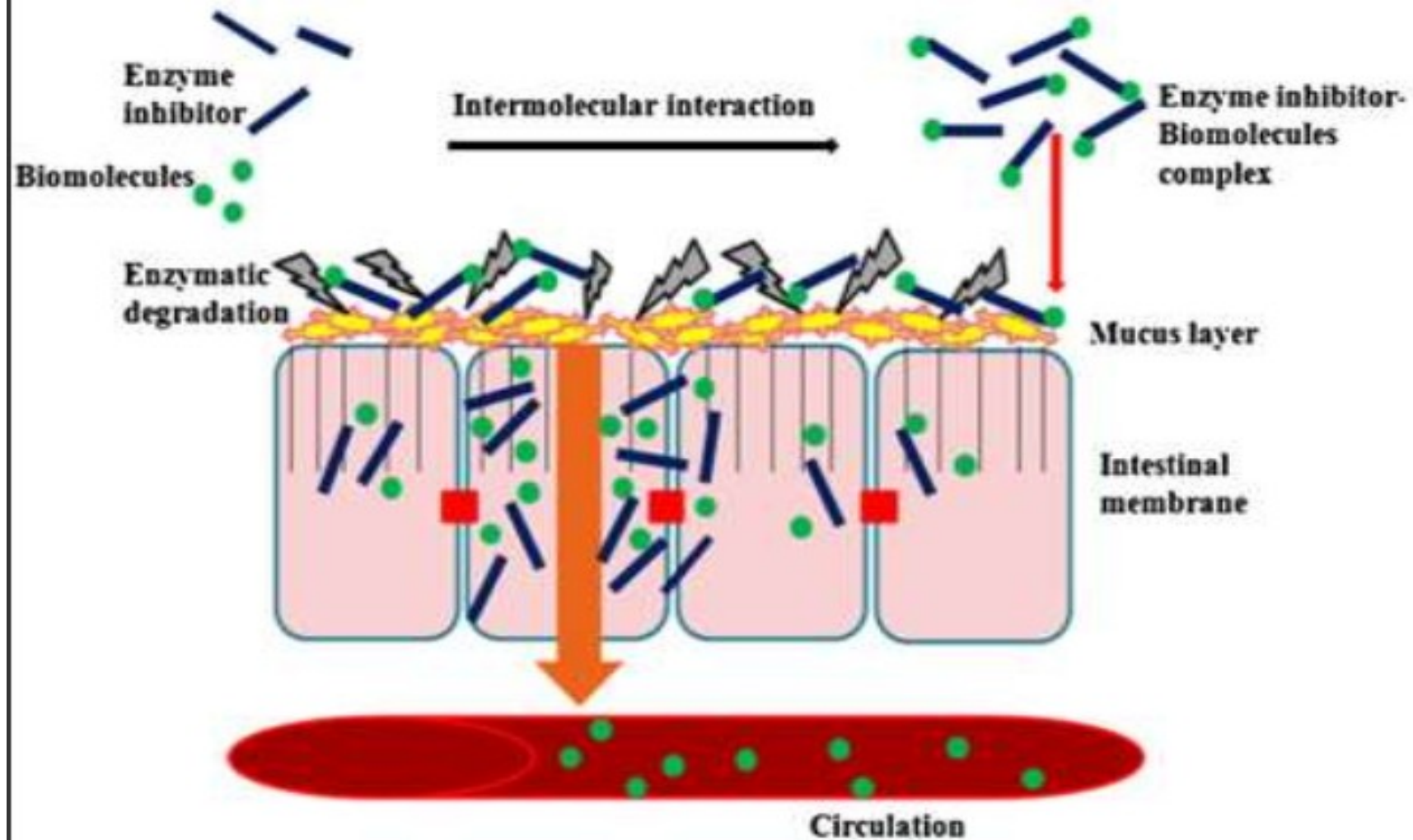
Penetration enhancers

Act by diverse mechanisms, including;

- Fluidization of the plasma membrane,
- inhibition of proteases and
- increasing the thermodynamic activity of proteins prone to self-association
- Loosening of the tight junctions between cells,



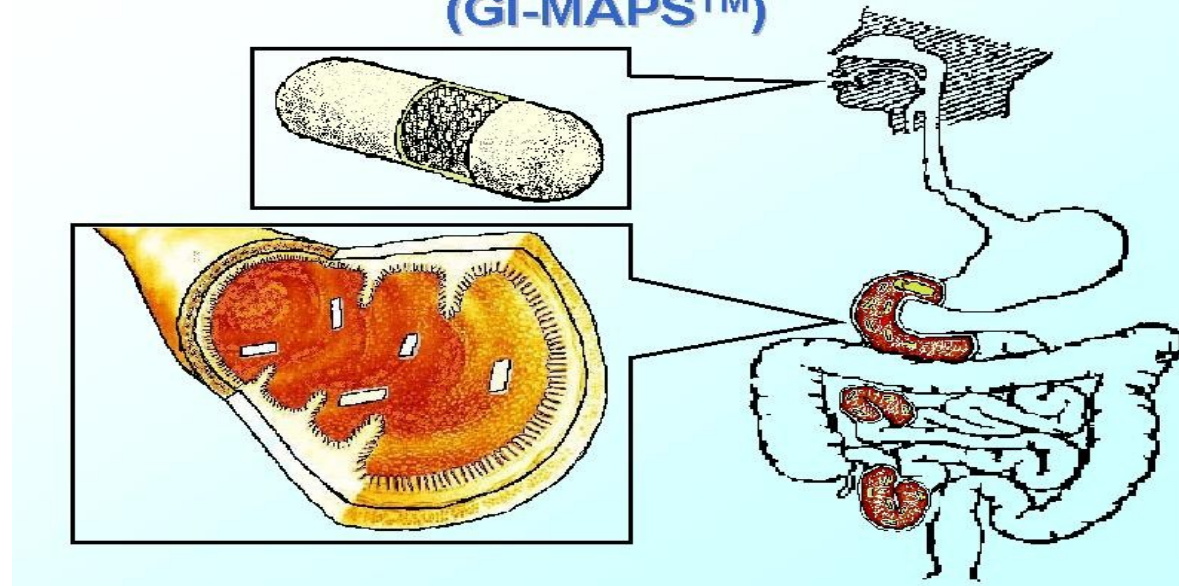




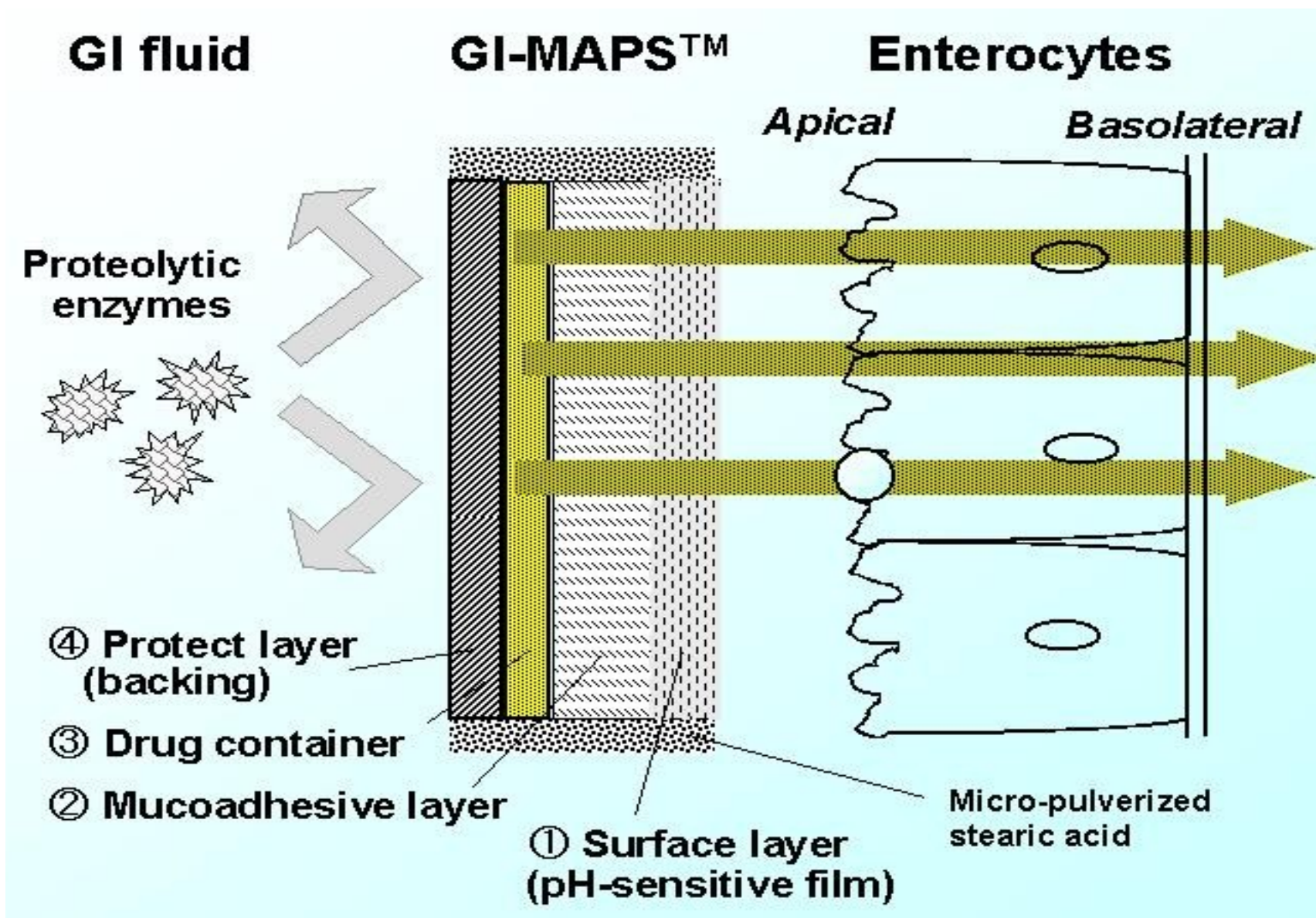
Sr. no.	Approach	Examples	Effects on bioavailability	Drawbacks
1	Absorption enhancers	Bile salts, fatty acids, Surfactants (anionic, cationic)	Enhanced bioavailability by increased membrane permeation	Available transport systems of both proteins/peptides and undesirable molecules in GIT
2	Enzyme inhibitors	Sodium glycocholate, camostatate mesilate	Resisted enzymes degradation in stomach and intestines	Produced severe side effects in the treatment of chronic diseases such as diabetes, etc.
3	Mucoadhesive polymeric systems	Thiolated polymer	Site-specific delivery and improved membrane permeation	Limitation due to the mucus turnover in absorption sites (intestine)
4	Prodrug strategies	Phenyl propionic acid	Prodrug permeability improved 1608fold than parent drug	Lack of methodology, structural complexity, stability problem of protein

Mucoadhesive patch delivery systems

Gastrointestinal mucoadhesive patch system (GI-MAPS™)



- The formulation adheres to specific region of the GI tract
- Ethylcellulose film protects drugs from proteolytic degradation in the system.
- The formulation is composed of 4 layers:
 - ✓ Ethylcellulose backing
 - ✓ Drug container (cellulose, citric acid)
 - ✓ Mucoadhesive glue (polyacrylic acid/PEG)
 - ✓ pH Surface layer (HP-55/Eudragit)



Pulmonary delivery of biopharmaceuticals

- Pulmonary delivery currently represents the most promising alternative to parenteral delivery systems

Advantages

- ✓ Very large surface area and low surface fluid volume
- ✓ Thin diffusional layer and slow cell surface clearance
- ✓ The presence of proteolytic inhibitors.
- ✓ Avoidance of first-pass metabolism;
- ✓ Availability of reliable, metered nebulizer-based delivery systems

Biological barriers to drug absorption

- ✓ Presence of a protective monolayer of insoluble phospholipid,
- ✓ The lung epithelial cells lining
- ✓ The interstitium and the basement membrane, composed of a layer of interstitial fibrous material
- ✓ The vascular endothelium, i.e. The monolayer of cells that constitute the walls of the blood vessels.

Nasal delivery of biopharmaceuticals

- A nasal-based biopharmaceutical delivery route is considered potentially attractive as:
 - ✓ It is easily accessible
 - ✓ Nasal cavities have high blood supply
 - ✓ Nasal microvilli generate a large surface area
 - ✓ Nasal delivery ensures bypasses for first-pass metabolism

Some disadvantages

- ✓ Clearance due to drug deposition upon the nasal mucous blanket and constantly cleared by ciliary action
- ✓ The existence of extracellular nasal proteases/peptidases
- ✓ Low uptake rates for larger peptides/polypeptides

Transdermal Patches delivery systems

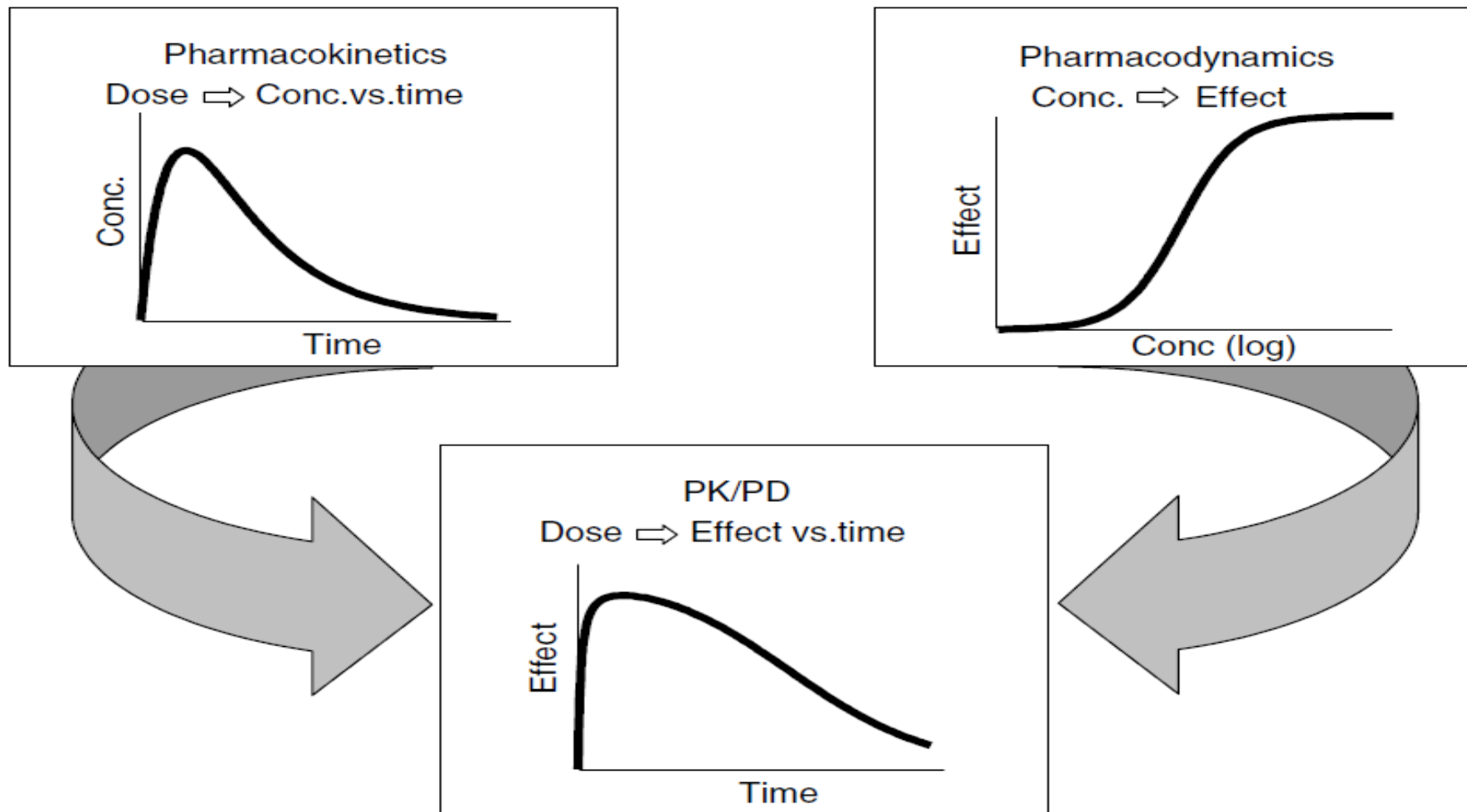
- ❖ **Proteins imbedded in a simple matrix** with appropriate additives
- ❖ **Patch is coated with small needles** that penetrate the dermal layer
- ❖ Proteins diffuse directly into the blood stream via capillaries
- ❖ Less painful form of parenteral drug delivery

Generally:

- ❖ Protein pharmaceuticals are (and will be) most rapidly growing in the pharmaceutical sector
- ❖ Most “cures” for difficult diseases (Alzheimers, cancer, Multiple Sclerosis, auto-immune diseases, etc.) will probably be found through protein drugs
- ❖ BUT Proteins are difficult to work with
- ❖ Most protein delivery is via injection and newer methods are appearing
- ❖ Oral delivery using “smart materials” is looking promising and hence many more protein drugs will be taken orally

Pharmacokinetics of biopharmaceuticals

- The development of biotech drugs generally rests on a fundamental understanding of the related disease and their clinical development has also proven to be more successful than for conventional small chemical molecules.
- The basis for the therapeutic use of biotech drugs is similar to that of small molecules; related with the drug concentration at its site of action.
- Pharmacokinetics comprises all processes that contribute to the **time course of drug concentrations in various body fluids: absorption, distribution, metabolism, and excretion.**



Pharmacokinetic/pharmacodynamic (PK/PD) modeling

- ❖ *Pharmacokinetic and pharmacodynamic principles are equally applicable to biotech drugs such as peptides, proteins, and oligonucleotides as they are to conventional small-molecule drugs.*

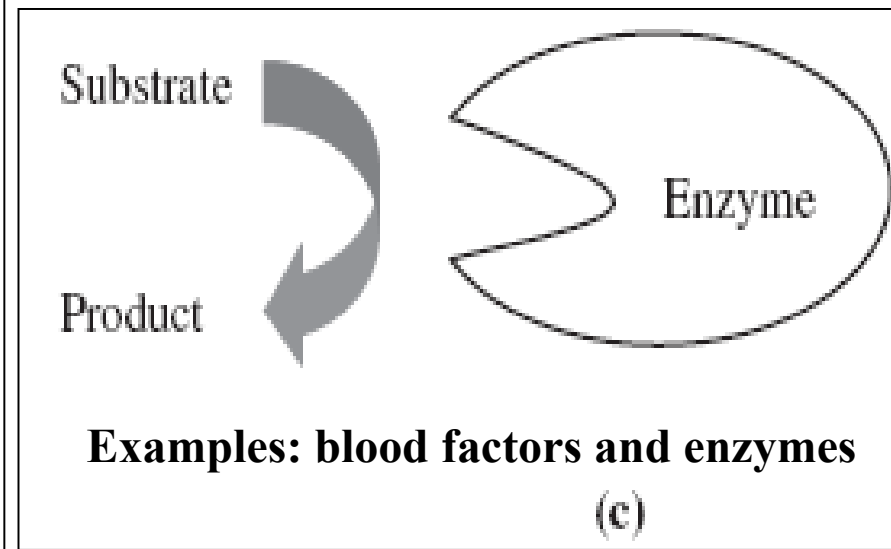
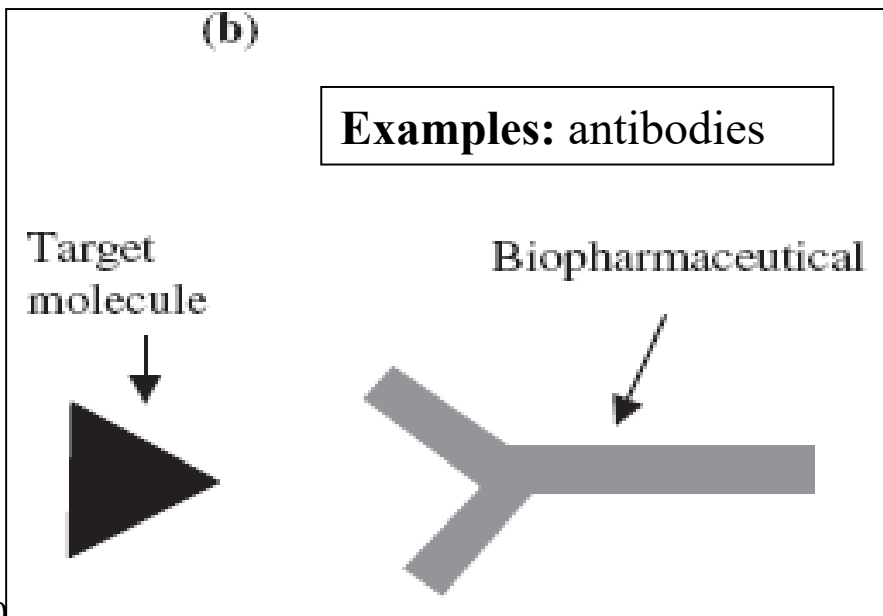
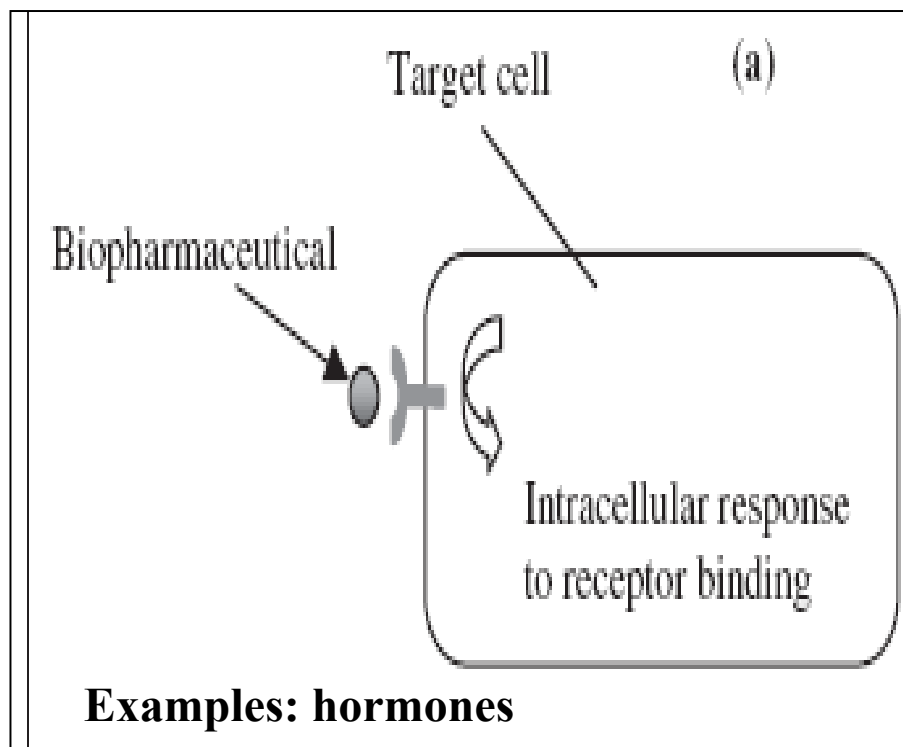


Fig. Overview of the mode of action of a biopharmaceutical

(a) binding to a specific cell surface receptor triggering the initiation of an intracellular response

(b) binding to and usually inactivating a target molecule, perhaps whose overexpression is causing/exacerbating a medical condition

(c) Catalysing the conversion of specific target substrate molecule into a product

Challenges for pharmacokinetic analysis of biopharmaceuticals

- ❖ *Biotech drugs are identical or similar to endogenous substances but usually exhibit unique pharmacokinetic and pharmacodynamic properties.*
- ❖ *The availability of an accurate, precise, and specific bioanalytical technique for the quantification of active drug moieties in plasma, blood, or other biological fluids is an essential prerequisite*
- ❖ *These methods have to meet pre-specified criteria for accuracy, precision, selectivity, sensitivity, reproducibility and stability.*
- ❖ *The level of complexity is often added to the bioanalytics of biotech drugs by the fact that **numerous of these are endogenous substances**; already in the body before drug administration.*
- ❖ *Thus, the analytical technique will detect a so-called baseline level prior to drug exposure which can either be constant or undergo complex changes*

Analytical considerations

- ❖ Assays for measuring and detecting peptides and proteins are often inaccurate, imprecise, and nonspecific
 - ✓ May lead to misleading results and erroneous pharmacokinetic evaluations for the compound of interest.
- ❖ Moreover, an additional level of complexity may be added when analyzing **glycosylated, PEGylated** or other conjugated biotech drugs
- ❖ **Techniques commonly utilized for the analysis of small molecules (i.e., HPLC and GC) are often not applicable for the analysis of biotech products.**
- ❖ Alternative methods for analysis of peptides and proteins usually include **bioassays, immunoassays, and mass spectrometry-based approaches.**

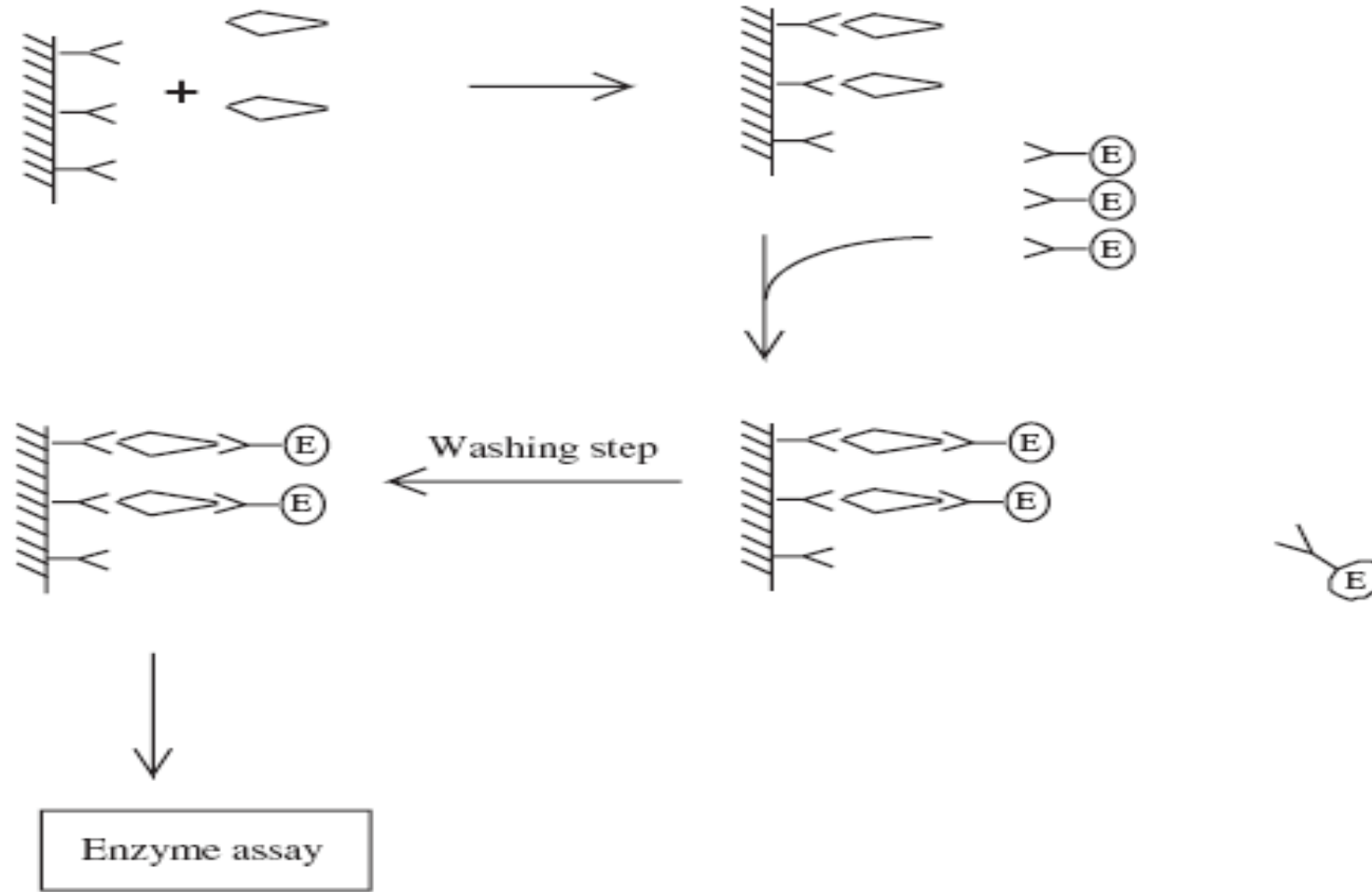
- ❖ **To characterize the clinical pharmacology of biotech drugs naturally present in the body, baseline values have to either be accounted for in the pharmacokinetic and pharmacodynamic analysis or be suppressed before exogenous drug administration.**
- ❖ **In contrast to the bioanalytics of small molecule drugs, immunoassays and bioassays are frequently applied to quantify peptides and proteins in biological samples.**
- ❖ **Immunoassays are considered the analytical method of choice for concentration determinations of peptides and proteins as they can relatively rapidly and easily be performed.**

Bioassays

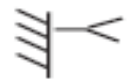
- ⊕ Are analytical procedures measuring a **biological activity of a test substance based on a specific, functional, and biological response of a test system**
- ⊕ The utility of bioassays lies in their ability to adequately characterize the **biologic product in terms of *consistency, purity, stability, and potency***.

Immunoassays

- ⊕ Because of their high **sensitivity, increased specificity, precision, availability, and relative ease of performance**, immunoassays have become the most frequently used analytical technique for quantifying peptides and proteins in biological fluids.
- ⊕ Immunoassays include **enzyme-linked immunosorbant assays (ELISA) and radioimmunoassays (RIA) with poly- and monoclonal antibodies**.
- ⊕ **ELISA is a standard methodology to detect and quantify biotech products.**



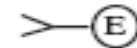
Key:



= Immobilized antibody



= Antigen
i.e. the therapeutic protein



= Antibody-enzyme conjugate

Basic principle of the ELISA system

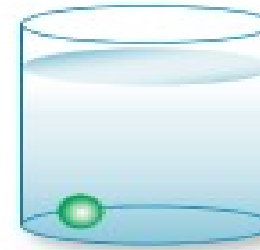
Sine G

Procedures of ELISA

- ✓ **Antibodies** raised against the **antigen of interest** are **first adsorbed** onto the **internal walls of microtitre plate wells**
- ✓ **The sample to be assayed** is then **incubated in the wells** and the **antigen present will bind to the immobilized antibodies**
- ✓ After equilibration, the wells are **washed**
- ✓ A preparation containing a second antibody, which also recognizes the antigen, is then added and also bind to the retained antigen and the enzyme label is conjugated to this second antibody
- ✓ Subsequent washing steps remove any unbound antibody-enzyme conjugate
- ✓ The activity of the enzyme retained is quantified by enzyme assay
- ✓ The activity recorded is proportional to the quantity of antigen present in the sample assayed

Coating

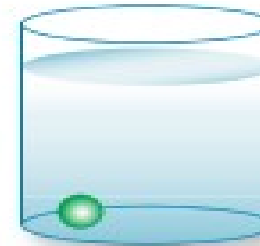
Antigen is adsorbed onto well in ELISA plate in coating buffer



Remove buffer and wash plate

Blocking

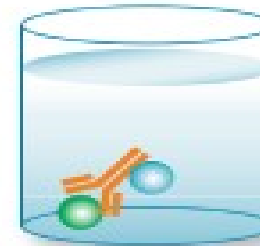
A buffer containing unrelated protein is used to block free sites in the wells



Remove buffer and wash plate

Detection

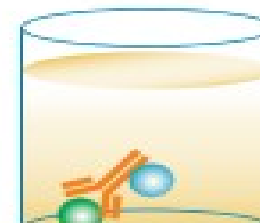
Enzyme conjugated detection antibody binds antigen



Remove buffer and wash plate

Readout

Substrate is catalyzed by enzyme to generate colored readout



Limitations of the ELISA methodology

- ❖ **Potential lack of specificity towards active (intact), inactive (denatured), and metabolized (partially degraded) forms of the biotech drug based on the epitope specificity of the antibody**
- ❖ **Lack of discrimination between endogenous and exogenous compounds**
 - ✓ May lead to an overestimation of concentration
- ❖ **Native antibodies directed against the biotech product may also interfere with ELISAs**

Pharmacokinetics of Peptides and Proteins

Absorption: the largest obstacle for a successful pharmacotherapy with peptide and protein drugs is their difficult delivery to the desired site of action.

- **Lack of systemic bioavailability** is mainly caused by two factors:
 - ✓ **High gastrointestinal enzyme activity**
 - ✓ **The function of the gastrointestinal mucosa as absorption barrier**
- **The apparent absorption rate constant** is thus the combination of **absorption into the systemic circulation** and **presystemic degradation at the absorption site**.
- **The true absorption rate constant k_a** can then be calculated as:

$$k_a = F \cdot k_{app}$$

Where F is the bioavailability and k_{app} is a rapid apparent absorption rate constant:

The result of a slow absorption and a fast presystemic degradation: *low systemic bioavailability*

Distribution: the catabolic degradation products of protein drugs are amino acids recycled in the endogenous amino acid pool.

- ✓ Biodistribution studies for peptides and proteins are primarily performed **to assess targeting to specific tissues as well as to identify the major elimination organs.**
- ✓ The volume of distribution of proteins is usually **small** and limited to the volume of the extracellular space *because of their;*
 - *High molecular weight and*
 - *Limited mobility to pass through biomembranes.*
- ✓ **Active tissue uptake and binding to intra- and extravascular proteins can substantially increase the volume of distribution.**

Protein Binding

- ❖ **Only the free, unbound fraction of a drug substance is accessible to distribution and elimination processes as well as interactions with its target receptor**
- ❖ **Interaction with binding proteins may facilitate cellular uptake processes and thus affect the drug's pharmacodynamics**
- ❖ **Apart from these specific bindings, peptides and proteins may also be nonspecifically bound to plasma proteins.**

Elimination

- Peptide and protein drugs are nearly exclusively metabolized via **the same catabolic pathways as endogenous or dietetic proteins**, leading to amino acids that are reutilized in the endogenous amino acid pool
- **Non-metabolic elimination pathways** such as **renal or biliary excretion are negligible** for most peptides and proteins
- **Amino acids** as well as some peptides and proteins such as **immunoglobulines** are excreted into the bile
 - ✓ If biliary excretion of peptides and proteins occurs, it generally results in subsequent metabolism of these compounds in the gastrointestinal tract.

- The elimination of peptides and proteins can **occur unspecifically nearly everywhere in the body or can be limited to a specific organ or tissue.**
- Locations of **intensive peptide and protein metabolism** are **liver, kidneys, gastrointestinal tissue, blood and other body tissues.**
- **Molecular weight** determines the major metabolism site as well as the **predominant degradation process.**
- The **metabolism rate** generally **increases** with **decreasing molecular weight**

Proteolysis: Proteolytic enzymes such as proteases and peptidases are especially localized in **blood, vascular endothelium and cell membranes and within cells.**

- ✓ *Thus, intracellular uptake is more of an elimination rather than a distribution process.*
- ✓ **Molecular weight** is considered as major determinant of the elimination mechanisms of peptides and proteins.

<i>Molecular weight</i>	<i>Elimination site</i>	<i>Predominant elimination mechanisms</i>	<i>Major determinant</i>
<500	Blood, liver	Extracellular hydrolysis Passive lipid diffusion	Structure, lipophilicity
500–1000	Liver	Carrier-mediated uptake Passive lipid diffusion	Structure, lipophilicity
1000–50 000	Kidney	Glomerular filtration and subsequent degradation processes (see Fig. 2)	Molecular weight
50 000–200 000	Kidney, liver	Receptor-mediated endocytosis	Sugar, charge
200 000–400 000		Opsonization	α_2 -macroglobulin, IgG
>400 000		Phagocytosis	Particle aggregation

Gastrointestinal Elimination

- For **orally** administered peptides and proteins, the gastrointestinal tract is the major site of metabolism
- **Parenterally** administered peptides and proteins, may also be metabolized in the intestinal mucosa following intestinal secretion.
 - ✓ Eg. About 20% of the degradation of endogenous albumin takes place in the gastrointestinal tract.

Renal Elimination

- For parenterally administered and endogenous peptides and proteins, the kidneys are the major elimination organ if they are smaller than the glomerular filtration limit of ~60 kD.

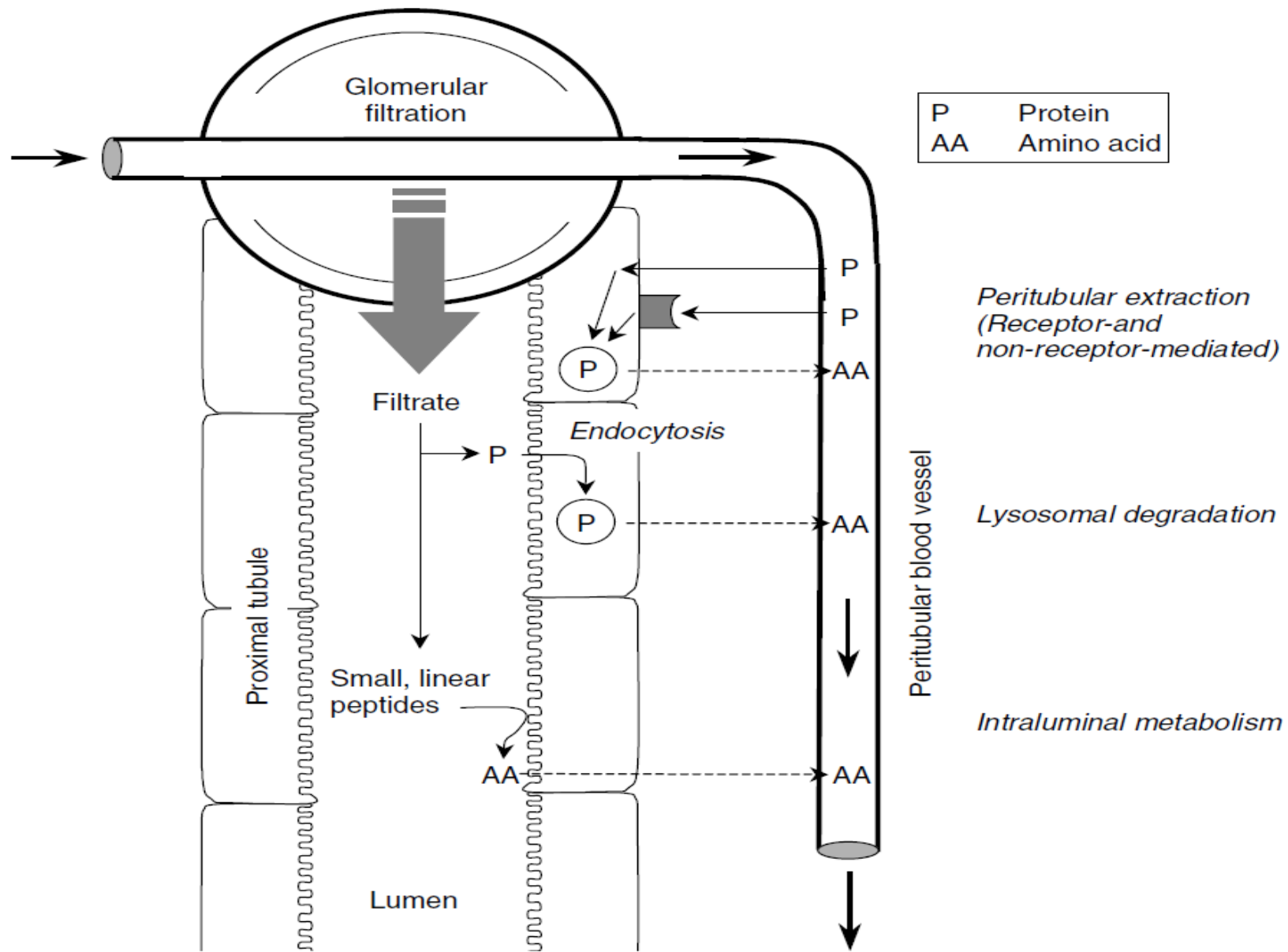


Fig 2. Renal elimination processes of peptides and proteins

- After glomerular filtration, **small linear peptides undergo intraluminal metabolism, predominantly by exopeptidases** in the luminal brush border membrane of the proximal tubules.
 - ✓ The resulting amino acids are transcellularly transported back into the systemic circulation.
- **Larger peptides and proteins are actively reabsorbed in the proximal tubules via endocytosis.**
 - ✓ This **cellular uptake** is followed by **addition of lysosomes** and hydrolysis to peptide fragments and amino acids, which are returned to the systemic circulation: therefore, only minute amounts of intact protein are detectable in urine.
 - ✓ An additional renal elimination mechanism is **peritubular extraction from post glomerular capillaries** with subsequent intracellular metabolism

Hepatic Elimination

- ◆ **The liver substantially contributes to the metabolism of peptide and protein drugs.**
- ◆ **Proteolysis usually starts with endopeptidases that attack in the middle part of the protein, and the resulting oligopeptides are then further degraded by exopeptidases.**
- ◆ **The rate of hepatic metabolism is largely dependent on specific amino acid sequences in the protein.**
- ◆ **An important first step in the hepatic metabolism of proteins and peptides is the uptake into hepatocytes.**
- ◆ **Small peptides may cross the hepatocyte membrane via passive diffusion if they have sufficient hydrophobicity.**
- ◆ **Uptake of larger peptides and proteins is facilitated via various carrier-mediated, energy dependent processes**

Immunogenicity

- ◆ Because of the **antigenic potential of proteins**, **formation of antibodies** is a **frequently observed** phenomenon during chronic therapy with protein drugs
- ◆ Most monoclonal antibodies are **murine** in nature and their systemic administration can lead to the development of **human antimouse immunoglobulin antibody (HAMA) response**
- ◆ Genetically, engineered mouse–human chimeric antibodies try to minimize this immunogenicity in man by joining variable domains of the mouse to the constant **regions of human immunoglobulins**
- ◆ The anti- EGFR (epidermal growth factor receptor) IgG monoclonal antibody **cetuximab** is an example of a murine–human chimeric antibody currently under clinical investigation for various cancer indications

- One approach to reduce the immunogenicity of biotech drugs involved their **conjugation with polyethylene glycol (PEG)**, called

PEGylation

- ✓ PEGylation can shield antigenic determinants on the drug through steric hindrance from detection by the immune system
- The other approach is to **alter the route of administration** as the administration route may affect immunogenicity profile
 - ✓ Extravascular injection is known to stimulate antibody formation more than intravenous application *due to the increased immunogenicity of protein aggregates and precipitates formed at the injection site*
 - ✓ Studies indicated that i.m. injection is less immunogenic compared to s.c. injection

Types of Protein and Peptide Therapeutic Agents

- ❑ Hormones are amongst the most important group of regulatory molecules produced by the body.

A. *Insulin*

- Insulin is a **polypeptide hormone produced by the beta cells of the pancreatic islets of Langerhans**.
- It plays a central role in **regulating blood glucose levels**, generally keeping it within narrow defined limits ($3.5\text{--}8.0\text{ mmol l}^{-1}$), irrespective of the nutritional status of the animal.
- It also has a profound effect on *the metabolism of proteins and lipids* and displays some mitogenic activity.
- Although many cells in the body express the **insulin receptor**, its most important targets are *skeletal muscle fibres, hepatocytes and adipocytes*.

Table: Some metabolic effects of insulin.

Metabolic pathway	Target tissue	Effect of insulin	Effect of glucagon
Glycogen synthesis	Liver	↑	↓
Glycogen degradation	Liver	↓	↑
Gluconeogenesis	Liver	↓	↑
Glycogen synthesis	Muscle	↑	—
Glycogen degradation	Muscle	—	—
Fatty acid synthesis	Adipose	↑	↓
Fatty acid degradation	Adipose	↓	↑

☐ Insulin coordinates metabolic response to the absorption of glucose and other nutrients in a number of ways:

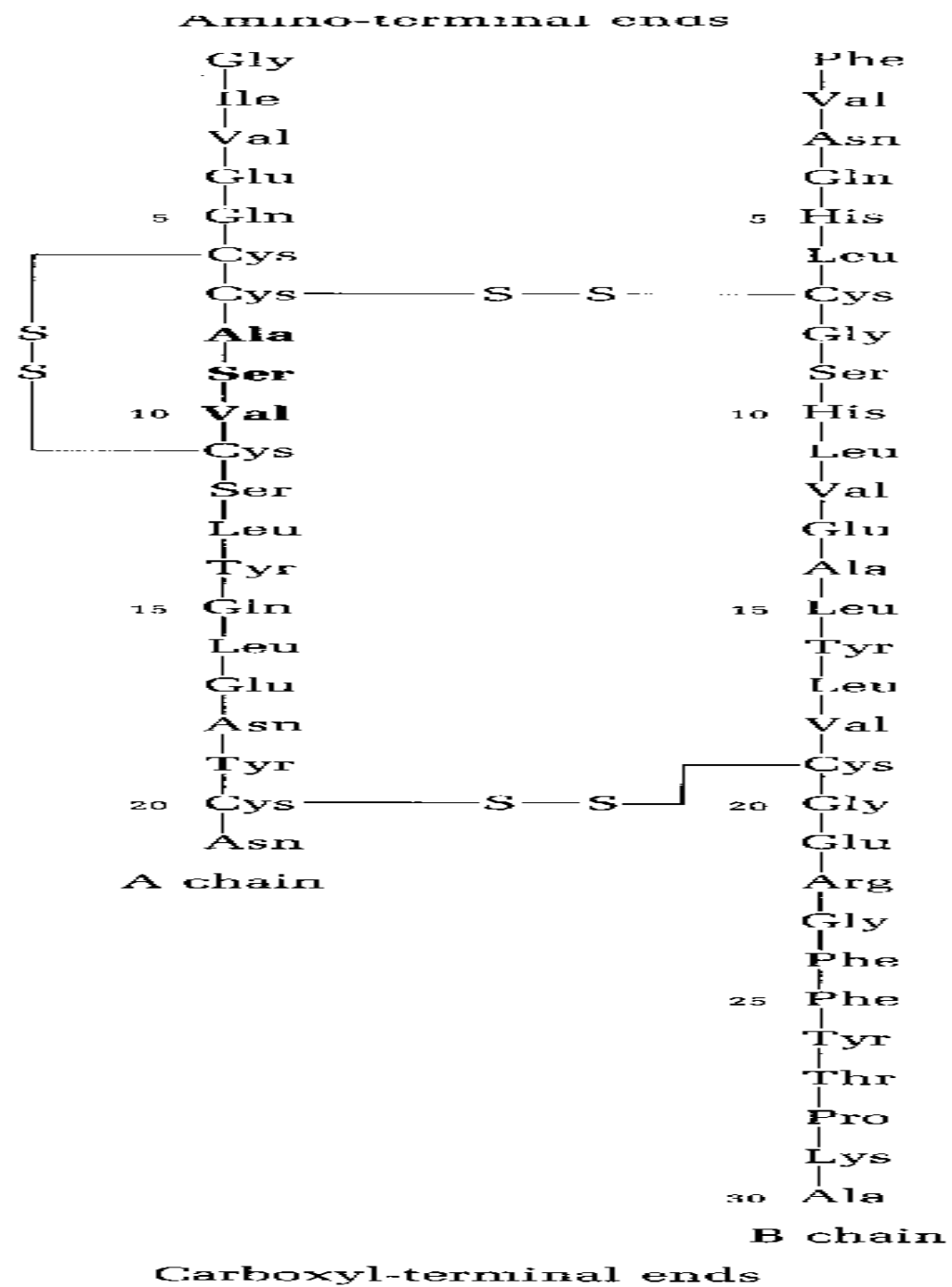
1. It stimulates glucose transport and transport of amino acids, K^+ ions and nutrients into cells
2. It stimulates intracellular biosynthetic (anabolic) pathways which helps to convert nutrients into a storage form in the cells
3. It inhibits catabolic pathways Eg. glycogenolysis
4. It stimulates protein and DNA synthesis; its growth-promoting activity

Diabetes mellitus: Failure of the body to synthesize sufficient insulin: insulin-dependent diabetes mellitus (IDDM)

- ◆ IDDM is caused by *T-cell-mediated autoimmune destruction of the insulin-producing β -pancreatic islet cells* in genetically predisposed individuals.
- ◆ This is probably due to the expression of a ‘super antigen’ on the surface of the β -cells in such individuals

The insulin molecule:

- ◆ First identified as an anti-diabetic factor in 1921
- ◆ Introduced clinically 1922
- ◆ Its complete amino acid sequence was determined in 1951
- ◆ **It is polypeptide with 51 amino acids arranged in two chains joined by two disulfide linkages**
 - ✓ **Chain A 21 amino acids chain B 30 amino acids**



- ◆ **Porcine insulin** (5777 Da) varies from the human form (5807 Da) by *a single amino acid*, whereas *bovine insulin* (5733 Da) differs by three residues
- ◆ The insulin receptor is a tetrameric integral membrane glycoprotein consisting of two 735 amino acid α -chains and two 620 amino acid β -chains
- ◆ Binding of insulin to its receptor promotes the autophosphorylation of three specific tyrosine residues which in turn unmask ATP (the phosphate donor) binding sites and substrate docking sites and activating its tyrosine kinase activity.
 - a. Activation of the ‘mitogenactivated protein kinase’ pathway is responsible for triggering insulin’s mitogenic effects
 - b. Activation of the PI-3 kinase pathway mediates insulin’s metabolic effects.



Insulin production

☐ Traditionally, insulin was produced by *direct extraction from pancreatic tissue of pigs and cattle followed by purification.*

☐ The use of animal-derived product had a number *of potential disadvantages:*

Immunogenicity: Particularly bovine insulin elicits immunological response in humans.

- ✓ ***Triggers insulin resistance*** (as anti-insulin antibodies neutralize some of the administered products)
- ✓ ***Affect the pharmacokinetics of the drug, as antibody-bound insulin molecules are largely resistant to insulin degradative process.***
- Worldwide, about 170 million people suffer from diabetes, a figure projected to double by 2030.
- The annual insulin requirement has surpassed 5000 kg and continues to grow, prompting concern of an insulin shortfall from slaughterhouse sources.

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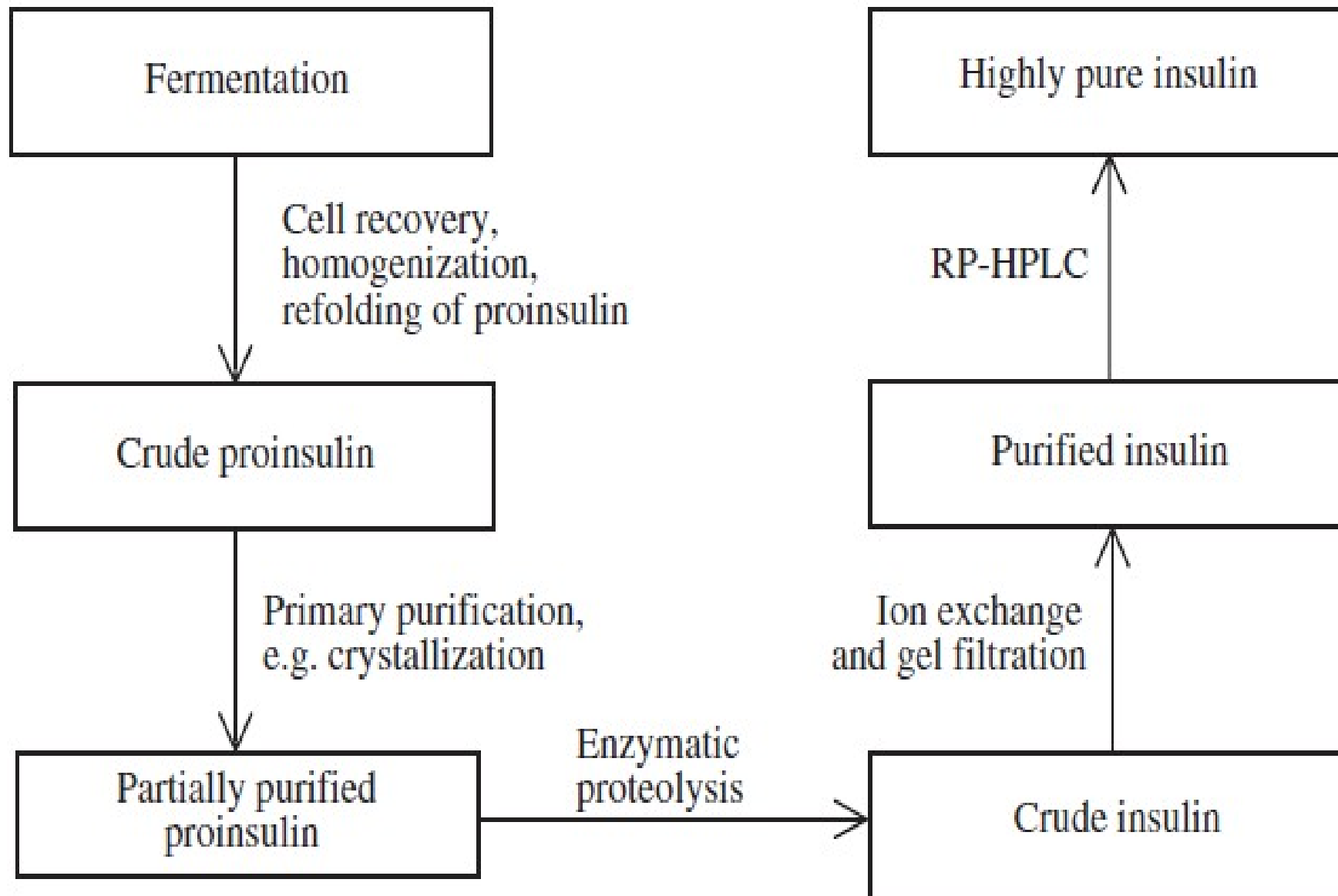
Production of human insulin by recombinant DNA technology

- Human insulin produced **by recombinant DNA technology** was first approved for general medical use in 1982 in USA, Germany, UK and The Netherlands.
- It was the first approved therapeutic product of recombinant DNA technology for use in humans.

Initial approach: *inserting the nucleotide sequence coding for the insulin A- and B-chains* into two different *E. coli* cells

- ***Culturing*** the cells separately in large-scale fermentation vessels with subsequent **chromatographic purification of the insulin chains produced.**
- The A- and B-chains are then ***incubated together under appropriate oxidizing*** conditions in order to promote interchain disulfide bond formation

The alternative and suitable method: inserting a ***nucleotide sequence coding for human proinsulin*** into recombinant *E. coli* followed by **purification** of the expressed proinsulin and subsequent ***proteolytic excision of the peptide in vitro.***



Purification scheme for human insulin *crb*

Product	Description	Structure	Company	Approved
<i>Recombinant products of native human insulin sequence</i>				
Humulin	Recombinant human insulin produced in <i>E. coli</i>	Identical to native human insulin	Eli Lilly	1982 (USA)
Novolin	Recombinant human insulin produced in <i>S. cerevisiae</i>	Identical to native human insulin	Novo Nordisk	1991 (USA)
Insuman	Recombinant human insulin produced in <i>E. coli</i>	Identical to native human insulin	Hoechst AG	1997 (EU)
Actrapid/Velosulin/Monotard/Insulatard/Protaphane/Mixtard/Actraphane/Ultratard	All contain recombinant human insulin produced in <i>S. cerevisiae</i> formulated as short/intermediate/long acting product)	Identical to native human insulin	Novo Nordisk	2002 (EU)
Exubera	Recombinant human insulin produced in <i>E. coli</i> but administered via the pulmonary route	Identical to native human insulin	Pfizer	2006 (USA)
<i>Engineered insulins</i>				
Humalog (Insulin lispro)	Recombinant short-acting human insulin analogue produced in <i>E. coli</i>	Engineered: inversion of native B28–B29 proline–lysine sequence	Eli Lilly	1996 (USA and EU)
Liprolog (Insulin lispro)	Recombinant short-acting human insulin analogue produced in <i>E. coli</i>	Engineered: inversion of native B28–B29 proline–lysine sequence	Eli Lilly	1997 (EU)
NovoRapid (Insulin Aspart)	Recombinant short-acting human insulin analogue produced in <i>S. cerevisiae</i>	Engineered: B28 proline replaced by aspartic acid	Novo Nordisk	1999 (EU)
Novolog (Insulin Aspart)	Recombinant short-acting human insulin analogue produced in <i>S. cerevisiae</i>	Engineered: B28 proline replaced by aspartic acid	Novo Nordisk	2001 (USA)

Formulation of insulin products

- Insulin may be formulated in a number of ways *to alter its pharmacokinetic profile*.
- *Fast (short)-acting insulins* yield an elevated blood insulin quickly
- *Slow-acting insulins* enter the circulation much more slowly from the depot (injection) site.

Pharmacokinetic characteristics of short, intermediate and long-acting insulin preparations

Category	Onset (hours after administration)	Peak activity (hours after administration)	Duration (h)
Short-acting	0.5–1	2–5	6–8
Intermediate-action	2	4–12	up to 24
Long-acting	4	10–20	up to 36

■ ***In order to prolong the duration of insulin action***, soluble insulin may be formulated to ***generate insulin suspensions***.

■ This is generally achieved in one of two ways:

1. Addition of zinc to promote Zn–insulin crystal growth that take longer to leak into the bloodstream from the injection depot site
 2. Addition of a protein to which the insulin will complex; from which the insulin will only be slowly released
- ✓ Protamines found in the sperm of several species of fish are commonly used

■ ***The long-acting insulins*** may be ***Protamine–Zn–insulin or isophane insulin***

■ ***Biphasic insulin*** include mixtures of short- and long-acting insulins, which attempt to mimic normal insulin rhythms in the body

Engineered insulins

- Recombinant DNA technology facilitates production of human insulin in microbial systems and generation of insulins of modified amino acid sequences
- The major aims of generating engineered insulin include:
 - a. Identification of insulins with altered pharmacokinetic properties, such as faster-acting or slower-acting insulins.*
 - b. Identification of super-potent insulin forms** (insulins with higher receptor affinities).
- ✓ **Economic benefits** from utilizing smaller quantities of insulin per therapeutic dose.

The amino acid residues that interact with the insulin receptor have been identified (A1, A5, A19, A21, B10, B16, B23-25)

Conversion of histidine to glutamate at B10, yields an analogue displaying **fivefold higher activity *in vitro***

Fast acting insulins: are absorbed from the site of injection into the bloodstream far more quickly than native soluble insulins

‘Insulin lispro’ (tradenname ‘Humalog’): was the first such engineered short-acting insulin to come to market

‘Insulin Aspart’: is a second fast-acting engineered human insulin analogue now approved for general medical use

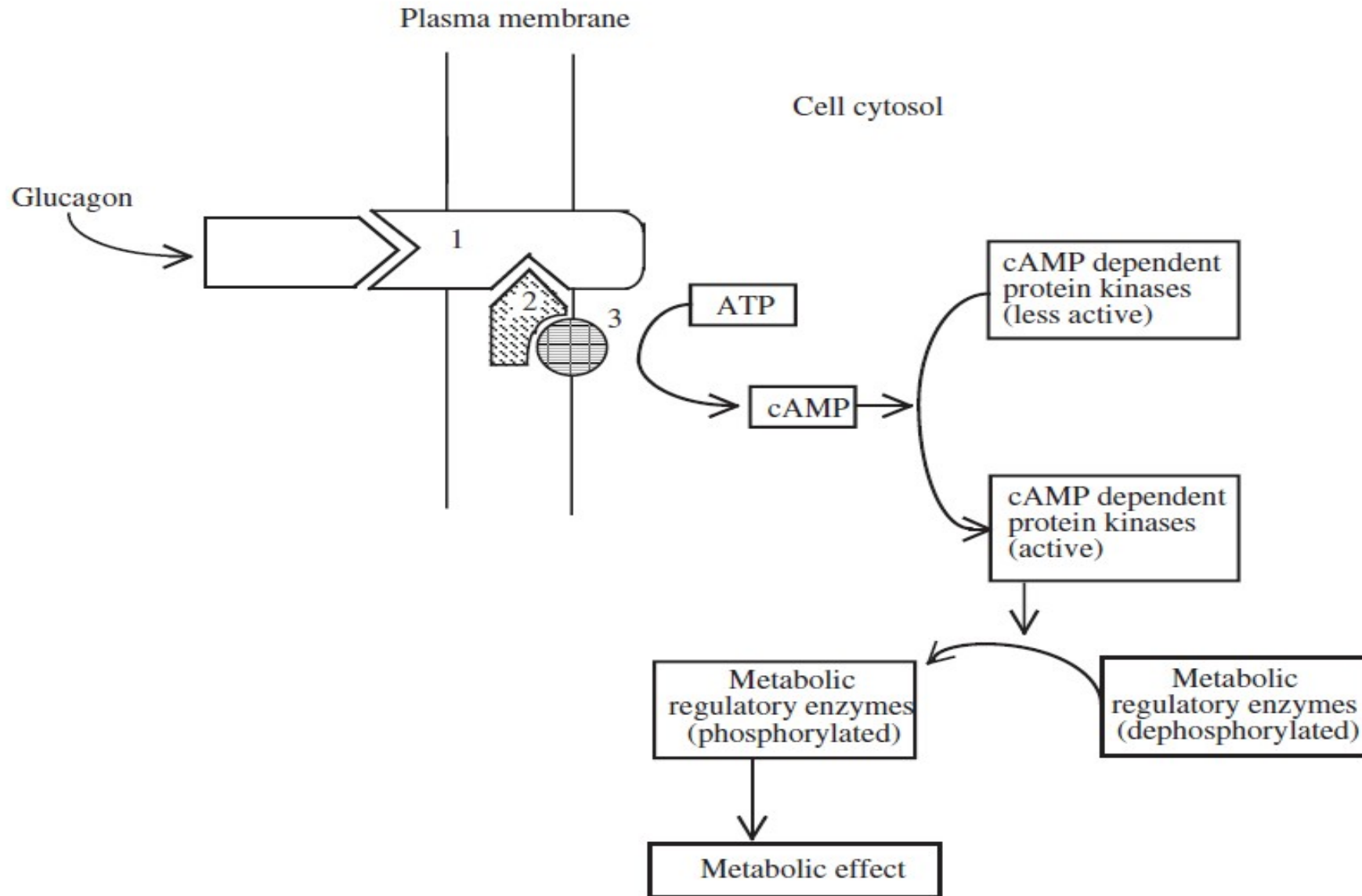
Longer-acting insulin analogues: The currently used Zn–insulin suspensions, or protamine–Zn–insulin suspensions, generally display a plasma half-life of 20–25 h.

✓ Selected amino acid substitutions have generated insulins that exhibit plasma half-lives of up to 35 h

“Optisulin or Lantus” is the tradenames given to one such analogue that gained general marketing approval in 2000.

B. Glucagon

- Glucagon is a **single-chain polypeptide of 29 amino acid residues** with a molecular mass of 3500 Da.
- It is synthesized by the **A-cells of the islets of Langerhans** and by related cells in the digestive tract
- It is synthesized as a high molecular mass from which the mature hormone is released by selective proteolysis
- The major biological actions of glucagon tend **to oppose those of insulin**, particularly with regard to regulation of metabolism
- Glucagon has an overall **catabolic effect**, stimulating the **breakdown of glycogen, lipid and protein**.
- A prominent metabolic effect is to **increase blood glucose levels (hyperglycaemic hormone)**.
- Its major physiological function is to **prevent hypoglycaemia**



Initiation of a metabolic response to the binding of glucagon to its receptor. (1) glucagon cell surface receptor; (2) G protein; (3) adenylate cyclase.

Traditionally, glucagon preparations utilized therapeutically are ***chromatographically purified from bovine or porcine pancreatic tissue***.

- The structure of **bovine, porcine and human glucagon** is **identical**

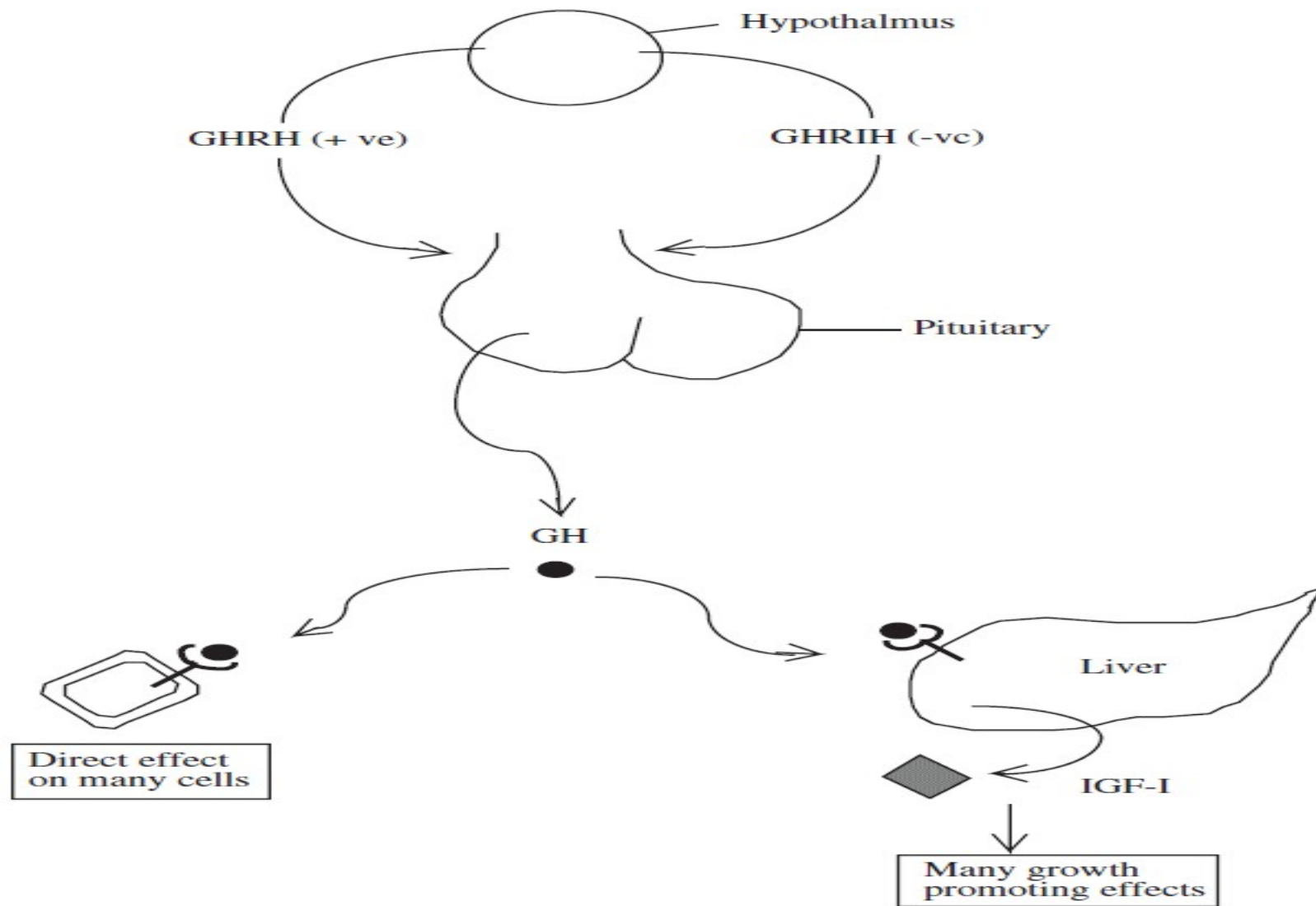
Recently, glucagon preparations produced via recombinant means have also become available

‘GlucaGen’ is the tradename given to one such product, produced by Novo Nordisk using an engineered *S. cerevisiae* strain.

- Upstream processing (aerobic batch-fed fermentation) is followed by an upward adjustment of media pH to dissolve precipitated product
- This facilitates the removal of the yeast by centrifugation
- Glucagon is then recovered and purified from the media by high-resolution chromatographic steps.

C. Human growth hormone

- ☐ *hGH (somatotrophin) is a polypeptide hormone synthesized in the anterior pituitary.***
- ☐ Mature hGH contains 191 amino acid residues and displays a molecular mass of 22 kDa. hGH displays significant species specificity.**
- ☐ It also contains two characteristic intrachain disulfide linkages.**
 - ➡ It promotes normal body growth and lactation and influences various aspects of cellular metabolism**
 - ➡ GHs isolated from other primates are the only preparations biologically active in humans.**
- ☐ GH synthesis and release is regulated by two peptide hypothalamic factors:**
 - ✓ GH releasing hormone (GHRH), somatorelin**
 - ✓ GH release inhibiting hormone (GHRIH), somatostatin**



Overview of the mechanisms by which GH induces its biological effects and how its secretion from the pituitary is regulated

Major biological effects promoted by growth hormone

- **Increased body growth** (particularly bone and skeletal muscle)
- **Stimulation of protein synthesis** in many tissues
- **Mobilization of depot lipids** from adipose tissue (lipolytic effect)
- **Elevation of blood glucose levels** (anti-insulin effect)
- **Increase of muscle and cardiac glycogen stores**
- **Increased kidney size and enhanced renal function**
- **Reticulocytosis** (increased reticulocyte production in the bone marrow)

Therapeutic uses of growth hormone

- ◆ hGH extracted from human pituitary glands was first used to treat pituitary dwarfism (i.e. caused by suboptimal pituitary GH secretion) in 1958
- ◆ It has subsequently proven effective in the **treatment of short stature** caused by a variety of other conditions

Note: The use of hGH extracted from the pituitaries of deceased human donors came to an abrupt end in 1985, when a link between treatment and Creutzfeld–Jacob disease (CJD, a rare, but fatal, neurological disorder) was discovered

- ◆ Fortunately, several recombinant hGH (rhGH) preparations were coming on-stream at that time
- ◆ Now, all hGH preparations used clinically are derived from recombinant sources.
- ◆ Currently, in excess of 20 000 people are in receipt of rhGH therapy.

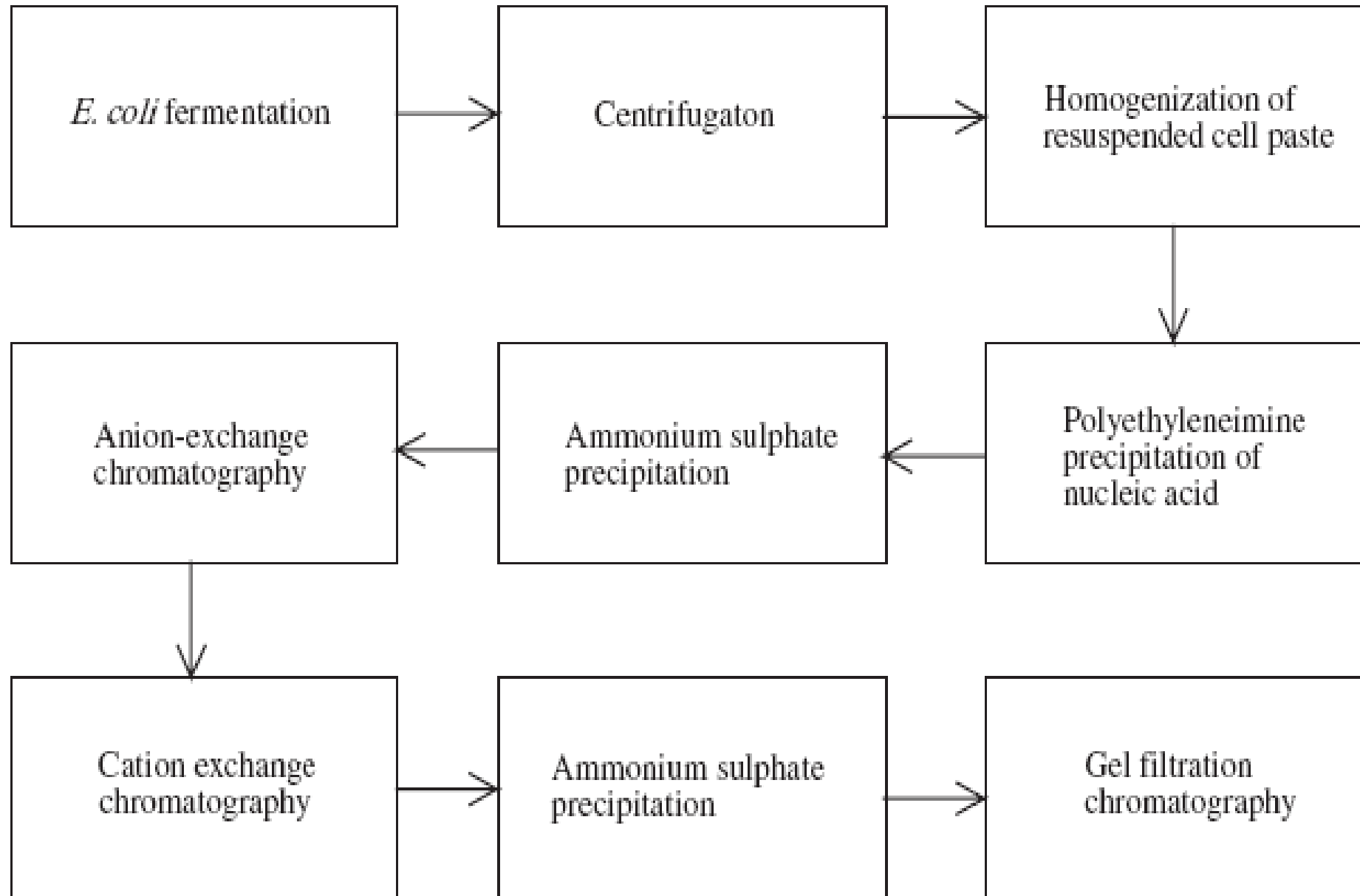
Actual or likely therapeutic uses of hGH

- Ⓢ Treatment of **short stature** caused by GH deficiency
- Ⓢ Treatment of **defective growth** caused by various diseases/medical conditions
- Ⓢ Induction of **lactation**
- Ⓢ Counteracting **ageing**
- Ⓢ Treatment of **obesity**
- Ⓢ Body building
- Ⓢ Induction of ovulation

■ rhGH was first produced in *E. coli* in the early 1980s.

■ The initial recombinant preparations differed from the native human hormone only in that they contained **an extra methionine residue**

■ Subsequently, a different cloning strategy allowed production in *E. coli* of products devoid of this terminal methionine.



rhGH preparations approved for general medical use

Product (tradename)	Company	Indication
Humatrope	Eli Lilly	hGH deficiency in children
Nutropin	Genentech	hGH deficiency in children
Nutropin AQ	Schwartz Pharma AG	Growth failure, Turner's syndrome
BioTropin	Biotechnology General	hGH deficiency in children
Genotropin	Pharmacia & Upjohn	hGH deficiency in children
Saizen	Serono Laboratories	hGH deficiency in children
Serostim	Serono Laboratories	Treatment of AIDS-associated catabolism/wasting
Norditropin	Novo Nordisk	Treatment of growth failure in children due to inadequate growth hormone secretion
Omnitrop	Sandoz	Treatment of children and adults with certain forms of growth disturbance
Valtropin	Biopartners	Treatment of children and adults with certain forms of growth disturbance
Somavert (hGH analogue, i.e. antagonist)	Pharmacia	Treatment of selected patients suffering from acromegaly

D. The gonadotropins

- ❑ The gonadotropins are a family of hormones for which the **gonads represent their primary target**
- ❑ They directly and indirectly **regulate reproductive function** and, in some cases, **the development of secondary sexual characteristics**
- ❑ Insufficient endogenous production of any member of this family will adversely affect reproductive function
 - ✓ Generally can be treated by administration of an **exogenous preparation of the hormone**
- ❑ Most gonadotrophins are synthesized by the **pituitary** and some are made by reproductive and associated tissues

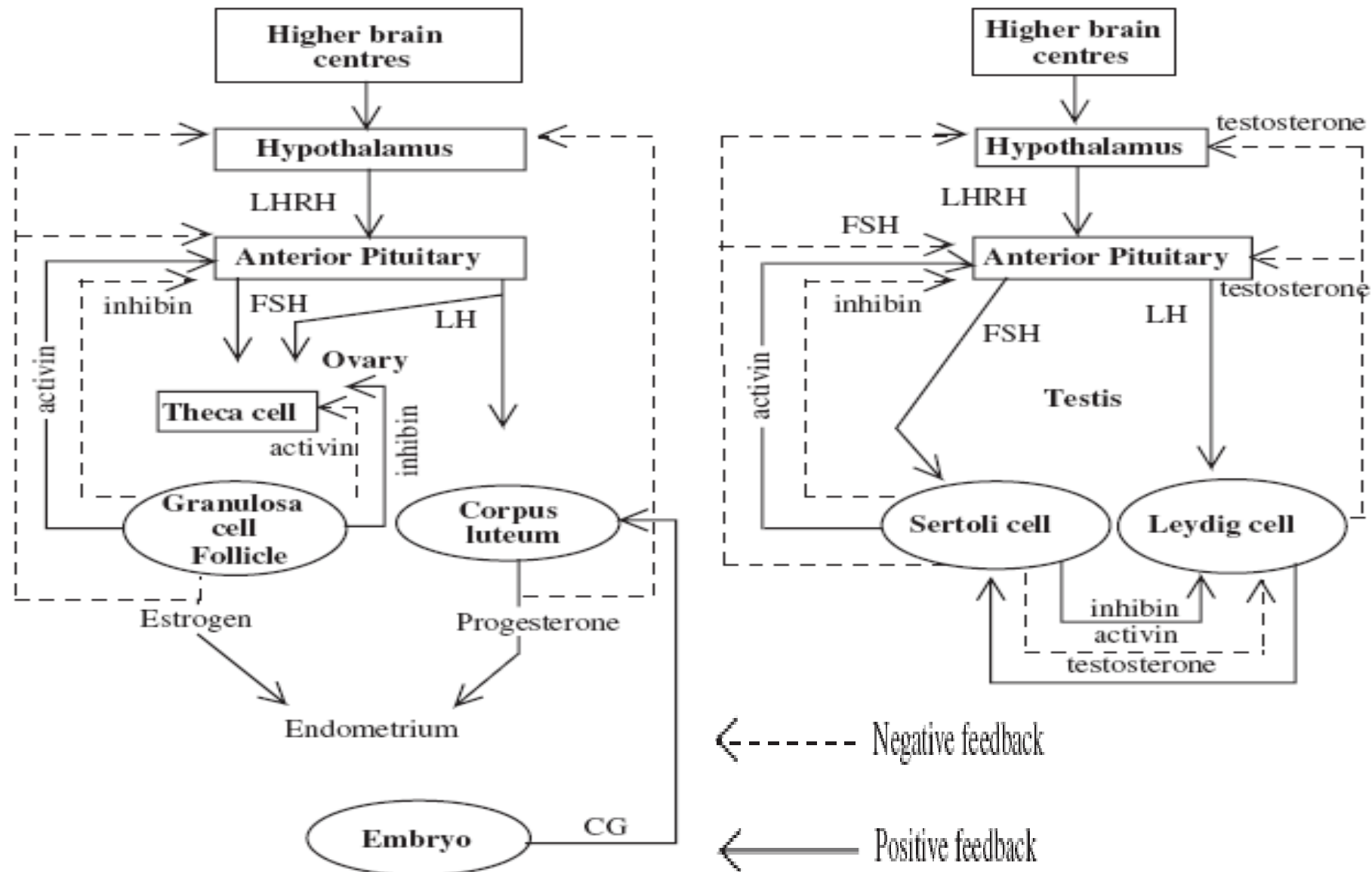
The gonadotrophins, their site of synthesis and their major biological effects

Gonadotrophin	Site of synthesis	Major effects
FSH	Pituitary	Stimulates follicular growth (female) Enhanced spermatogenesis (male)
LH	Pituitary	Induction of ovulation (female) Synthesis of testosterone (male)
CG		Maintenance of the corpus luteum in pregnant females
PMSG (horses only)	Endometrial cups	Maintenance of pregnancy in equids
Inhibin	Gonads	Inhibition of FSH synthesis Probably tumour repressor
Activin	Gonads	Stimulation of FSH synthesis

▣ **FSH and LH** play critical roles in the development and maintenance of male and, particularly, Female reproductive function

▣ **Human chorionic gonadotrophin(hCG)**, produced by pregnant women, plays a central role in maintaining support systems for the developing embryo during early pregnancy.

▣ All three are heterodimeric hormones containing an identical α -polypeptide subunit and a unique β -polypeptide subunit that confers biological specificity to each gonadotrophin



Interrelationships between various hormones regulating reproductive function in the male and female

Medical and veterinary applications of gonadotrophins

- ◆ Because of their central role in maintaining reproductive function, the therapeutic potential of gonadotrophins in treating subfertility/some forms of infertility was obvious.
- ◆ Gonadotrophins are also used to induce a superovulatory response

Recombinant gonadotrophins approved for general medical use

Product (tradename)	Company	Indication
Gonal F (rhFSH)	Serono	Anovulation and Superovulation
Puregon (rhFSH)	N.V. Organon	Anovulation and Superovulation
Follistim (rhFSH)	Organon	Some forms of infertility
Luveris (rhLH)	Ares-Serono	Some forms of infertility
Ovitrelle (rhCG)	Serono	Used in selected assisted reproductive techniques

2. Biological Products

I. Cytokines

- Cytokines are group of **regulatory proteins** or glycoproteins produced in small quantities by the body
- They act as **chemical communicators** between various cells, inducing their effect by **binding to specific cell surface receptors**, thereby **triggering various intracellular signal transduction**
- Are soluble mediators or glycoproteins **that aid in the communication between cells**: primarily cells of the immunological, hematological and neurological system
- About 60 endogenous cytokines are produced in the body and help the maturation and development.

Differences between cytokine and hormones

1. Unlike hormones, an individual cytokine can be secreted by a **number of different types of cells**
 2. Cytokines are usually targeted to produce a **local effect** while hormones are targeted to affect cells at **distant sites**.
 3. The release of a cytokine is often a brief and self limited effect i.e. have **short half life** compared to hormones.
- The soluble mediators classified as cytokines are made up of a wide range of glycoproteins including **interleukins, interferons, colony stimulating factors, chemokines, inflammatory cytokines, anti-inflammatory factors**, etc.

Classification

Cytokines can be classified based on:

1.Their source

Lymphokines: obtained from T/B lymphocytes

Eg. IL-2 and INF- γ

Monokines: produced by **mononuclear cells** like **monocytes** in matured WBCs, eg. TNF- α

2.Their function

Interleukins: cytokines which communicate between WBCs

Interferons: cytokines that interfere with the viral infection of a cell;
can be used as antiviral agents

Interleukins (ILs)

⊕ There are about 33 ILs produced by our immune cells
(IL1-IL33)

- ✓ **Isolation and purification is difficult** to use them as therapeutic agents
- ✓ **The yield is also very less** unless using rDNA technology
- Many of the interleukins are in clinical studies, only two products are approved by US FDA and are commercially available by the brand names “**Proleukin**”, an **IL-2 product** and ‘**Neumega**’, an **IL-1**

Many cell types are capable of producing interleukins

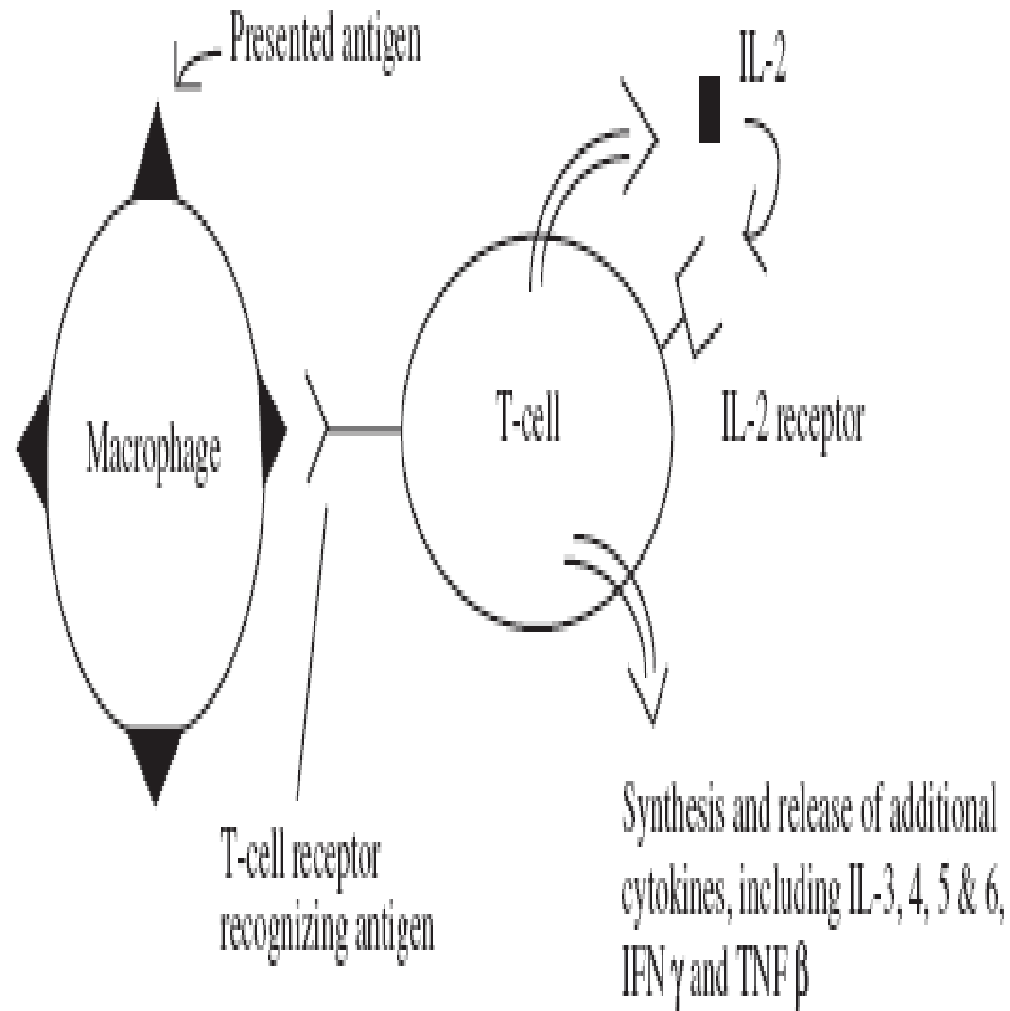
Cell type	Interleukins produced
Macrophages	IL-1, IL-6, IL-10, IL-12
Eosinophils	IL-3, IL-5
Vascular endothelial cells	IL-1, IL-6, IL-8
Fibroblasts	IL-1, IL-6, IL-8, IL-11
Keratinocytes	IL-1, IL-6, IL-8, IL-10

These interleukins regulate a variety of **physiological and pathological conditions:**

- ✓ Normal and malignant cell growth
- ✓ All aspects of the immune response
- ✓ Regulation of inflammation

Interleukin-2

- IL-2, also known as **T-cell growth factor** and plays a central role in the immune response.
- It is produced exclusively by **T-lymphocytes** (especially T-helper cells), in response to activation by **antigen and mitogens**
- Human IL-2 is a single-chain polypeptide containing 133 aas
- IL-2 induces its characteristic biological activities by binding a specific receptor on the surface of sensitive cells
- The high-affinity receptor complex consists of three membrane-spanning polypeptide chains (α , β and γ)



- ✓ **Activation of T-cells by interaction with macrophage-displayed antigen**
- ✓ **Activation results in IL-2 production, which acts in an autocrine manner to stimulate further T-cell growth and division**
- ✓ **IL-2 thus represents the major regulatory molecule responsible for stimulation of cell-mediated immunity.**

Medical uses of interleukin-2

1. For the treatment of renal cell carcinoma
 2. Treatment other cancers
 3. T-cell and other forms of immunodeficiency
 4. Treatment of infectious diseases
 5. To stimulate the growth, differentiation and activation of B-lymphocytes, T-lymphocytes and NK cells
- Produce immunological effect by binding to the cellular IL-2 receptor on the surface of these cells.

• .

“Proleukin”: is the marketed recombinant IL-2 preparation manufactured by Chiron and approved for the treatment of certain cancers

- It is produced in engineered *E. coli* and differs from native human IL-2 in that it is **non-glycosylated**
- Lacks an N-terminal alanine residue, and **cysteine 125** has been replaced by a **serine residue**
- The **absence of glycosylation** on the recombinant product **does not alter its biological activity**

Interleukin-1

- ◆ IL-1 is also known as **lymphocyte-activating factor (LAF)**
- ◆ It displays a wide variety of biological activities and has been appraised clinically in several trials
- ◆ **Two distinct forms of IL-1** exist: IL-1 α and IL-1 β
 - Although different gene products, and exhibiting only 20 percent amino acid sequence homology, both of these molecules bind the same receptor and induce similar biological activities
- ◆ The genes coding for IL-1 α and -1 β both reside on human chromosome number 2, and display similar molecular organization

Cells capable of producing IL-1

T-lymphocytes

B-lymphocytes

Monocytes/macrophages

NK cells

Large granular lymphocytes

Keratinocytes

Chondrocytes

Vascular endothelial cells

Fibroblasts

Astrocytes

Dendritic cells

Microglia

Glioma cells

IL1 has been associated with various biological activities:

- ✓ Activation of T-helper cells: **lymphocyte activating factor**
- ✓ Stimulation of B-cell differentiation: **B cell differentiation factor**
- ✓ Augmentation of NK cell cytotoxicity
- ✓ Induction of fever: **Endogenous pyrogen**
- ✓ Enhancement of collagen production
- ✓ Release of other cytokines

Proposed medicinal uses of IL-1

- ◆ For immunostimulatory effect
- ◆ Anti-proliferative effects against various human tumour cell lines
- ◆ Treatment of bone marrow suppression induced by chemo/radiotherapy;
- ◆ Treatment of various cancers

The interferons

- Interferons were the first family of cytokines to be discovered.
- In 1957, researchers observed that susceptible animal cells, if they were exposed to a colonizing virus, immediately became resistant to attack by other viruses.
- This resistance was induced by a substance secreted by virally infected cells which was named **interferon**
- It has been shown that most species actually produce a whole range of interferons
- Humans produce at least three distinct classes, IFN- α , IFN- β and IFN- γ
- There are also IFN- τ and IFN- ω

Human interferons and the cells that produce them

Interferon family	Additional name	No. distinct interferons in family	Producing cells
IFN- α	Leukocyte interferon	>15	Lymphocytes
	B cell interferon		Monocytes
	Lymphoblast interferon		Macrophages
IFN- β	Fibroblast interferon	1	Fibroblasts
	IFN- β -1 ^a		Some epithelial cells
IFN- γ	Immune interferon	1	T-lymphocytes
	T cell interferon		NK cells

Biological effects of interferons

- ⊕ Induction of cellular resistance to viral attack
- ⊕ Regulation of most aspects of immune function
- ⊕ Regulation of growth and differentiation of many cell types

IFN- γ (Type II INF)

- ❑ Represents the main **macrophage-activating factor**, thus enhancing macrophage-mediated effects including:
 - Destruction of invading microorganisms
 - Destruction of intracellular pathogens
 - Tumour cell cytotoxicity
 - Increased major histocompatibility complex (MHC) antigen expression, leading to **enhanced activation of lymphocytes via antigen presentation**

Medical uses of interferons

- ⊕ Augmentation of the immune response against infectious agents (viral, bacterial, protozoal, etc.)
- ⊕ Treatment of some autoimmune conditions
- ⊕ Treatment of certain cancer types

II. rBlood products and therapeutic enzymes

Recombinant blood products

- ⊕ **Blood** and **blood products** constitute a major group of traditional biologics
- ⊕ The main components are **RBCs** and **WBCs** along with **platelets** and the **plasma** in which these cellular elements are suspended
- ⊕ Whole blood remains in routine therapeutic use, as do red blood cell and platelet concentrates
- ⊕ A variety of therapeutically important **blood proteins** are **being purified from plasma for therapeutic purpose**

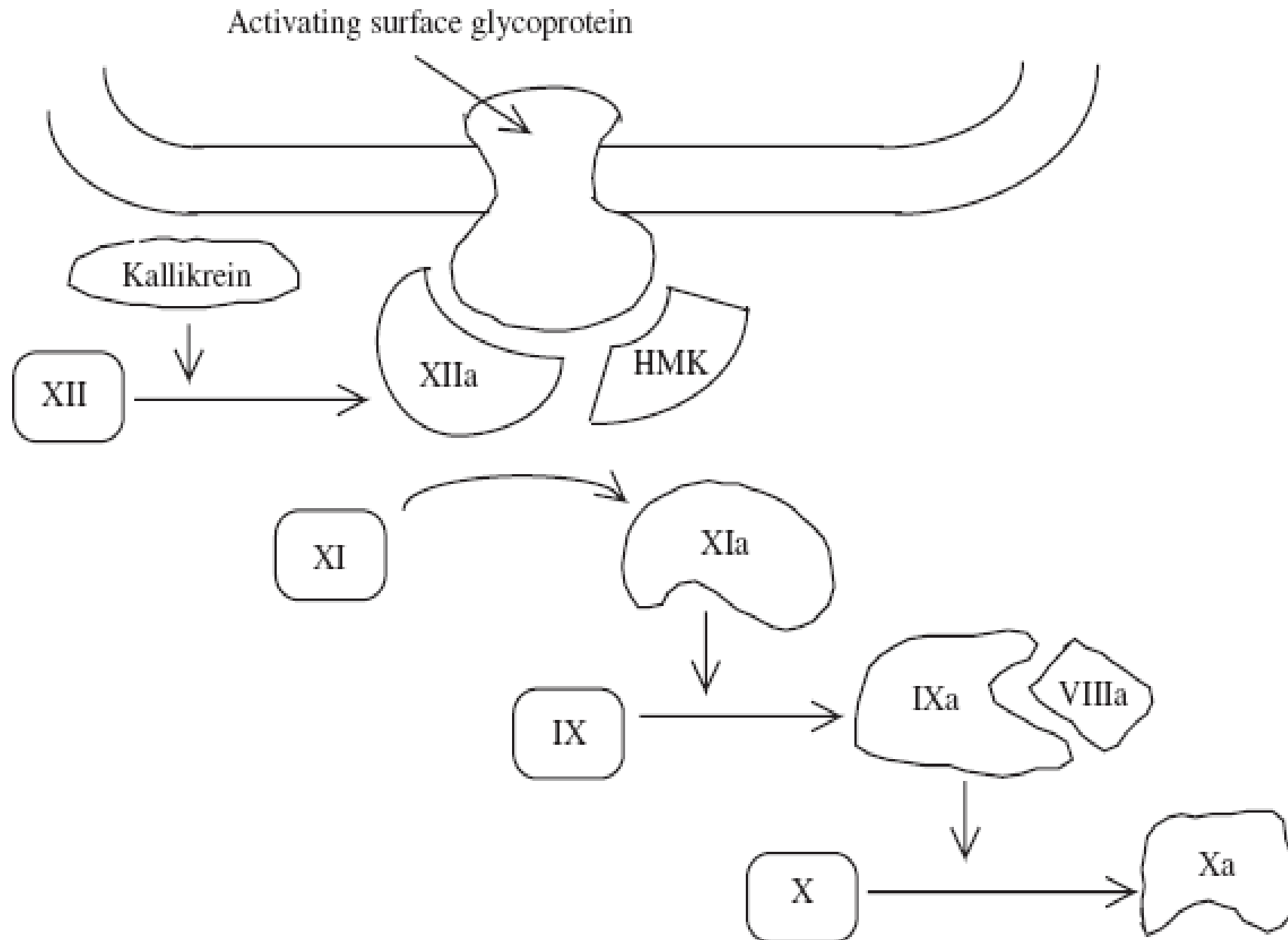
Haemostasis: the rapid arrest of blood loss upon vascular damage, in order to maintain a relatively constant blood volume

■ In humans, three main mechanisms underline the haemostatic process:

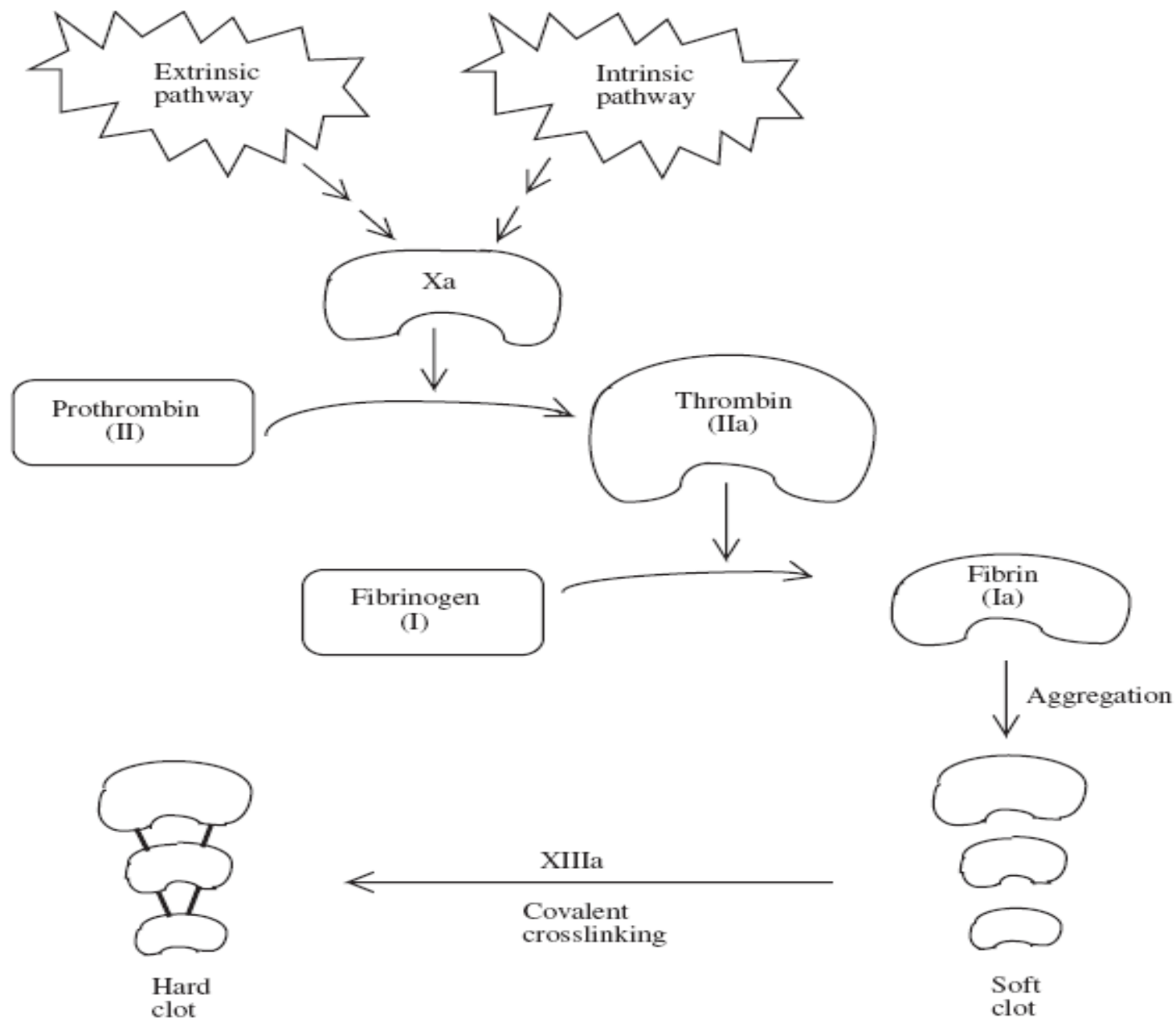
1. The **congregation and clumping of blood platelets at the site of vascular injury**, thus effectively **plugging the site of blood leakage**
2. **Localized constriction of the blood vessel**, which minimizes further blood flow through the area
3. Induction of the **blood coagulation cascade**
 - ✓ This culminates in the conversion of a soluble serum protein, fibrinogen, into insoluble fibrin

The coagulation factors that promote the blood clotting process

Factor No.	Common name	Pathway	Function
I	Fibrinogen	Both	Forms structural basis of clot after its conversion to fibrin
II	Prothrombin	Both	Precursor of thrombin, which activates factors I, V, VII, VIII and XIII
III	Tissue factor (thromboplastin)	Extrinsic	Accessory tissue protein which initiates extrinsic pathway
IV	Calcium ions	Both	Required for activation of factor XIII and stabilizes some factors
V	Proaccelerin	Both	Accessory protein, enhances rate of activation of X
VII	Proconvertin	Extrinsic	Precursor of convertin (VIIa) which activates X (extrinsic system)
VIII	Antihæmophilic factor	Intrinsic	Accessory protein, enhances activation of X (intrinsic system)
IX	Christmas factor	Intrinsic	Activated IX directly activates X (intrinsic system)
X	Stuart factor	Both	Activated form (Xa) converts prothrombin to thrombin
XI	Plasma thromboplastin antecedent	Intrinsic	Activated form (XIa) serves to activate IX
XII	Hageman factor	Intrinsic	Activated by surface contact or the kallikrein system. XIIa helps initiate intrinsic system
XIII	Fibrin-stabilizing factor	Both	Activated form cross-links fibrin, forming a hard clot



The steps unique to the intrinsic coagulation pathway



Recombinant blood coagulation factors approved for the management of coagulation disorders

Product(Trade name)	Company	Indication
Advate (rhFactor VIII)	Baxter	Haemophilia A
Bioclata (rhFactor VIII)	Aventis	Haemophilia A
Benefi x (rhFactor IX) Institute	Genetics	Haemophilia B
Kogenate (rhFactor VIII)	Bayer	Haemophilia A
Helixate NexGen (rhFactor VIII)	Bayer	Haemophilia A
NovoSeven (rhFactor VIIa)	Novo-Nordisk	Some forms of haemophilia
Recombinate (rhFactor VIII)	Baxter Healthcare/ Genetics Institute	Haemophilia A
ReFacto (B-domain deleted rhFactor VIII)	Genetics Institute	Haemophilia A

Anticoagulants

- Although blood clot formation is essential to maintaining haemostasis, inappropriate clotting can lead to serious, sometimes fatal medical conditions
- The formation of a blood clot (thrombus) often occurs inappropriately within diseased blood vessels
 - ✓ This partially or completely obstructs the flow of blood (and hence oxygen transport)
- Anticoagulants **prevent blood from clotting** and, hence, are of therapeutic use in cases where a high risk of coagulation is diagnosed
 - ✓ They are often administered to patients with **coronary heart disease** and **heart attack**

Heparin

- ◆ **Heparin** is a **carbohydrate-based** (glycosaminoglycan) anticoagulant mainly found stored intracellularly as granules in mast cells
- ◆ Upon release into the bloodstream, heparin binds to and thereby **activates an additional plasma protein; antithrombin**
- ◆ The **heparin–antithrombin complex** then binds to activated clotting factors (including IIa, IXa, Xa, XIa and XIIa) thereby **inactivating them**
- ◆ Heparin was originally extracted from liver but commercial preparations are now obtained **from beef lung or porcine gastric mucosa**

■ Although it has proven to be an effective and inexpensive anticoagulant it has some clinical **disadvantages**:

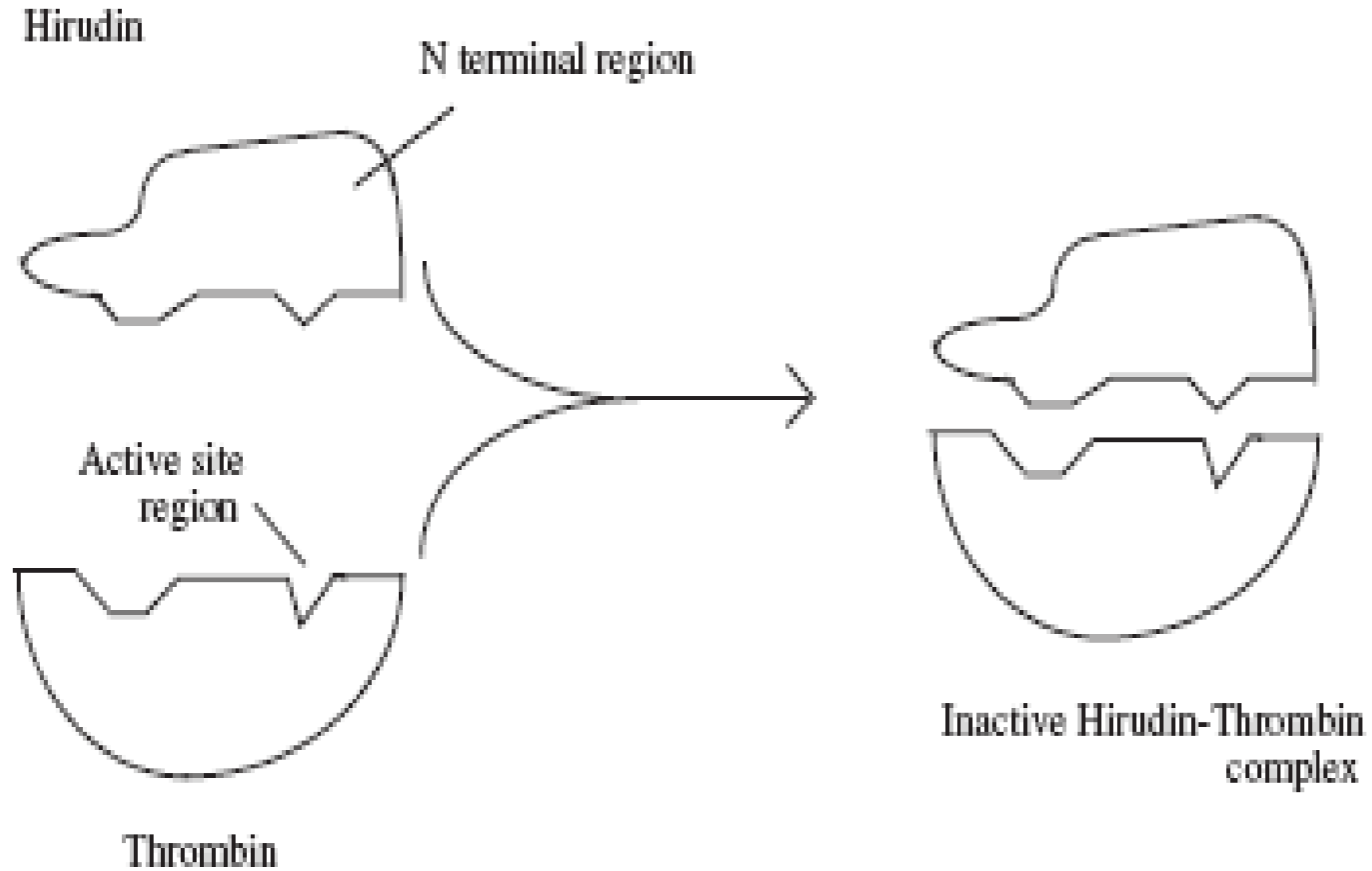
- The need for a **cofactor (antithrombin III)**
- Poorly predictable dose responses profile

■ **The vitamin K anti-metabolites: Dicoumarol and warfarin**

- ✓ Are related **coumarin-based anticoagulants** which unlike heparin may be administered **orally**
- These compounds induce their anticoagulant effect by preventing the **vitamin K-dependent γ -carboxylation of certain blood factors**, specifically **factors II, VII, IX and X**

Hirudin

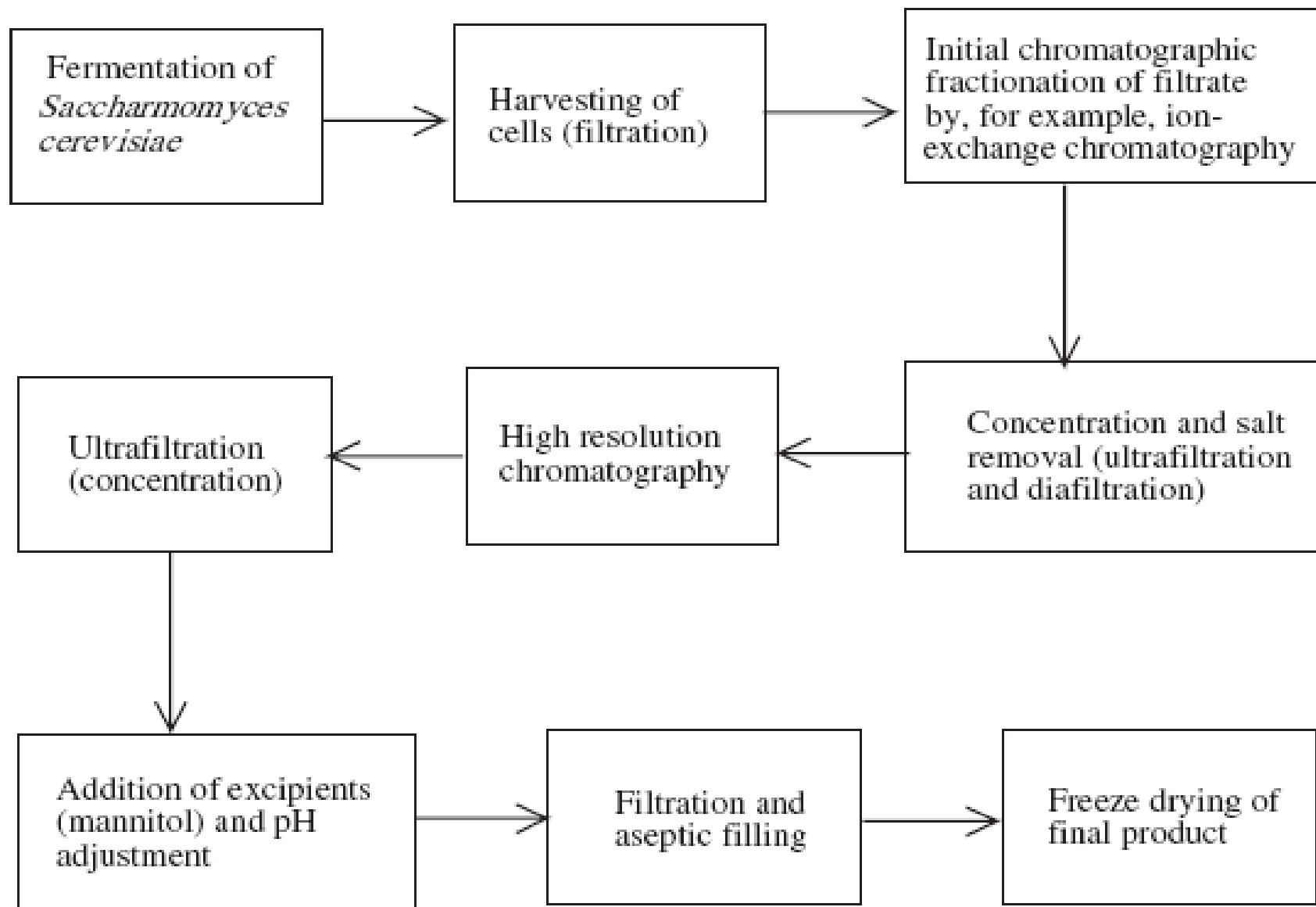
- **Hirudin** is a **leech-derived anticoagulant** that functions by **directly inhibiting thrombin**
- A range of blood-sucking animals contain substances in their saliva that inhibit some element of the blood coagulation system
 - ✓ A bite from any such parasite is characterized by **prolonged host bleeding**
- The presence of an anticoagulant in the saliva of the leech, **Hirudo medicinalis**, was first described in 1884
- The major anticoagulant **hirudin** was purified in 1957
 - ✓ **Hirudin** is a short (65 aa) polypeptide , molecular mass 7000 Da.



Binding of hirudin to thrombin, thus inactivating this activated coagulation factor

Potential therapeutic advantages Hirudin

- ✓ It **acts directly upon thrombin**
- ✓ It does not require a cofactor
- ✓ It is less likely to **induce unintentional haemorrhage**
- ✓ **It is a weak immunogen**
- The hirudin gene was cloned in the 1980s, and it has subsequently been expressed in a number of recombinant systems, including *E. coli*, *Bacillus subtilis* and *Saccharomyces cerevisiae*
- A recombinant hirudin (“**Refludan**”) was first approved for general medical use in 1997



Overview of the production of Refludan (recombinant hirudin)

Antithrombin

- ◆ It is a single-chain 432 amino acid glycoprotein displaying four oligosaccharide side chains with Mol. mass of 58 kDa
- ◆ Is a **potent inhibitor of thrombin** (factor IIa), as well as of factors IXa and Xa
- ◆ Recombinant antithrombin has been expressed in the milk of transgenic goats
- ◆ This product (**Atryn**) was approved for general medical use in Europe in 2006
- ◆ A related product (**Xigiris**, also known as **drotrecogin** alfa) has also been approved for medical use

Thrombolytic agents

- These are **clot-degrading agents** and used to **prevent blood thrombosis in blood vessels**
- In the USA alone, it is estimated that 1.5 m ach year, and there are another 0.5 million suffer from **strokes**

Tissue plasminogen activator

- **Plasmin** is a protease that catalyses the proteolytic degradation of **fibrin** present in clots, thus effectively dissolve blood clot
- Plasminogen is synthesized in the kidneys
- It is a single-chain 90 kDa glycoprotein stabilized by several disulfide linkages.

- ◆ Plasmin is derived from plasminogen by **tissue plasminogen activator (tPA)**
- ◆ tPA is the most important physiological **activator of plasminogen**
- ◆ tPA is a 527 amino acids and is a serine protease synthesized predominantly in vascular endothelial cells (cells lining blood vessels) and displays five structural domains each of with a specific function
- ◆ The tPA gene was cloned from **Bowes melanoma cell** line in 1983 and subsequently production in cell lines by recombinant DNA technology

- ***Alteplase:*** is the first marketed product licenced by Genentech in the USA in 1987 for the treatment of **acute myocardial infarction**

Engineered tissue plasminogen activator

- **Modified forms of tPA** have been generated to develop a product with an improved therapeutic profile
 - ✓ **Faster acting or exhibiting a prolonged plasma half-life)**

Reteplase: is such a **modified human tPA** produced in recombinant *E. coli cells* and is sold under the trade names **Ecokinase, Retavase and Rapilysin**

Hematopoietic growth factors

🧬 **Hematopoietic growth** is the process by which blood cells can be formed from stem cells.

🧬 The **growth and survival of blood cells** require the presence of specific factors which are called **hematopoietic growth factors (colony stimulating factors)**.

🧬 These factors regulate both **hematopoiesis** and **functional** activities of matured cells.

There are three important factors:

1. **Granulocyte colony stimulating factor(G-CSF)**
2. **Granulocyte macrophage colony stimulating factor (GM-CSF)**
3. **Erythropoietin (EPO)**

🧬 Chemically, they are all glycoproteins but differ each other based on amino acid sequence and their glycosilation.

🧬 They have cysteine-cysteine disulfide bridge and disturbance of this bridge affects their biological activity.

🧬 There are specific gene fragments that express these factors.

🧬 There are commercially available **rDNA derived forms**.

Filgrastim and *Lenograstim* are the **two** G-CSF derivatives available for clinical use.

Malgramostim and *sargramostim* are GM-CSF forms in clinical use and under clinical trial respectively.

Erythropoietin

- A glycoprotein with 165 aa, two disulfide bonds, three nitrogen and one oxygen linked to carbohydrate chains
- **Epoitein α** and **Epoitein β** are the two recombinant versions
- Erythropoietin maintains a normal **RBC count** by causing **committed erythroid progenitor cells** to proliferate and differentiate

Growth factors approved for general medical use

Product ^a	Indication	Company
Neupogen (filgrastim; G-CSF)	Neutropenia caused by chemotherapy Bone marrow transplants	Amgen Inc.
Leukine (sargramostim, GM-CSF)	Autologous bone marrow transplantation Neutrophil recovery after bone marrow transplantation	Berlex Labs
Neulasta (PEGylated filgrastim, see above)	Neutropenia	Amgen
Epogen (epoetin alfa, rEPO)	Anaemia associated with various medical conditions	Amgen
Procrit (epoetin alfa, rEPO)	Anaemia associated with various medical conditions	Ortho Biotech
Neorecormon (epoetin beta, rEPO)	Anaemia associated with various medical conditions	Boehringer-Mannheim
Aranesp (darbepoetin alfa, a rEPO analogue)	Anaemia associated with various medical conditions	Amgen
Nespo (darbepoetin alfa, a rEPO analogue)	Anaemia associated with various medical conditions	Dompe Biotec
Regranex (becaplermin, rPDGF)	Neuropathic diabetic ulcers	Janssen & Ortho-McNeil
Kepivance (palifermin, rKGF)	Severe oral mucositis	Amgen
Increlex (mecasermin, rh IGF-1)	Growth failure in children	Tercica/Baxter
Iplex (mecasermin rinfabate, complex of rhIGF-1 and IGFBP-3)	Growth failure in children	Insmed
GEM 21S (implantable product containing rhPDGF-BB)	Periodontally related defects	Luitpold Pharmaceuticals and Biomimetic Pharmaceuticals

^aIGFBP: insulin like growth factor binding protein.

Blood and Blood Products

Whole Blood

Contents

RBC's	WBC's
Platelets	Plasma
Clotting factors	

Indications

- ✓ Acute loss of whole blood

Packed Cells

Contents

- RBC's
- 20% Plasma

Indications

- Replace O₂ carrying capacity with less volume
- Severe anemia, slow blood loss, CHF

Granulocytes

Contents

- WBC's
- 20% Plasma

Indications

- Life-threatening decreases in WBC count

Platelets

Contents

- Platelets
- WBC's
- Plasma

Indications

- Low platelet counts

Plasma

Contents

Clotting factors Fibrinogen

Prothrombin Albumin

Globulins

Indications

- ❖ Clotting factor deficiency
- ❖ Volume expansion

Plasma Protein Fraction

Contents

- 5% Albumin/Globin in saline

Indications

- Expand volume in burns
- Hemorrhage
- Hypoproteinemia

Albumin

Contents

- 5% or 25% albumin

Indications

- Replace volume in shock
- Burns
- Hypoproteinemia

Cryoprecipitate

Contents

- Factors VIII and XIII, Fibrinogen

Indications

- Hemophilia A
- Fibrinogen deficiency
- Factor XIII deficiency

Prothrombin

Contents

- Factors II, VII, IX, and X

Indications

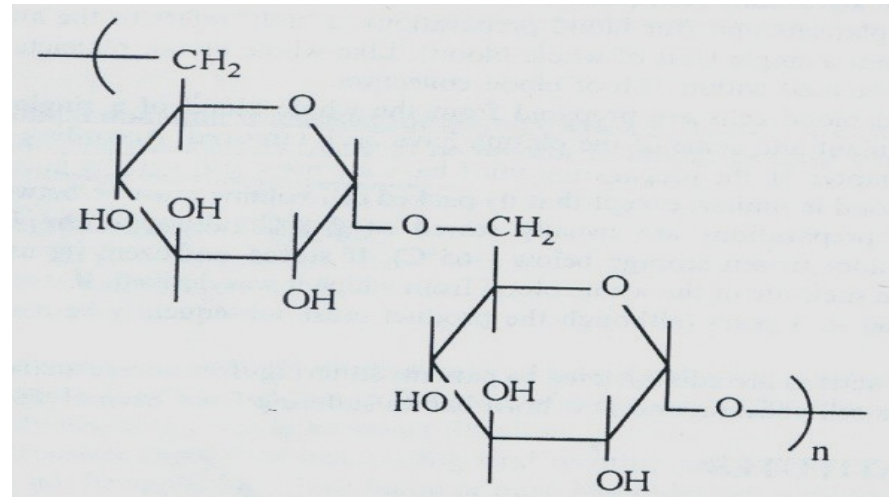
- Hemophilia B
- Liver disease

- Replace or maintain blood volume
- Ensure oxygen delivery to body tissues
- Function only as plasma expanders
 - ✓ Maintaining blood pressure by providing vascular fluid volume after hemorrhage, burns, sepsis or shock
 - ✓ Relatively high cost
 - ✓ The risk of prompting a hypersensitivity reaction

Blood substitutes....

Dextrans

- Polysaccharides produced naturally by various microorganisms
- Native dextran : high molecular mass
- Clinical dextran
 - ✓ By depolymerization of the native dextran
 - ✓ By direct synthesis



Dextrans...

- Higher molecular mass dextran (70, 75, 110, 6% w/v)
 - ✓ Promotes short-term expansion of plasma volume (preventing/treating shock due to blood loss)
 - ✓ Inhibit the aggregation of red blood cells (preventing/treating post-operative thrombo-embolic disorders and improving blood flow)
- Lower molecular mass dextran (40 kDa, 10% w/v)
 - ✓ Similar therapeutic effects to the higher dextran

Negative effects

- Dextran 40 and 70 : acute renal failure
- Infusion of dextran : severe anaphylactic shock
- These negative effects are due to presence of anti-dextran antibodies

Albumin (Human serum albumin (HSA))

- The single most abundant protein in blood (42 g/ liter, 60 % of total plasma protein)
- 585 amino acid, 65.5 kDa polypeptide, unglycosylated
- Synthesized and secreted from the liver

Uses

Osmo-regulatory function:

- ✓ Responsible for over 80 % of the colloidal osmotic pressure of human blood

Transport function:

- ✓ Various metabolites bound to HAS (fatty acids, amino acids, steroid hormones and heavy metals and many drugs)
- ✓ Prepared by fractionation from normal plasma or serum, or purified from placentas

As a plasma expander

- ✓ For the management of hemorrhage, shock, burns and oedma and administration after surgery
- ✓ HSA derived from native blood sometimes harbour pathogens

Screening pathogens

- *Filtration, storage into sterile container, heat treatment (60 °C for 10 h)*
- ✓ rDNA technology can overcome the concern
- HSA gene and cDNA have been expressed in a wide variety of microbial systems.
- HSA's relatively large size and the presence of so many disulphide bonds => complicate recombinant HSA of high levels
- The main stumbling block is an economic one.

Gelatin

- Produced by partial acid or partial alkaline hydrolysis of animal collagen
- The manufacture of hard and soft capsule shells, suppositories, tablets, sponge during surgery
- 4 % solution of gelatin : plasma expander
- Infusion of gelatin solution : sometimes initiate hypersensitivity reactions
- Excreted relatively quickly and mostly via the urine

Blood Transfusion

✓ Blood must be typed prior to administration

ABO Antigens

- | | |
|--------------------|---------|
| ✓ A Antigen | Type A |
| ✓ B Antigens | Type B |
| ✓ A and B Antigens | Type AB |
| ✓ No Antigens | Type O |

❖ Plasma Antibodies Agglutinate (Clump) Cells of other Types

Blood Transfusions

Type A = B Antibodies

- (Clumps B or AB)

Type B = A Antibodies

- (Clumps A or AB)

Type AB = No Antibodies

- (Clumps Nothing)

Type O = A and B Antibodies

- (Clumps everything except O)

O Negative = Universal Donor

AB Positive = Universal Recipient

Patient and donor RBC selection by ABO and Rh type

Patient	Donor
A	A, O
B	B, O
AB	A, B, AB, O
O	O
Rh(+)	Rh(+), Rh(-)
Rh(-)	Rh(-)

Adverse effects of transfusion

Immunologic

1. Alloimmunization
2. Hemolytic transfusion reactions
 - immediate
 - delayed
3. Allergy
4. Immunosuppressive effects
5. Transfusion-related acute lung injury
6. Post-transfusion purpura(subcutaneous bleeding)
7. Graft-versus-host disease

Nonimmunologic

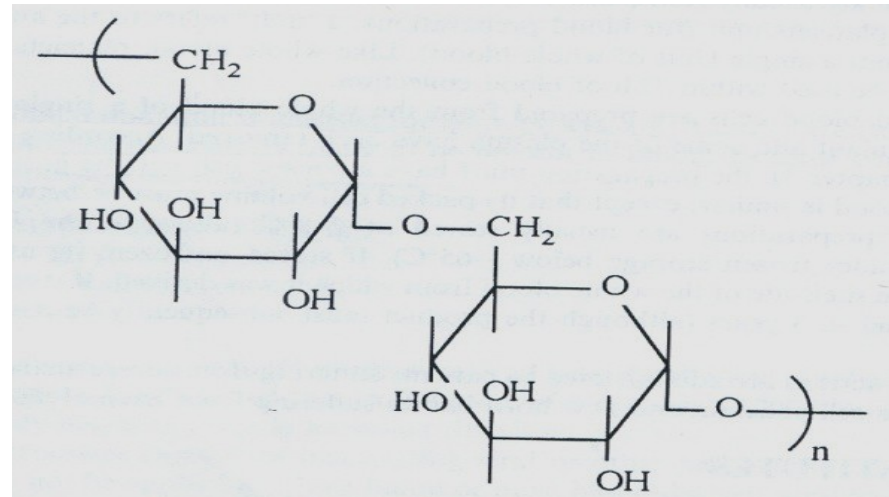
1. Volume overload
2. Massive transfusion: metabolic, hypothermia, dilutional, pulmonary microembolism
3. Transfusion **hemosiderosis**(Abnormal deposit of hemosiderin; a granular brown substance composed of ferric oxide; left from the breakdown of hemoglobin; can be a sign of disturbed iron metabolism)
4. Infections
 - ✓ Hepatitis(A, B, C)
 - ✓ HIV
 - ✓ Bacterial contamination
 - ✓ Malaria
 - ✓ other organisms

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Assignment

ENZYMES OF THERAPEUTIC VALUES

- Most enzymes are administered intravenously

Problems

1. **Immunological response** (antibody formation)
2. **Immediate allergic/anaphylactic response**
3. **Removed from the blood stream and rapidly degraded**

=> Antigenicity and short plasma half-lives

Solutions

1. **Encapsulation** : non-antigenic substance (albumin, nylon, polyacrylamide or cellulose nitrate), proper pore size
 2. **Liposome**: target molecule incorporation to liposome membrane
 3. **Chemical modification** : succinylation, coupling to polyglycols (polyethylene glycol [PEG])
- ✓ PEG-asparaginase was approved by the FDA
 - ✓ PEG-superoxide mutase : increase of half-life from 5 h to 30 h

Enzymes used therapeutically

Enzyme	Application
Tissue plasminogen activator	Thrombolytic agent
Urokinase	Thrombolytic agent
Ancrod	Anticoagulant
Factor IXa	Haemophilia B
Asparaginase	Anti-cancer agent
Nuclease (DNase)	Cystic fibrosis
Glucocerebrosidase	Gaucher's disease
α -Galactosidase	Fabry disease
Urate oxidase	Hyperuricaemia
Laronidase	Mucopolysaccharidosis
Superoxide dismutase (SOD)	Oxygen toxicity
Acid α -glucosidase	Pompe disease
α -L-Iduronidase	Mucopolysaccharidosis I (MPS I)
N-Acetylgalactosamine-4-sulfactase	Mucopolysaccharidosis IV
Trypsin/papain/collagenase	Debriding/anti-inflammatory agents
Lactase/pepsin/papain/pancrelipase	Digestive aids

Streptokinase:

- Is a 48 kDa **extracellular bacterial protein** produced by several strains of *Streptococcus haemolyticus group C*
- It directly **convert or activate plasminogen to plasmin** and is a widely employed **thrombolytic agent**
- It is administered to treat a variety of **thrombo-embolic disorders**, including:
 - ✓ **Pulmonary embolism (blockage of the pulmonary artery)**
 - ✓ **Deep-vein thrombosis (thrombus formation in deep veins)**
 - ✓ **Arterial occlusions (obstruction of an artery)**
 - ✓ **Acute myocardial infarction**

Urokinase:

- Is a serine protease produced by the **kidney** and found in both the **plasma and urine**
- It forms *a one-to-one complex with plasminogen and the complex formed transforms the free plasminogen to plasmin*: can be obtained from human urine but difficulty of purification and low yield
- Urokinase is used clinically under the same circumstances as streptokinase; **because of its human origin, adverse immunological responses are less likely**
- Urokinase can also be produced by techniques of animal cell culture utilizing human kidney cells or by recombinant DNA technology

Asparaginase

- Catalyze the **hydrolysis of L-asparagine**
 - => aspartic acid and ammonia
- **Inhibit the proliferation of the leukaemic cells**
 - PEG-L-asparaginase was approved for the treatment of refractory childhood **acute lymphoblastic leukaemia**
- *Side effects*
 - Severe **nausea**, vomiting, diarrhoea and compromised liver and kidney function
 - Due to a transient asparagine deficiency in various tissues

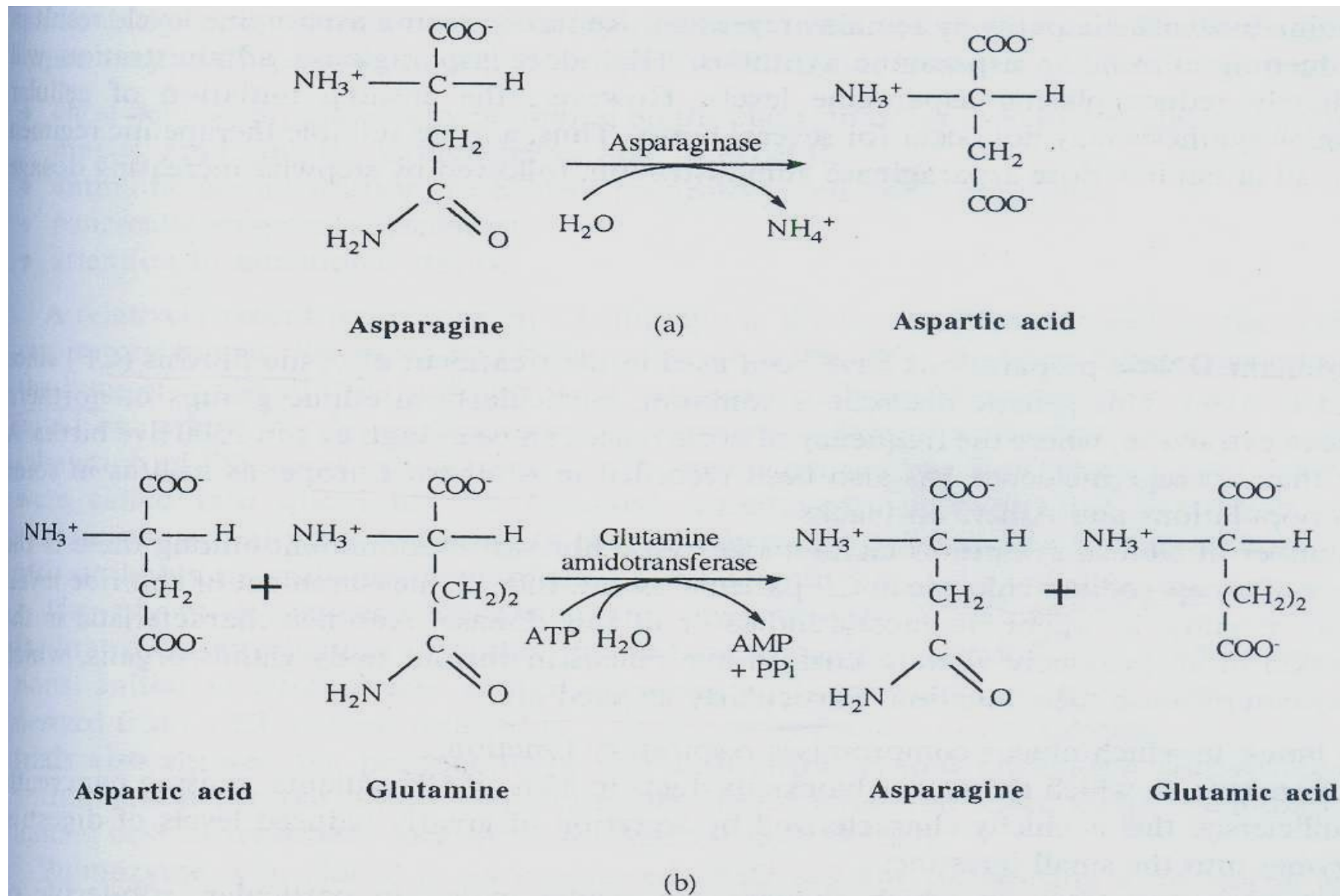


Fig. (a) Hydrolytic reaction catalyzed by L-asparaginase; (b) Reaction by which asparagine is synthesized in most mammalian cells

DNase

- An enzyme which cleaves the free DNA as high molecular mass DNA
- ✓ **Pulmozyme** is rDNase product of Genentech approved in 1993 by FDA
- It is clinically very important for the treatment of **cystic fibrosis** (CF; a disease of the lung due to excessive viscous secretion containing different chemicals)
- CF is common, particularly in ethnic groups of northern Europe, southern Europe, Jewish population and American blacks.
 - ❖ The **presence of excess sodium chloride in sweat**
 - ❖ The **production of an extremely viscous**, custard-like mucus in various body glands/organs (lungs, pancreas, reproductive tract, liver and small intestine)
-

CF management

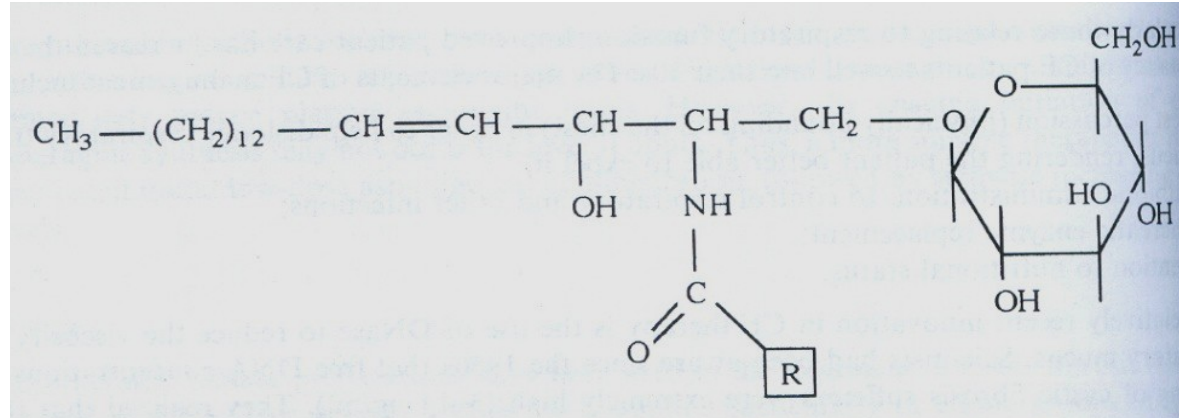
- **Chest percussion** in order to help remove respiratory tract mucus, rendering the patient better able to expel it.
- **Antibiotic administration**, to control respiratory and other infections
- **Pancreatic enzyme replacement**
- Attention to nutritional status
- ✓ ***Recent innovation in CF therapy is the use of Dnase to reduce the viscosity of respiratory mucus***

DNase treatment

- Free DNA conc. in the lung of cystic fibrosis sufferers is extremely high
- ✓ **Free DNA contributes to the mucus viscosity**
- Pulmozyme is produced in an engineered CHO cell line
- ✓ **Administered directly into the lungs by inhalation of an aerosol vapor**

Glucocerebrosidase

- Degrade the glucocerebroside



- Relieve the symptoms of **Gaucher's disease**
 - ✓ **Inborn error of metabolism characterized by lack of the enzyme glucocerebrosidase, with consequent accumulation of glucocerebrosides, particularly in tissue-based macrophages**
 - ✓ **Damage to Liver, spleen, long bones and mental retardation**

Glucocerebrosidase...

▪ *Ceredase*

- ✓ Extracted from **placentas** (afterbirths)
- ✓ Extremely expensive

▪ *Cerezyme (extracted from CHO cells,)*

- ✓ Shelf-life : 2 years at 2-8°C
- ✓ In excess of 95 % of **injected glucocerebrosidase is removed from the circulation by the liver** via binding to hepatocyte surface lectins
- ✓ To make the enzyme specific to the tissue macrophages, the enzyme was **mannose** engineered

- Human a-galactosidase
 - ✓ 100 kDa homodimeric glycoprotein
 - ✓ Each 398 amino acid monomer displays 45.3 kDa
 - ✓ Glycosylated at three positions (asparagines 108, 161, 184)
- **Recombinant a-galactosidase : long-term enzyme replacement therapy in patients with Fabry's disease**
 - ✓ Genetic disease of **lipid metabolism**
 - ✓ Lack of liposomal a-galactosidase A activity
 - ❖ *Resulting in the progressive accumulation of glycosphingolipids in several body cell types*

a-Galactosidase.....

Fabrazyme

- ✓ Produced by Genzyme
- ✓ CHO cell line
- ✓ Five chromatographic purification steps

Replagal

- ✓ Produced by TKT Europe
- ✓ Human cell line
- ✓ Five chromatographic purification steps

Urate oxidase

- Tetramer composed of four identical polypeptide subunits
- Each subunit : 301 amino acids, 34 kDa, N-terminal acetylated
- **Used for treatment of acute hyperuricaemia** (elevated plasma uric acid levels), associated with various tumours
- **Oxidizes uric acid to allantoin**, more water-soluble than **uric acid** and freely excreted via renal route

Uric acid

- End product of purine metabolism in humans, other primates, birds and reptiles
- Produced by the **oxidation of xanthine and hypoxanthine**
- ***Due to low solubility, an increase in serum uric acid levels often triggers ;***
- ***the formation and precipitation of uric acid crystals (gout, urate stones in the urinary tract)***

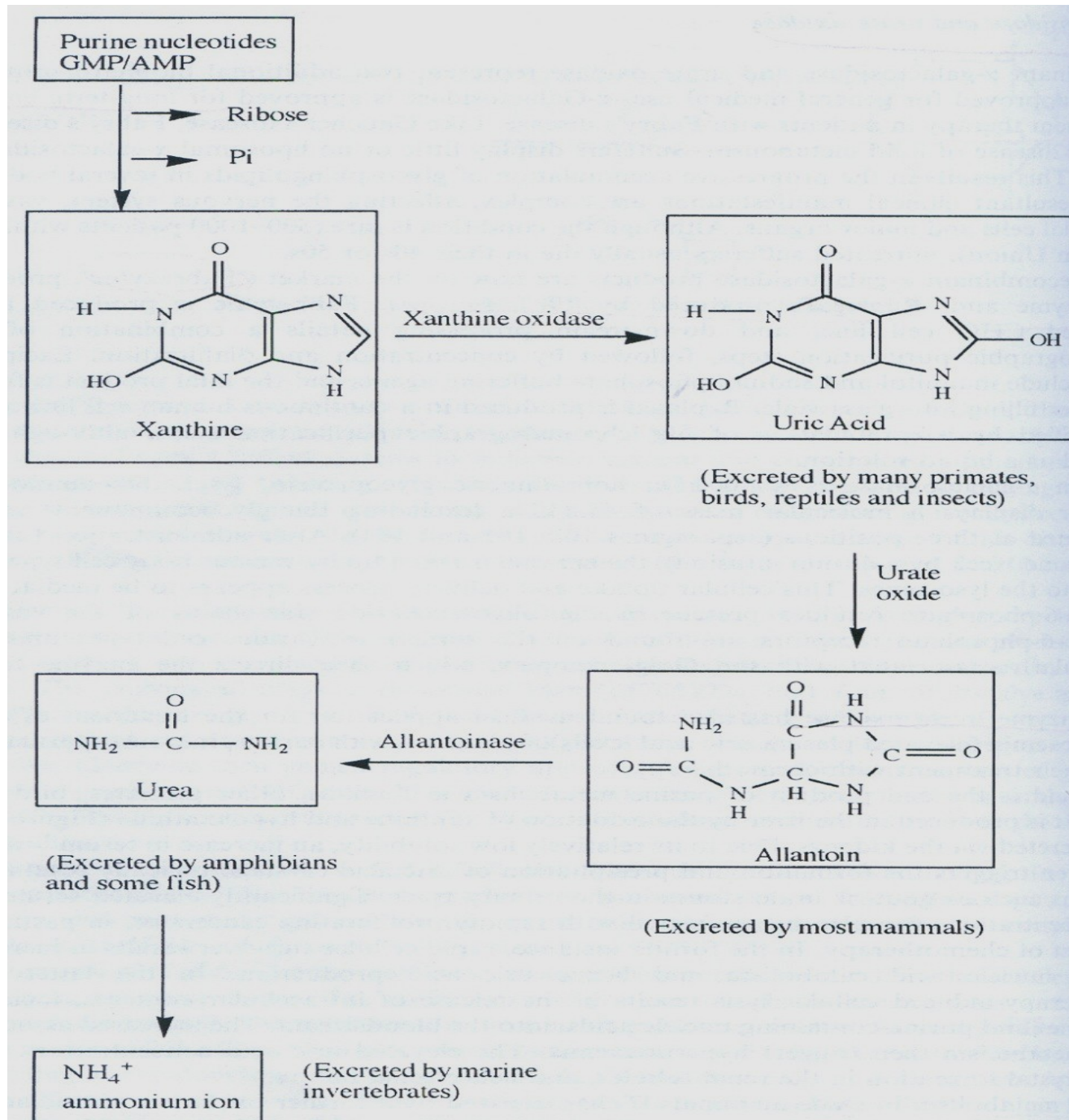


Fig. Schematics of purine metabolism

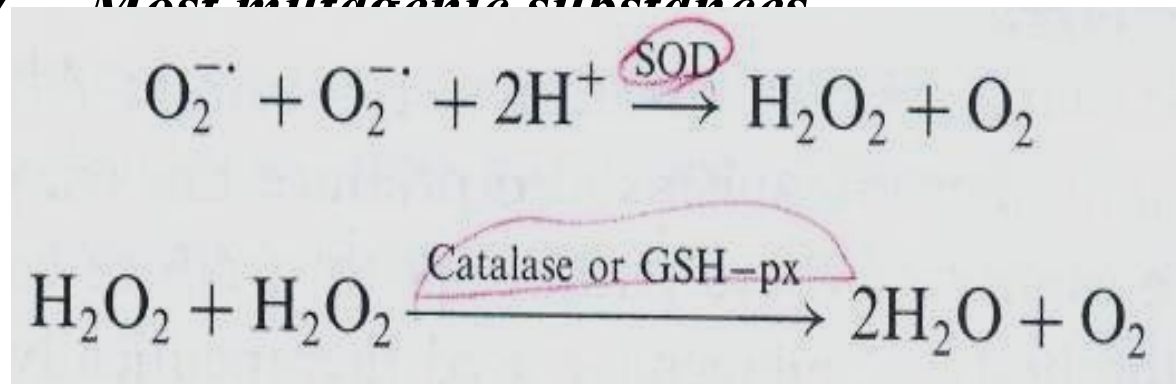
Uric acid

- Significantly elevated serum uric acid conc
 - ❖ Rapidly proliferating cancers
 - ✓ *Rapid cellular turnover results in increased rates of nucleic acid catabolism and hence uric acid production*
- Onset of chemotherapy
 - ✓ Chemotherapy-induced cellular lysis results in the release of intracellular contents, including free purines and purine-containing nucleic acids
 - ✓ Increased purine metabolism then triggers hyperuricaemia

Superoxide dismutase

- **In aerobic metabolism, oxygen** is reduced by four electrons, forming H_2O
- Incomplete reduction results in the **generation of oxygen radicals** and other reactive species (O_2^- , H_2O_2 , $\text{OH}^\cdot$)
- The radicals are **reactive** and can **attack membrane components** (nucleic acids and other cellular macromolecules)

✓ *Most mutagenic substances*



Superoxide dismutase...

- All aerobic organisms : oxygen-defence system (three kinds of Superoxide dismutase)

In human,

- **O_2^- increase and SOD decrease** : ageing, asthma, accelerated tumour growth, neurodegenerative diseases and inflammatory tissue necrosis
- Administration of SOD reduces tissue damage due to irradiation or other conditions (O_2^-)
- SOD from bovine liver or erythrocytes has been used medically

Debriding agents

- Debridement: The process of cleaning a wound by removal of foreign material and dead tissue
- Cleansing facilitates **rapid healing** and **minimize the risk of infection**

Physical means: cutting away dead tissue, washing/cleaning the wound

Enzymatic means: proteolytic enzyme

- ✓ Aqueous-based cream, bandage
- ✓ Trypsin, papain, collagenase and various microbial enzymes

Debriding agents...

Trypsin

- ✓ 24 kDa proteolytic enzyme
- ✓ Synthesized by the mammalian pancreas in an inactive zymogen form (trypsinogen)
- ✓ In small intestine, trypsinogen is activated into trypsin

Active trypsin: hydrolyze peptide bonds of **arginine** or **lysine**

Trypsinogen: is extracted from the pancreatic tissue of slaughterhouse animals

Collagenase

- ✓ Protease that can utilize collagen as a substrate
- ✓ **Applied topically to promote debridement of wounds, skin ulcers and burns**

Debridging agents...

Chymotrypsin

- Debridement, the reduction of soft tissue inflammation, facilitating cataract extraction
- Chymotrypsin is prepared by activation of chymotrypsinogen (bovine pancreatic tissue)
- Tyr, trypt ,phe

Papain

- Cysteine protease extracted from the latex of the immature fruit and leaves of the plant, Carica papaya
- 23.4 kDa, 212 amino acid polypeptide

Use: Debridging, meat tendering, clarification of beverage

Digestive aids

- To supplement normal digestive activity, or to confer upon an individual a new digestive capability
 - Powder (particularly tablet) form
 - As the enzyme never enters the blood stream, the purity need not be stringent (semi-pure preparation)
 - Reduced efficacy
 - ✓ Acidic pH condition of stomach
 - ✓ Endogenous proteolytic activities (stomach and small intestine)
- ⇒ *Partially overcome by formulating the product as a tablet*

Enzymes used as digestive aids

Enzyme	Application
α -Amylase	Aids in digestion of starch
Cellulase	Promotes partial digestion of cellulose
α -Galactosidase	Promotes degradation of flatulence factors
Lactase	Counteracts lactose intolerance
Papain	Enhanced degradation of dietary protein
Pepsin	
Bromelains	
Pancreatin	Enhanced degradation of dietary carbohydrate, fat and protein

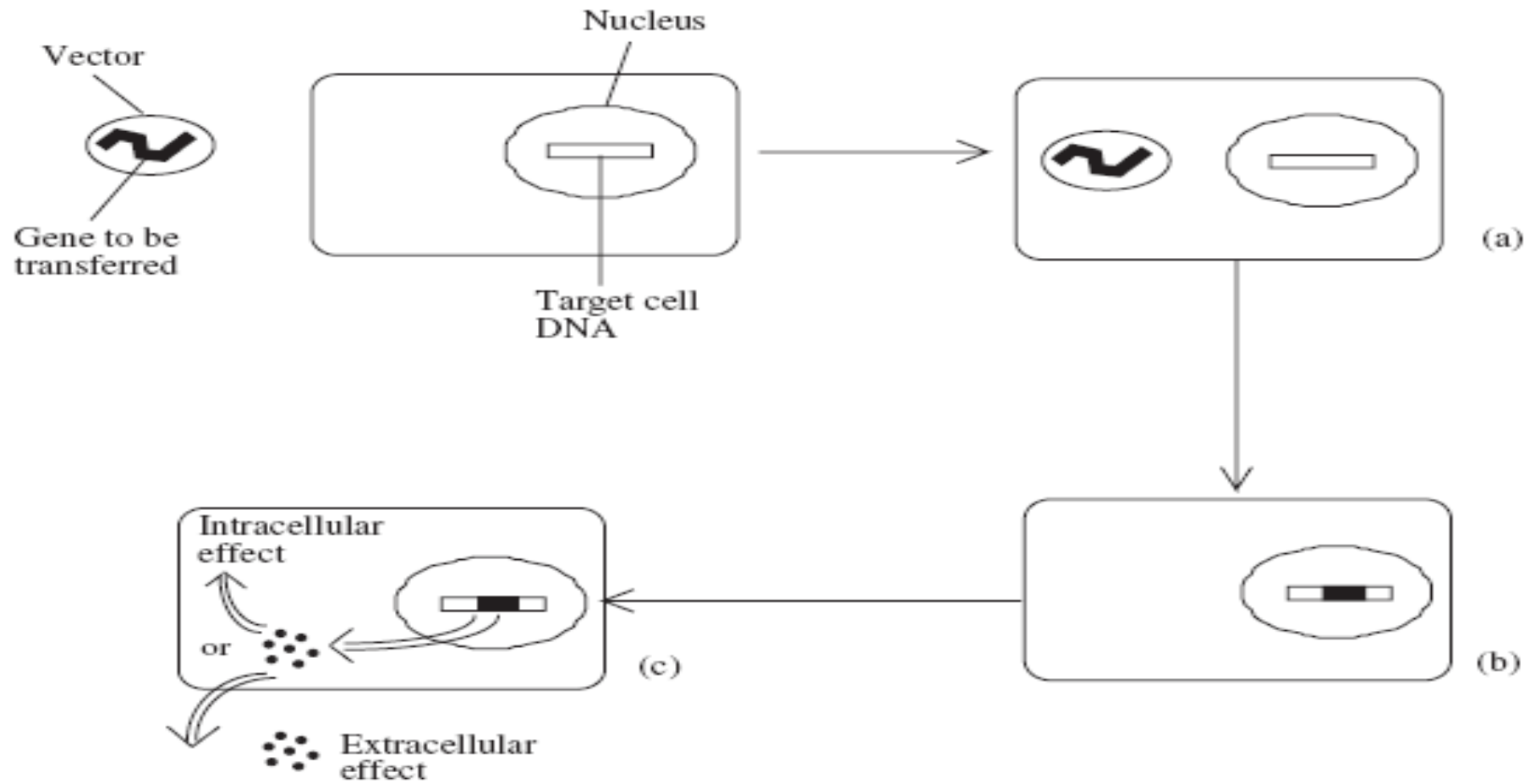
Nucleic-acid- and cell-based therapeutics

- @Current developments *in nucleic-acid-based* therapeutics centre around gene therapy, as well as **antisense technology (including RNAi)** and **aptamer technology**
- @The medical consequences of inappropriate gene (over)expression could be ameliorated/prevented by **downregulating the expression**; a nucleic-acid-based approach to achieve is termed “*antisense technology*”
- @By the early 2007 only three nucleic-acid-based products had gained approval worldwide:
 - @antisense-based product (*Vitravene*),
 - @aptamer (*Macugen*) and
 - @gene therapy product (*Gendicine*, approved only in China).

- The potential of gene therapy as a curative approach for **inborn errors of metabolism and other conditions induced by the presence of a defective copy of a specific gene (or genes)**

Diseases for which gene-based therapeutic approaches are currently being appraised in clinical trials

Cancer, various forms	AIDS
Cystic fibrosis	Haemophilia
Familial hyper-cholesterolaemia	Severe combined immunodeficiency diseases (SCID)
Gaucher's disease	α_1 -antitrypsin deficiency
Purine nucleoside phosphorylase deficiency	CGD
Rheumatoid arthritis	Peripheral vascular disease



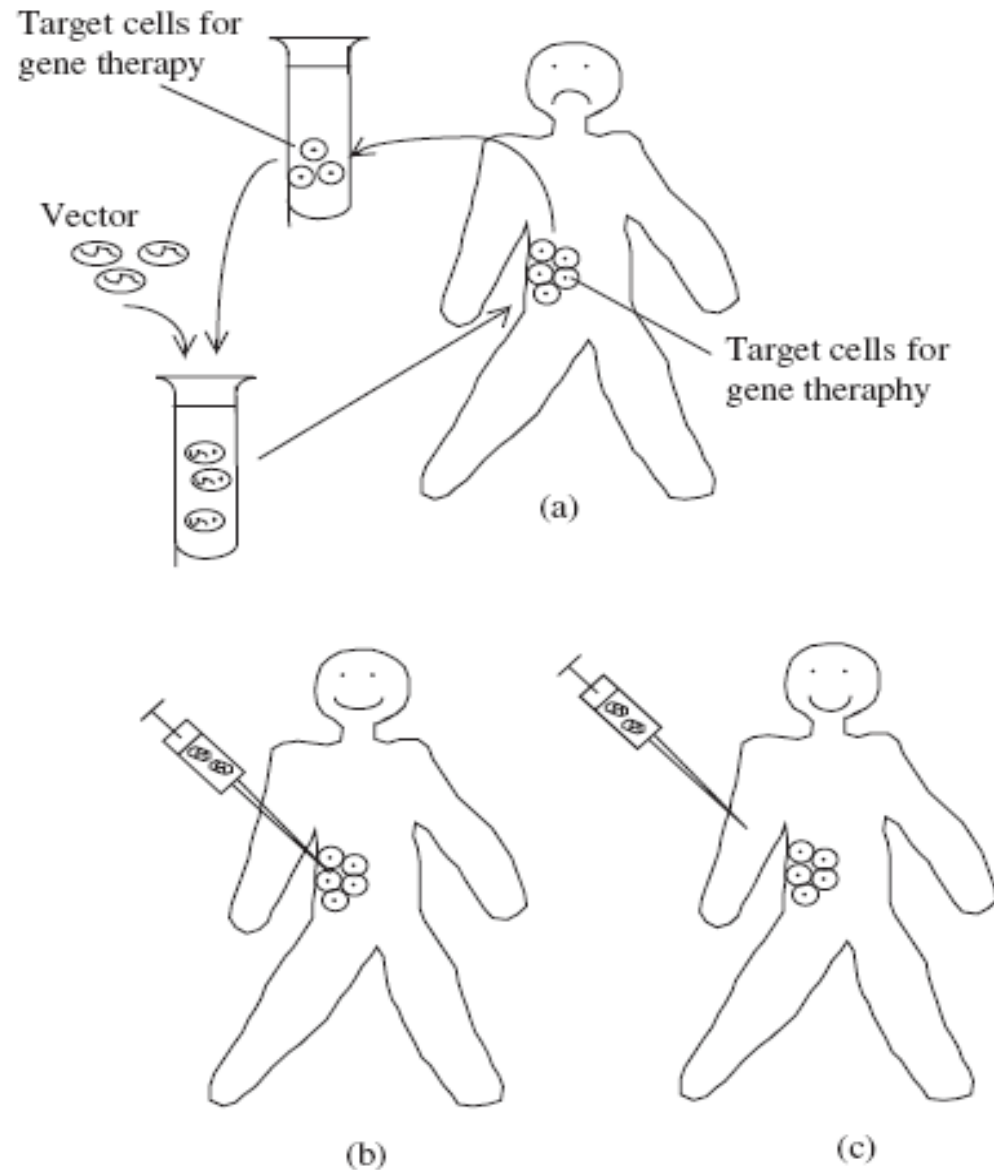
Simplified schematic representation of the basis of gene therapy

- (a) Entry of the therapeutic nucleic acid, with its vector, into the cell cytoplasm.*
- (b) Transfer of the nucleic acid into the nucleus of the recipient followed by integration of the foreign genetic material into the cellular DNA.*
- (c) The foreign gene is expressed, resulting in the synthesis of the desired protein product.*

(a) In vitro gene therapy: removal of target cells from the body followed by their incubation with nucleic acid-containing vector. After the vector delivers the nucleic acid into the human cells, they are placed back in the body.

(b) In situ gene therapy: entails direct injection of the vector immediately adjacent to the body target cells.

(c) In vivo gene therapy: involves intravenous administration of the vector. The vector has been designed such that it will only recognize and bind the intended target cells.



The various practical approaches for gene therapy.

Retroviral vectors

- **Retroviruses are enveloped viruses**; their genome consists of **single-stranded RNA (ssRNA)** of approximately 5–8 kb
- Upon entry into sensitive cells, the viral RNA is reverse transcribed and eventually yields double-stranded DNA which subsequently **integrates into the host cell genome**

(MoMuLV): is the most commonly employed (recombinant) retrovirus derived from the Maloney murine leukaemia virus

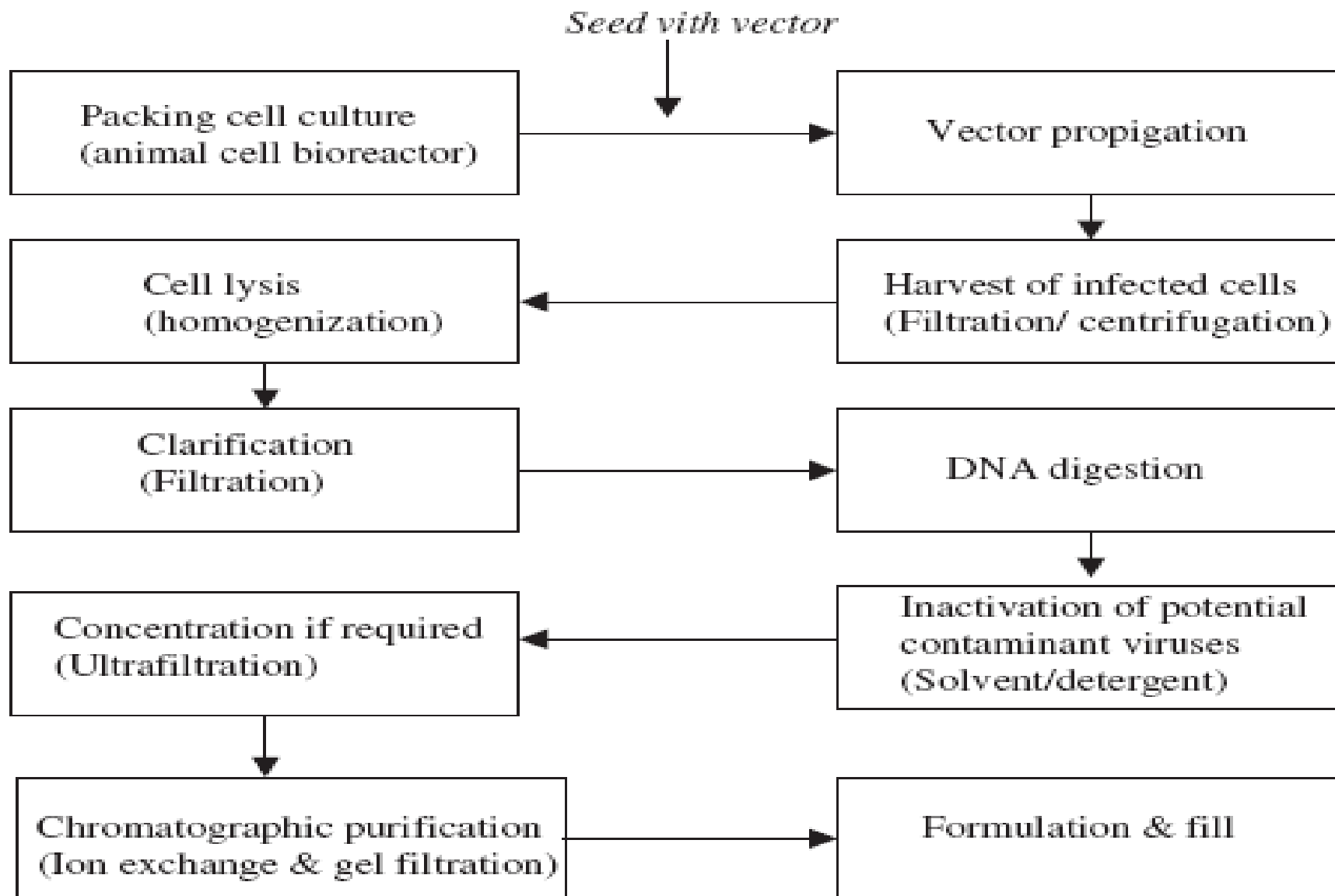
Retroviruses properties influence their potential as vectors in gene therapy:

- **Have been studied in detail and their biochemistry and molecular biology are well understood**
- **Can integrate their proviral DNA** only into actively replicating cells
- **Integrated DNA can be subject to long-term**, relatively high-level expression
- Proviral DNA integrates randomly into the host chromosomes
- Complete copies of the proviral DNA are passed on to daughter cells if the original recipient cell divides;
- Safety studies using retroviral vectors have already been carried out on various animal species

Adenoviral-based vectors

- Adeno-associated virus, the herpes virus, and a number of other viruses, are also being considered in gene therapy
- Adenoviruses are relatively large, non-enveloped structures, housing double-stranded DNA as their genetic material
- Their major advantage relates to their ability to infect non-dividing cells efficiently and the usually observed expression of large quantities of the desired gene products

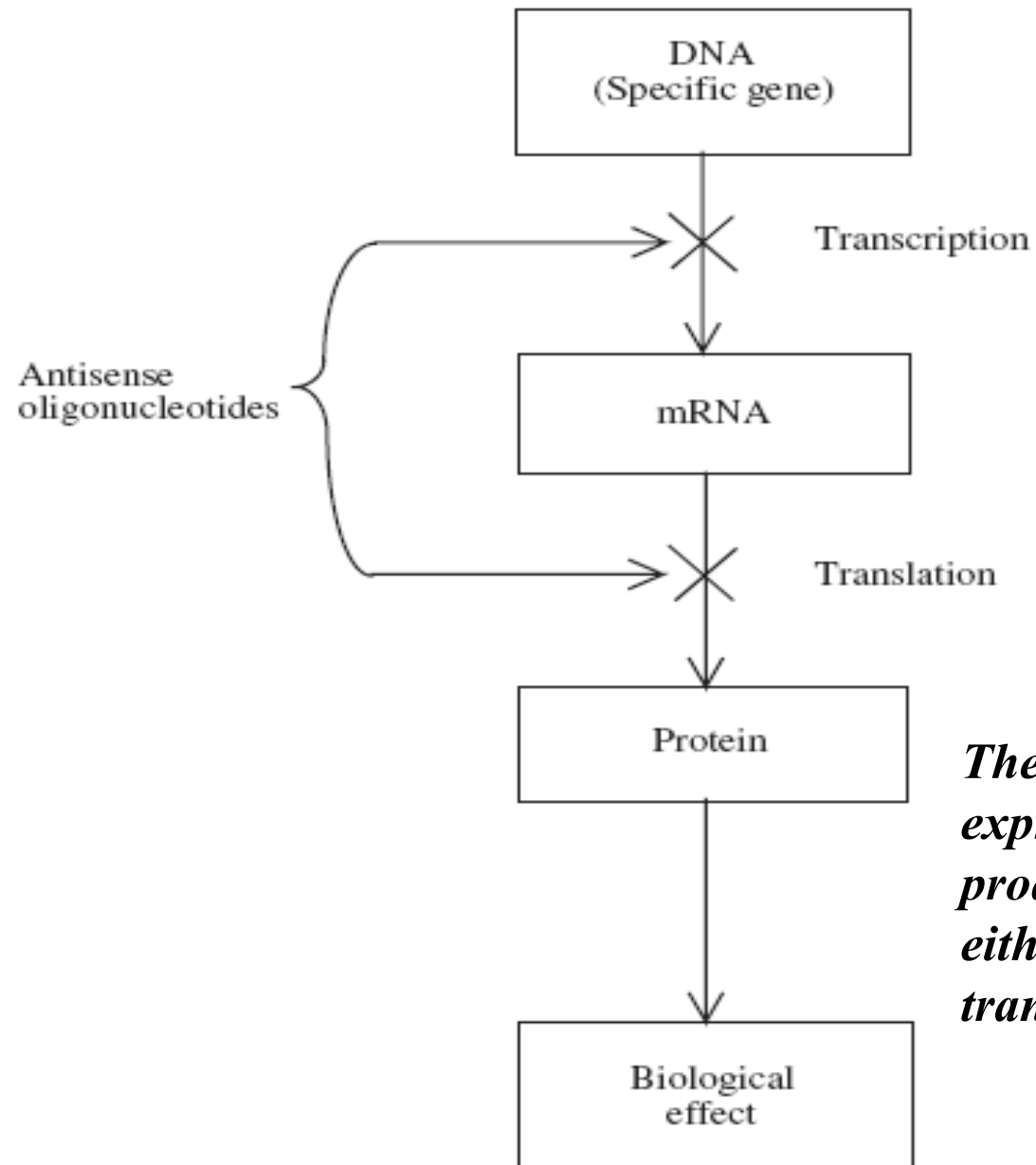
Advantages	Disadvantages
Adenoviruses are capable of gene transfer to non-dividing cells	Adenoviruses are highly immunogenic in man
They are easy to propagate in large quantities	The duration of expression of transferred genes can vary, and is usually transient
High levels of gene expression are usually recorded	Infection of permissive cells with wild-type adenovirus usually results in cell lysis
They are relatively stable viruses	Adenoviruses display a broad selectivity in the cell types they can infect



Large-scale manufacture of adenoviral vectors for use in gene-therapy

Antisense technology

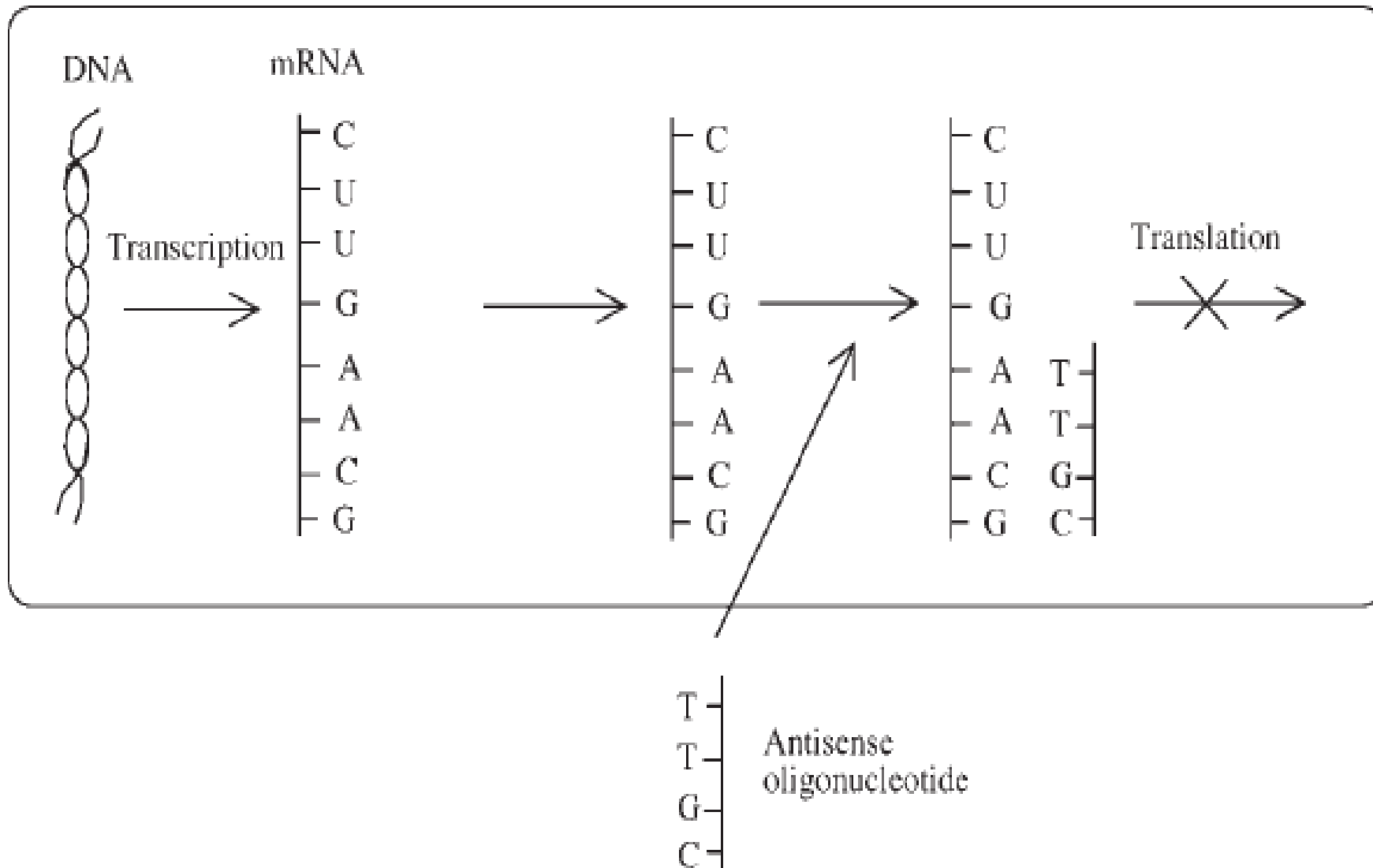
- Various disease states are associated with the inappropriate production/overproduction of gene products
 - ✓ **Expression of oncogenes**, leading to the transformed state
 - ✓ **Overexpression of cytokines** during some disease states with associated worsening of disease symptoms
 - ✓ **Overproduction of angiotensinogen**, - which ultimately results in hypertension
 - ✓ Intracellular transcription and translation of virally encoded genes during intracellular viral replication.
- The medical consequences of such inappropriate gene (over) expression could be ameliorated/prevented if this expression could be downregulated
- A nucleic-acid-based approach to achieve just this is termed '**antisense technology**'.



The end goal is the prevention of expression of a particular gene product (invariably a protein) by either blocking the transcription or translation of that gene

Antisense oligonucleotides

- The **nucleotide sequence of mRNA molecule** contains the encoded blueprint that dictates the amino acid sequence of a protein (make 'sense'), the mRNA sequence is complementary to an 'antisense' DNA strand, i.e. it is the antisense strand of DNA in a given gene that serves as template for the mRNA synthesis)
- As long as such an 'antisense' oligonucleotide can enter the cell, the complementarity of sequences can **promote hybridization between the mRNA and the antisense oligonucleotide**



Outline of how an antisense oligonucleotide can prevent synthesis of a gene product by blocking translation

Major advantages of antisense oligonucleotides

- ✓ *Relatively low toxicity and few significant side effects due to the highly specific nature and the they are 'natural' biomolecules*
- ✓ *The requirement for only low levels as target mRNA is itself present in nanomolar concentrations*
- ✓ *Easier and straightforward manufacturing of specified nucleotide sequence using automated synthesizers*

Disadvantages of antisense oligonucleotides

- ✓ *Sensitivity to nucleases*
- ✓ *Very low serum half lives*
- ✓ *Poor rate of cellular uptake*
- ✓ *Orally inactive*

Aptamers

- Aptamers are *single-stranded DNA or RNA-based sequences* that fold up to adopt a unique three dimensional structure, allowing them to bind a specific target molecule
- Because of their **high binding specificity and affinity**, aptamers (like antibodies) are/may prove useful for **affinity-based purification, target validation and drug discovery, diagnostics and therapeutics**

Macugen is one such product has been approved for general medical use to date

- A number of additional aptamers are in clinical trials, aimed at treating conditions including **infectious diseases, cancer and haemophilia**

THANK YOU